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**ENG509 Midterm Short Notes BY
PIN2 and MUHAMMAD
(MAS All Rounder)**

Lesson 01

What is Morphology?

Topic: 01: Introduction:

Some basic notions in morphology.

- What does morphology study?
- What are the goals of morphological analysis?
- What is its history?
- The term morphology
- Johann Wolfgang von Goethe (1749– 1832)
- **Greek:** morph- means ‘shape, form’,
- Morphology is the study of form or forms

In linguistics “morphology refers to

- The mental system involved in word formation

Or

- The branch of linguistics that deals with words, their internal structure, and how they are formed.

According to Fabregas & Scalise (2012):

- Morphology is the part of linguistics that studies grammatical properties of words and how words relate to each other in a language.

The chief task of morphology:

(1) a. deep

b. deepen

(2) a. dog

b. dogs

Deep and dog are morpheme that denote a property having to do with a physical dimension or referring to

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an animal. -en is a morpheme that forms verb and is associated with the meaning (causes to) become x. -s is morpheme that makes -s plural.

- The central question in morphology

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- How words are related to each other?

Are there their some aspects of morphological research which have something to do with morphology?

Topic: 02: Morpheme:

Morphologists investigate words:

- Their internal structure
- How they are formed.

‘The smallest linguistic pieces with a grammatical function.’

Not inclusive definition

- A morpheme may consist of a word, such as hand,
- A meaningful piece of a word, such as the -ed of looked,

Another way in which morphemes have been defined is as:

- ‘a pairing between sound and meaning.’
- Some morphemes have no concrete form or no continuous form,
- Some do not have meanings in the conventional sense of the term

The term morph: specifically to the phonological realization of a morpheme.

- [h]and
- [t] jumped
- [r]ooted
- [ed] rooted and wedded

Let's consider

Reconsideration.

- Re + consider + ation.

Stem: ‘A stem is a base unit to which another morphological piece is attached.

Root: consider a root is like a stem in constituting the core of the word to which other pieces attach, but the term refers only to morphologically simple units.

For example, disagree is the stem of disagreement,.

- **Disagree:** agree is both the stem the root of the entire word.
- **Affixes:** re- and -ation

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- -ation: -ate and -ion, (cf. *reconsiderate),

Consider the English words

- lovely and quickly. the suffix -ly.
- Two other sorts of affixes
- infixes and circumfixes.

(1) Root -um /sulat/ /s-um-ulat/ 'one who wrote'

(2) /gradwet/ /gr-um-adwet/ 'one who graduated'

Circumfixes

- stem besar 'big' to form a noun ke-besar-an meaning 'bigness, greatness' (MacDonald 1976: 63; Beard 1998: 62).

Topic: 03: Morphology in Action:

These are examples of morphology in action –

Morphological facts of everyday life.

- Novel words and word play
- "Rebop,"

Consider now the following phrases, taken from a Toni Braxton song:

- Unbreak my heart, uncry these tears

All human beings have this capacity for generating and understanding novel words. Sometimes someone creates an entirely new word, as J. R. R. Tolkien 'hobbit'. But more often than not, we build new words from pre-existing pieces, as with unbreak and uncry, or as with hobbitish and hobbit-like, built by adding suffixes to the stem hobbit. We could easily go on to create more words on these patterns. Novel words are all around us.

Topic: 04: Abstract Morphological Facts:

1. I eat one melon a day.
2. We eat two melons a day.
3. *We eat two melon a day.

No morphological plural

- Indonesian or Japanese

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- Two melon (three melon, four melon, etc.)

Indonesian:

Saiga makan dua buah semangka (se) tiaphari

I eat two fruit melon every day

'I eat two melons every day.'

Japanese:

mainichi futatsu-no meron-o tabemasu

every.day two- gen melon-obj eat.imp

'I eat two melons every day.'

- The evil giant at the top of the beanstalk eats two melons, three fish, and four children a day.

Children

Consider the following:

- Today they claim that they will fix the clock tower by Friday, but yesterday they claimed that it would take at least a month
- speaking Vietnamese
- speaking Chinese
- Today they say ... but yesterday they said ... tell us told us know knew
- That these verbs and others do not add - about English morphology.

It is a fact about English that there is a morphological distinction among universal quantifiers between the one designating all of two (both) or all of more than two (all) of a particular type of entity.

Ancient Greek

English

Singular	stratiô:tes	One horse
Dual	stratió:ta	Two horse
Plural three or more than	stratiô:tai	Three or more horses

Manam (Papua New Guinea: Gregersen 1976) and Larike (Central Maluku, Indonesia: Laidig and Laidig 1990) that distinguish not only singular, dual, and plural, but also trial

Topic: 05: Background & Beliefs:

A general introduction to morphology and morphological analysis from the point of view of a morphologist. Some of our foundational beliefs about linguistics and linguistic methodology. **First**, we believe that languages differ from one another. **Second** that languages, which we can write with a small l, are different from Language, with a capital L.

Language vs language

Individual languages have features that are not characteristic of Language in general. Linguists need to pay equal attention to both small-l language and capital-L Language.

Our next belief

- Morphology is a distinct component of languages or grammars.

We also have some that pertain to methodology. The first is that we should take an attitude of skeptical realism. Albert Einstein said that a physicist must be both a realist and a nominalist. Our other methodological belief can be summed up as a motto: **anything goes**(**Paul Feyerabend, a twentieth-century philosopher**).

- A noholds-barred approach to linguistics
- Our tools are not theory-based in that way.

Lesson 02

Words and Lexemes

Topic: 06: Intro to Morphological Analysis:

Two basic approaches:

- Analysis

- Synthesis

Principle 1

Forms with the same meaning and the same sound shape in all their occurrences are instances of the same morpheme.

Principle 2

Forms with the same meaning but different sound shapes may be instances of the same morpheme if their distributions do not overlap.

Principle 3

Not all morphemes are segmental.

Principle 4

A morpheme may have zero as one of its allomorphs provided it has a non-zero allomorph

Topic: 07: What is a Word:

What is a Word?

A single word can have multiple uses and interpretations.

- MINERS REFUSE TO WORK AFTER DEATH
- EYE DROPS OFF SHELF
- LOCAL HIGH SCHOOL DROPOUTS CUT IN HALF

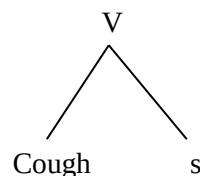
Words like noun, verb, adjective, and adverb refer to what linguists call lexical category. They are labels that tell us how a word is generally used in a sentence. A noun can be the subject of a sentence, but not so a verb. In many cases, identical-sounding or identical-looking words can belong to multiple categories, and that is what is going on in some of these sentences.

What is a Word?

Various ways to define a word, but no definition is entirely satisfactory

Defining words syntactically

- Harry coughs every time he steps outside.



“What is syntax?” If we think of syntax as the component of the human grammar that governs the

ordering of items, then -s should be a word.

Another characteristic the smallest unit of language that can stand alone:

A. When are you going to the store?

B. Tomorrow.

Affixes: units incapable of standing alone

A. Whose book is this?

B. *My.

In the musical Camelot, Queen Guenevere sings the following lines:

(5) It's May, it's May, the month of "yes, you may" The time for every frivolous whim, proper or im ... When all the world is brimming with fun, wholesome or un

Defining words phonologically

- Words tend to be important units phonologically as well as syntactically.
- The word is typically the domain of stress assignment.

(6) a. o ánthropos 'the person'

b. o ánthropòs mas 'our person'

Grammatical words

Despite the elusiveness of a definition of word, speakers – literate and illiterate – have clear intuitions about what is and what isn't a word. The term grammatical word or morphosyntactic word is virtually synonymous with word but is generally used to refer specifically to different forms of a single word that occur depending on the syntactic context.

Topic: 08: Types of Words:

Words may be defined in different ways from different perspectives, with each perspective picking out a somewhat different object. Linguists distinguish phonological words, grammatical words and lexemes.

Phonological words

A string of sounds that behaves as a unit for certain kinds of phonological processes, especially stress or accent. Every phonological word has a main stress. Separate elements written as e words but do not have their own stress are not phonological words in English.

- The white dogs ran for the lake.
- white dogs ran lake

No stress on *the* or *for*. The string *for the lake*, written as three separate words, is a single phonological clitics.

word. Items like *the* and *for*, phonologically dependent on adjacent words, are called

Syntactically, **clitics** pattern like distinct words, but they cannot stand alone phonologically. **Proclitics** precede their host and enclitics follow it. Well-known examples of clitics are the contracted form of the English auxiliary verb **be**.

- Mary's here or We're in this together.

Distinguish between content words and function words

Content words	Function words
Nouns: baby, bargain, Josianne	Pronouns: I, him, our
Verbs: publicize, hurtle, sleep	Verbs: am, was, should
Adjectives: peaceful, quick, bright	Demonstratives: this, those Adverbs:3 very, not Preposi
Adverbs: readily, carefully	by

If we took the function words out of speech, it would be hard to figure out what was going on:

- took function words speech hard figure going on

Lexemes

- dog1: [noun], a canine
- dog2: [noun], a hooked or U-shaped device used for gripping heavy objects
- dog3: [verb], to follow closely and persistently

When we want to distinguish among phonologically similar forms on the basis of their differing meanings, as in, we call each a separate lexeme. A lexeme is a theoretical construct that corresponds roughly to one of the • common senses of the term word.

Topic: 09: Empirical Tests for Word hood:

Empirical Tests for Wordhood

Difficult to come up with a definition that tells us whether something is a word that can tell us whether something is or isn't a word.

Fixed order of elements

- unbreakable *breakableun or *unablebreak

The same doesn't hold for sentences.

- “I see what I eat”
- “I eat what I see,”
- “I like what I get”
- “I get what I like”

When we change the order of morphemes in a word, we generally end up with something ungrammatical. In English we cannot change the order of words in a sentence any way and still have a grammatical result. *get like I what I.

- Non-separability and integrity
- non-separability

Words differ from larger units, such as phrases, in that they cannot be broken up by the insertion of segmental or

- A shirt
- A *white* shirt

Integrity

Likewise, syntactic processes cannot be applied to pieces of words.

- a. *Possible, it’s im-. * Which school did you see bus?

Non-*school bus* consist of a single word, rather than a pair of words.

Stress

The diagnostics given in the preceding section, non-separability and integrity, establish that hot dog (the edible kind) is a compound.

- The hot dog you are eating is hotter than mine,
- *You were eating a hotter dog
- *very hot dog.

Topic: 10: Inflection Vs. Derivation

Inflection vs. Derivation

Inflection involves the formation of grammatical forms – past, present, future; singular, plural; masculine, feminine, neuter; and so on – of a single lexeme. The use of these grammatical

forms is generally dictated by sentence structure. Thus is, are, and being are examples of inflected forms of the lexeme **be**,

Regular verb lexemes in English have a lexical stem, which is the bare form with no affixes (e.g., select) and three more inflected forms, one each with the suffixes -s, -ed, and -ing

Noun lexemes in English have a singular and plural form. Adjectives, adverbs, prepositions, and other parts of speech typically have only one form in English. Inflection can be realized through affixes.

Examples of words + inflectional morphemes

- **Nouns:** wombat + s ox + en
- **Verbs:** brainwash + es
- dig + s
- escape + d
- rain + ing

Derivation involves the creation of one lexeme from another, Compounding is a special type of derivation, since it involves the creation of one lexeme from two or more other lexemes.

- Bookshop,
- greenhouse

Derivation generally results in a change in lexical meaning or the lexical category of a particular word, while inflection does not.

Examples of words + derivational affixes

Noun to noun	New York + ese /fish + ery /Boston + ian /auto + biography/vice + president
Verb to verb	un + tie /re + surface/pre + register /under + estimate
Adjective to adjective	gray + ish/a + moral/sub + human/il +legible

Noun to adjective	hawk + ish /poison + ous/soul + ful /iron + like
Verbs to nouns:	acquitt + al /digg + er
Adjective to adverb	sad + ly/ efficient + ly

Topic: 11: Two Approaches to Morphology

Hockett (1954) Two Approaches to Morphology:

- Item-and-Arrangement
- Item-and-Process

Developed by structuralists I&A proceeds from a picture of each language as a set of elements and the patterns in which those elements occur. with word analysis, using techniques for breaking words down into their component morphemes, which are the items.

Morphology is the arrangement of these morphemes into a particular order or structure.

For example, books results from the concatenation of the two morphemes

book and -s.

I&P is an approach in which complex words result from the operation of processes on simpler words. Working in an I&P model, we might say that ‘books’ results when the lexeme book undergoes the function ‘make plural’

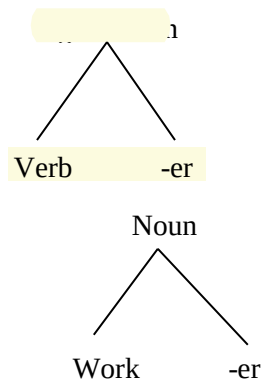
In regular cases, this function will add the segment which is realized as /-s/ after most voiceless segments (cf. giraffes), and as /e z/ after sibilants and affricates (cf. roses).

[]. Everything you can express in I&A can be expressed in I&P and almost anything you can express in I&P can be expressed in I&A.

- Affixation in the I &P and I& A
-) X]V (er)@]N
- think]V er]N, runn]V er]N, fli]V er]N, hunt]V er]N

We generally think of lexeme-formation functions as having a phonological, a syntactic, and a semantic component.

Additive functions like this one are easily recast in the I&A model.



- Non-affixal phenomena and the I&P model
- Many languages have morphological phenomena that do not involve affixes.
- These can be represented within the I&A model and morphological frameworks based on I&A.

Verb

- Overflow
- conduct
- insert
- reject
- convict

Noun

- overflow
- conduct
- insert
- reject
- convict

• Verb	Noun
• Over□flow	□overflow
• con□duct	□conduct
• in□sert	□insert
• re□ject	□reject
• con□vict	□convict

Topic: 12: The Lexicon

Greek *lexikós* ‘pertaining to words’ and often designates a book containing a list of words in a language along with their definitions. Linguists use the term in particular to refer to the mental dictionary. Within linguistics, lexicon has taken on multiple definitions. Two widely accepted views of the lexicon. According to one, the lexicon is a list of the indivisible morphological units, or morphemes, in a language. The second (Bloomfield 1933), is a list of irregular or arbitrary forms. As irregular or arbitrary, they must be memorized

- [arbr]
- slide

It would be an error to assume that the first definition is equivalent to the second and that the list of irregular forms is a list of morphemes, which is to say a list of indivisible units. But where natural language is concerned, this position is too extreme. Evidence suggests that even morphologically complex forms are present in a speaker’s lexicon.

- Representative
- re-,present, and -ative

Most words ending in -ative are adjectives

“Meaning of a complex word is the sum of the meaning of its parts, because the difference between the meaning that we expect a word to have on the basis of the meanings of its parts and the meaning that it actually has is quite subtle.

The lexicon contains more than words. Affixes, such as English re

lexicon. Speakers know and understand such affixes and readily attach them to new stems. Some affixed

inflected forms, like *says*, must also be in a lexicon. We know this because *says* is an exception to the general rule “Add /-z/ to the basic stem of a verb to form the third person singular present.”

Say *say* and *says* out loud: *say* [sei] has a tense vowel, but *says* [sɛz] has a lax one.

Antidisestablishmentarianism

- anti-, dis-, establish, -ment, -ary, -an, and -ism

We need to list some compounds

- Fixed phrases (with respect to ...,
- names of people and places
- proverbs all need to be memorized, too, and are arguably in the lexicon.

The most accurate conception of the lexicon

- a list of forms that you know.
- *ir*

It cannot simply be a list of indivisible morphological elements. Instead, it contains irregular forms, forms that are in some way unpredictable. Some are indivisible or unanalyzable morphologically, and others are not.

Lesson 03

Morphology and Phonology I

Topic: 13: Morphology and Phonology

Some of the many interactions that take place between morphology and phonology. These interactions and the grammar that describes them are often called morphophonology or morphophonemics. By looking at phonological processes such as assimilation and the effect they have on the shapes of morphemes

as words and stems. Move on to two general types of affixes that are distinguished, in part, by phonological criteria. Their phonological behavior reveals details about their underlying structure and the point at which they attach to their bases.

A look at secret languages in which morphology and phonology interact to disguise the shapes of words. Readers to have the rudimentary knowledge of linguistics. You need to know three terms that are often not introduced in such courses:

- onset, nucleus, and coda
- *cat*

Topic: 14: Allomorphs

Kujamaat Jóola, the stem /baj- in the presence of a morpheme with an underlying tense vowel, and [baj-] elsewhere. In English the plural marker comes in several shapes, among them [s] as in lips, [z] as in balls, [e z] as in roses, [ŋ] as in oxen, and null as in sheep. Alternations based purely on phonological context, as with the Kujamaat Jóola facts.

In the English data, only the first three allomorphs of the plural suffix depend on phonological context. The last two, as in oxen and sheep, are lexical, and of no concern here.

Like the English plural suffix -s, the Eng [ed]:

- [d] blamed [blejmd],
- [t] jumped [dʒAmpt],
- [ed] aided [eIdId]

The

We can formulate the distribution of the allomorphs in even simpler terms: the English past tense suffix is /d/. Where we find [t], the /d/ has assimilated to the preceding segment in voicing.

- [e d], the [e] has been added by an automatic phonological rule of epenthesis that is triggered by the fact that the final segment in the verb and the suffix itself agree in both place and continuancy.
- /d/ the basic form or the basic allomorph of the English past tense suffix.

It is not always easy, or even possible, to determine the basic form of a morpheme. To do so, we must decide which form of a morpheme best accounts for the full range of data. The forms coming below are in the nominative case, used for subjects, and those on the right are in the genitive case.

- (2) aithiops 'Ethiopian' aithiopos 'of an Ethiopian' phleps 'vein' phlebos 'of a vein'
- phlep-, the stem of the nominative, or phleb-, of the genitive

Epenthesis:

The insertion of a sound or letter within a word, e.g. the b in thimble. Similarly if you say film as "FIL-um,"

As morphologists, we are used to dealing with cases like this. Unlike syntax, which tends to be very regular, morphology is full of irregularities and exceptions.

Topic: 15: Prosodic Morphology

Prosodic morphology deals with the interaction of morphology and prosodic structure. Prosodic structure is particularly concerned with the timing units of languages, for example, the word and syllable,

and vowel length.

From this general category we present three phenomena:

- phonotactic constraints,
- root-and-pattern morphology,
- reduplication

Phonotactic constraints

Phonotactics is a branch of phonology that deals with restrictions in a language on the permissible combinations of phonemes.

Phonotactic constraints

- twelfths /twelfθs/
- s/ + /t/ + /j/ (not in most accents of American English)
- /s/ + /p/ + /j ɪ l/
- /s/ + /k/ + /j ɪ l w/

At their most basic, phonotactic constraints determine the minimum length of content words in particular languages. For example, in Mohawk, each content word contains at least two syllables (Michelson 1988, cited by Hayes 1995: 47). Other languages require a heavy syllable. In Mayan languages, roots are predominantly of the shape CVC and in Bantu they are generally CVCV. In Semitic languages, roots consist of three consonants: CCC.

Root-and-pattern morphology

In Semitic languages such as Hebrew and Arabic, roots generally consist of three consonants. To form words, vowels are superimposed on this consonantal pattern. We call this type of ‘morphology root-and-pattern’.

- Triconsonantal root, M-L-K./k/ is realized as the fricative [x].
- **M-L-K:** melex ‘king’ malkah ‘queen’
- malax ‘he reigned’ yimlox ‘he reigns, he will reign’
- malxut ‘royalty, royal power, reign, kingdom’
- mamlaxah ‘kingdom, sovereignty,

Reduplication

A continuous substring from either the beginning or the end of a word is copied. Languages may use

- kaldín ‘goat’ kal-kaldín ‘goats’

- púsa ‘cat’ pus-púsa ‘cats’
- kláse ‘class’ klas-kláse ‘classes’
- jyánitor ‘janitor’ jyan-jyánitor ‘janitors’ ró?ot
- ‘litter’ ro:-ró?ot ‘litter (pl)’
- trák ‘truck’ tra:-trák ‘trucks’

Topic: 16: Primary and Secondary Affixes

Over the years, our knowledge of morphological structure has been enhanced by work in phonology. By observing the phonological processes that take place or do not take place within particular sets of morphologically complex words. One distinction that has come out of work that pairs morphology and phonology is between primary and secondary affixes, known as level 1 and level 2 affixes or class 1 and class 2 affixes. In English, this distinction is intimately connected with language history. Primary affixes in English are often of Latin-Romance origin, secondary origin. Etymology can only take us so far in morphological analysis. The primary-¹ is a living process, regardless of its history, and in English, as in other languages of the world, it cannot be explained away as etymological residue.

Some example³ of words bearing -(i)an, a primary affix and ones bearing -ism, a secondary affix

			Primary
a.	Mendel	→ Mendelian	
b.	Mongol	→ Mongolian	
c.	c. Parkinson	→ Parkinsonian	
d.	d. Shakespeare	→ Shakespearian	
e.	grammar	→ grammarian	Secondary
f.	Mendel	→ Mendelism	
g.	Mongol	→ Mongolism	
h.	Parkinson	→ Parkinsonism	
i.	national	→ nationalism	

j. **capital** → **capitalism**

Primary affixes cause a stress shift, while secondary affixes do not. If primary and secondary affixes both occur in the same word, we can make a second prediction. The primary affix will occur closer to the stem than the secondary affix. Traditional usage among morphologists is to use the symbol ‘+’ to mark the juncture between a stem and a primary affix and to use ‘#’ to mark the juncture between a stem and a secondary affix.

Same stem (primary suffix) +able		Same stem(secondary suffix) ‘#’ able	
reparable	capable of being repaired’(liable to be paidback or recovered’)	repairable	capable of being repaired’(a broken appliance)
+able	Base	#able	
prefer+able [prɛˈf(e)r e bəl]		prefer#able [pr e fˈeːr e bəl]	
compar+able [kɑˈmp(e)r e bəl]		compar#able [kɑmpəˈr e bəl].	
a. cultivable	Base	cultivable	Base
b. educable	Cultiv	Educatable	cultivate
c. irrigable	Educ irrig	Irrigatable	Educate
d. navigable	navig	Navigatable	irrigate
e. demonstrable	demonstr	demonstratable	navigate demonstrate

Prefixes can be primary or secondary as well.

Primary Prefix

In+ (not)

(in-

a. irregular

b. inviolable

c. imperceptible

d. indivisible

Secondary Prefix

n#(r

unregulatable

unviolatable

unperceivable

undividable

Important difference

Primary	Secondary
Often of Latin-Romance origin	Often of Germanic origin
Cause a stress shift	Do not cause a stress shift
Usually occur closer to the stem	Usually occur outside of primary affixes secondary affixes
The semantics of the derived form tend to be less compositional	The semantics of the derived form tend to be compositional
May attach to a non-basic allomorph	Attach to a stem's basic allomorph of the stem
The affix itself may have allomorphs	The affix itself does not have allomorphs
May attach to non-lexical stems	Attach to lexical stems (i.e., words from lexical categories N, V, Adj/Adv)

Topic: 17: Linguistic Exaptation, Leveling, and Analog

Rudes (1980) and Lass (1990) have both raised the question of what to do with “linguistic left-overs” (Rudes’s term) or “linguistic junk” (Lass’s term). In both cases, it has to do with morphemes that lose their semantic content or morphosyntactic function as a result of language change and are left as contentless, functionless strings of phonemes floating around in the system. They show that languages are in general intolerant of useless elements, and speakers reanalyze them as having a new role. Lass calls this process linguistic exaptation, extending a term of evolutionary biology to the study of language change. Cameron- classes and gender, and stem alternation, respectively, suggest that linguistic exaptation is pervasive.

A natural consequence of a core psycholinguistic mechanism that makes it easier for speakers to master complex inflectional systems or to learn the meanings of new vocabulary items. It plays an important role in the evolution of inflectional systems cross linguistically.

Germanic family has exaptation

Indo-European languages have a distinction (e.g., English write, wrote). originally been used to encode aspectual distinctions. The case: the verbal suffix -esc in Romance. Latin, where the suffix -sc attached to sequences of verb stem plus theme vowel to form the inchoative aspect (which has the general meaning 'to begin to').

Uninchoative aspect		inchoative aspect	
palleo	I am pale	- pallesco.	'I begin to pale'.
amo -	'I love',	amasco	'I am beginning to love',

English doesn't have a productive inchoative aspect, but we do have pairs like white and whiten 'become pale; begin to be pale'. The tendency for languages to prefer regular paradigms over irregular ones sometimes leads to **leveling**, the elimination of differences in meaning.

Grammatical Function	Prehistoric Latin	Old Latin	Classical Latin
Nominative	col-os	col -os	color
Genitive	es *col-os-	col -or-is	col-or- is
Dative	*col-os-ei	col -or-ei/-e	col-or- i
Accusative	em *col-os-	col -or-em	col-or- em

Ablative	*col-os-i	col -or-e	col-or- e
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Hock gives the example of English plurals. It is because of four-part **analogy** cow is cows, replacing the earlier form kine. The new plural cows generalizes the plural formation familiar from other words, such as stone, stones. stone :

- stone-s cow : X = cow-s

Leveling and analogy are powerful forces in the development of languages over time driven by a seemingly innate preference in speakers for phonological and morphological similarity between members of a paradigm or a class of words.

Topic: 18: Morphophonology and Secret Languages

Permutations of existing languages

Secret languages are found around the world and have been attested Dutch, Thai, Cuna (Sherzer 1970), and Haitian Creole, to name only a few.

Examples of creative language use

Considered external to the mental grammar. Speakers go from the existing language to the secret language through the regular application of phonological rules-a morphological derivation. Secret languages also exploit notions that are independently motivated in phonology and morphology, notably the syllable. One secret language is Pig Latin. In one variation, words that start with vowels are suffixed with way [wej]. A resonant or consonant cluster shift the entire onset

sequence to the end and are suffixed with ay [ej]:

- **Pig Latin:** igpay atinlay ‘‘
- ‘free form’ eefray ormfay
- inflectionway ‘inflection’

Manipulate words in this fashion, they make use of their subconscious knowledge of linguistic entities such as onset and nucleus

- Verlan [vɛʁlɑ̃], based on French.

The word verlan is derived by reversing the syllables of l’envers [lɑ̃vɛʁ]. Verlan works best with words of two syllables, because in these the two syllables can simply be reversed:

--	--	--

pourri [puri]

ripou [ʁipu]

rotten' (generally refers to corrupt

		police)
. branché [brɑ̃ʃe]	. chébran [ʃebʁɑ̃]	plugged in, informed'
pétard [petɑʁ]	tarpé [tapʁe]	cannabis joint'

Monosyllables are verlanized differently depending on whether they are open or closed.

Open monosyllables	Closed monosyllables (treated as if they end in a schwa underlyingly bisyllabic)
pue [py] 'stinks', the order of consonant and vowel is reversed, yielding forms like [yp]. Closed	femme [fam] 'woman', The syllables are reversed, and the final vowel is dropped: [fam e] → [fa.m e] → [m e .fa] → [mœf].

According to Bullock, the 'rules' of verlan are artificial compared to those of the standard – and natural – language. In Pig Latin, words with a consonant cluster postpose the consonant, then add the suffix -ay. (e.g., losetcay for closet). Any rule that operates on letters rather than on phonological entities such as onsets is unnatural. Languages are first and foremost oral, and orthographies are systems imposed on them by people. Secret languages must be looked upon as somewhat artificial and should not on their own be used to draw conclusions about the workings of natural languages.

Lesson 04

Derivation and the Lexicon

Topic: 19: The Saussurean Sign

Ferdinand de Saussure (1857–1913), one of the first modern linguists, believed that language was a system of signs. He defined a linguistic sign as an arbitrary pairing between what he called the *signifiant* 'signifier', a particular sequence of sounds, and the *signifié* 'signified', the concept that is denoted by the sound sequence. These three terms, sign, signifier, and signified, are still standard in linguistics. Saussure (1969) distinguished between motivated and unmotivated signs. A inspection you can get clues as to what it means. A walk signal at a crosswalk is an example of a motivated sign, because the stylized image of a person walking indicates whether you should or should not cross the street. A stop sign is partially motivated.

Signs can lose their motivation: consider the name of the basketball team, the Los Angeles Lakers. Motivation is not all-or-nothing, and signs can be partially motivated.

Topic: 20: Motivation and Compositionality

Motivation is related to the logical notion of composition or compositionality. We say that something is logically compositional if it is defined entirely in terms of its parts.

- Doghouse

Its meaning is derivable from its two components. Another example of a compositional form is the Kujamaat Jóola dubitive-incompletive,



a. ni- ɲar -ɛ:n
 1sg.sub- take -inc

c. ni- ɲar -ɛ:n -ɛ:n
 1sg.sub- take -inc -inc

e. 'I was taking (emphasis on incompletive aspect of action)' Expressing emphasis through doubling should be considered compositional.

A set of English words in light of partial motivation and compositionality: **These words show partial motivation**

- behead 'to remove someone's head'
- befriend 'to make yourself a friend to someone'
- besiege 'to lay siege to'
- bewitch 'to place under one's power as if by magic'

Compounding

One derivational process. Here are some basic examples: English compounds tool + bar, amusement + park, puppy + love coffee + house

An extreme example,

- What Violet does for a living?, we might respond:
- She's a high voltage electricity grid systems supervisor.
- [a good Noun]

The whole word shows a single noun which shows compositionality
High voltage electricity grid systems supervisor behaves as a single unit for the purposes of wh-

movement.

Q: Which electricity grid systems supervisor did you see?

A: ?The high voltage one. ✓

Q: Which systems supervisor did you see? A:

*The high voltage electricity grid one. c.Q: ✗

Which supervisor did you see?

A: *The high voltage electricity grid systems one. ✗

Contrast these with syntactic strings of modifier plus noun which are easily broken up, as shown in the following:

Q: Which supervisor did you see?

A: The tall one.

? A very high voltage electricity grid systems supervisor

The structure of high voltage electricity grid systems supervisor as evidence that it is a single noun formed by compounding. Words in English are generally head -final, meaning that the lexical category of the form as a whole matches that of its final constituent. e. ○

- A dogsled is a kind of sled, not a kind of dog; the Red River Valley is a valley

“Is this noun in my lexicon?” Probably not. This compound is formed by a very productive process and there is nothing irregular involved in it. It is absolutely compositional and fully motivated. Words like this that are used but not stored as **hapax legomena**. Nonce means ‘a particular occasion’, and hapax legomena is a Greek term meaning ‘said once’ that is used to refer to words that occur only once in the recorded corpus of a given language. In discussing compounds, linguists sometimes use the terms endocentric and exocentric. These terms are related to the notions of motivation and compositionality. An endocentric compound is one that has a head. The head expresses the core meaning of the compound, and it belongs to the same lexical category as the compound as a whole. For example, **exocentric compounds** meaning are **endocentric compounds** c.

Endocentric compounds

Exocentric compounds

jackknife	funny farm
board game	lazybones
bluebird	loony bin
high chair	scarecrow
sailboat	pickpocket

Topic: 21 : Zero-derivation

- English-speaking adults vocabularies of 20,000 to 50,000 words,
- Children ranging from about 50–600 words at age 2 to about 14,000 at age 6.
- Children frequently coin new words (Clark 1995: 393, 399–401).

One way children **derivation**, or conversion, a productive derivational process in English. **Zero derivation**, is a kind of word formation involving the creation of a word (of a new word class) from an existing word (of a different word class) without any change in form, which is to say, derivation using only zero. **Zero-derivation changes the lexic** without changing its phonological shape.

Examples of novel verbs formed by 2- to 5-year-olds by zero- derivation

a. SC (2;4, as his mother prepared to brush his hair): **Don't hair me.**

b. JA (2;6, seated in a rocking chair): **Rocker me, mommy.**

c. SC (2;7, hitting baby sister with toy broom): **I broomed her.**

d. SC (2;9, playing with toy lawnmower): **I'm lawning.**

e. DM (3;0, pretending to be Superman): **I'm supermanning.**

f. FR (3;3, of a doll that disappeared): **I guess she magicked.**

j. KA (5;0): **Will you chocolate my milk?**

A word formed by zero-derivation or any other productive derivational process becomes lexicalized:

The English verbs: chair, leaf, ship, table, and weather.

- **Mail** from the French is another example: *I'm going to mailbox this parcel.

Affixation

The next type of derivational process we consider here is affixation. We have already looked at a number of affixes, the type of affixation. The affixation process is the process of adding affixes to a base word to form a new word.

Prefixes:	un + do hemi + sphere
Suffixes:	rough + age arachno + phobia cut + ie
Infixes:	abso + effing + lutely
Circumfix: Kujamaat Jóola	u - ... al (first person plural inclusive subject): ub c p ε:n-ε:n c rut-al c 'We had not yet sent him'

We can characterize the stem an affix attaches to as bound or free and as belonging to a particular lexical category. Take the following words formed via prefixation with re-:

- reignite d.reanalyze
- reboot e. recertify

Re-atta Verb. It produces words that are also verbs. All of the stems to which it attaches in the preceding examples are free. Ignite, boot, read, analyze, certify, The same cannot be said for all stems that seem to bear the prefix re-, based

- repeat
- resuscitate
- recognize

Topic: 22: Other Derivational Processes

Derivational processes is not limited to compounding, zero-derivation, and affixation.

Blending: a common derivational process also called portmanteau form word by combining parts of more than one word.

- a. smog < smoke + fog
- b. chunnel < channel + tunnel
- c. bit < binary + digit

These processes are not limited to English.

Common in Japanese

Rare or languages but common in Hebrew

Common example in Hebrew, and and Japanese: (Bat-El 1996)

a. prígurt 'fruit yogurt' < prí 'fruit' + yógurt 'yogurt'

b. kadurégel 'football' < kadúr 'ball' + régel 'foot'

c. maškár 'cold drink' < mašké 'drink' + kár 'cold'

d. kalcéfet 'easy-to-make' < kál 'easy, light' + kacéfet 'ice cream' 'whipped cream'

e. ramzór 'traffic light' < ramáz 'to hint' + ?or 'light'

f. pomelít 'hybrid of pomelo < poméla 'pomelo' + and grapefruit' ?eškolít 'grapefruit'

g. šmanmúx 'dumpy' < šmanmán 'plump' + namúx 'short'

Blending is an example of creative language use. Not adhere to strict constraints, as does affixation.

Acronyms: formed by initial letters of a string of words and to form a new one.

Dependent on orthography rather than pronunciation

It is external to the general phenomenon of lexeme formation or a special case of lexeme formation that is among literate users of a language. If acronym formation is orthographically based, just say the NATO aloud the pronunciation of the A and O reflects the name of the letters, not the initial sounds of the words.

How many acronyms can you recall?

a. self-contained underwater breathing apparatus

CUBA

b. radio detecting and ranging

RADAR

c. North Atlantic Treaty Organization

NATO

d. Acquired Immune-Deficiency Syndrome

AIDS

e. Federal Bureau of Investigation

FBI

FBI differs from the other acronyms in being pronounced letter by letter, rather than as a word *[fbi], *[fbaj]. While it is still an acronym, some people prefer to call it an abbreviation or alphabetism for this reason. A characteristic of acronyms in some languages is that they can serve as the base for further morphological operations, particularly affixation, as in the following French examples:

- cégétiste someone from the CGT

Topic: 23: Clipping & Backformation

Cutting

Clipping: creation of a new word by truncation of an existing one.

Many nicknames are formed via this process:

- Rob (< Robert) Trish (< Patricia) Sue (< Susan)

Other nouns,

- ad, co-ed, typo. or fan

Folk etymology

This happens when speakers reinterpret a form – typically a borrowing from another language – on the basis

And form words using morpheme in the language

Examples of folk etymologies:

- cockroach < Spanish cucaracha
- witch hazel < wych [= weak] hazel

A historical process of reanalysis and a source of words, but not a true derivational process.

- **Backformation:** the creation of a word by removing what appears to be an affix.

An example of historical reanalysis and not a productive derivational process responsible for the occurrence of words such as surveil, from surveillance, or liposuct from liposuction

Some very familiar words

- Peddle: peddler

- Edit editor,

Historically, cherry comes from Old Norman French cherise, minus the s, which English speakers mistook for the plural affix.

- Pea from pease,

In order to recognize whether a form has resulted from backformation or clipping, it is often necessary to know its history.

A	B	word-formation processes
garage, sale	garage sale	a. compounding,
refuse + nik	refusenik	b. affixation (suffixation)
d. baby-sitter	baby-sit	d. backformation
e. blanket (n)	blanket (v)	e. zero-derivation or conversion
f. cheese, hot dog	cheese dog	f. blending
g. Asparagus	sparrow grass	g. folk etymology

Topic: 24: Derivation and Structure

Having considered the issue of lexical storage of derived forms, let's go back to the notion of derivation and structure. We can schematize derivation as follows:

Input → Output Lexeme X Lexeme Y

If we can have lexeme X as an input and lexeme Y as an output, then it should also be possible to take _____ ion:

Input → Output Lexeme Y Lexeme Z

This _____ like unfriendly:

- **Function 1:** add -ly friend → friendly
- **Function 2:** add un friendly → unfriendly

Even unfriendly → unfriendliness via a function that adds -ness.

Ir. _____. To determine the order of functions leading to a form, it helps to consider other words that contain the same parts.

Un- attaches to nouns only in exceptional cases: uncola,

It regularly attaches to adjectives. We use this fact in determining that the function 'Add -ly', which forms adjectives, must come before the function 'Add un-'. The fact that speakers of many languages can add phonological material to either end of a word sometimes leads to complex structures.

Take the two **English words** below:

a. reinterpretation

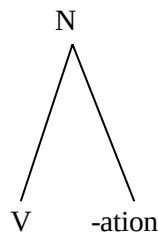
b. poststructuralist

These words have the following structures:

a. [[re- [interpret]V]V -ation]N

b. [post- [[[structur]N -al]A -]A

(a) tells us that reinterpretation is the act of reinterpreting, from reinterpret, not re- the act of interpreting. We start out with a verb, interpret, form a new verb via the prefix re-, and finally form a noun by adding the suffix -ation.



re- interpret

Combining prefixation and suffixation leads to other potentially ambiguous forms in English.

Three famous examples are given in below:

a. undressed

b. unpacked

c. unzipped

Prefix un- has at least two distinct roles in English, depending on what it attaches to. When prefixed to a verb, un- is a so-called reversative with the basic meaning '*undo the action of the verb*'. If you unpack a suitcase, you return the suitcase to the state it was in before the packing action took place. When attached to adjectives, including participial adjectives like wounded or stressed, un- means 'not'. If you untie a package, you return it to the state it was in prior to being tied. If a soldier leaves the battlefield unwounded, it is not the case that he was first wounded and then unwounded, because it is impossible to unwind a person

Words & its Forms: Inflections

Topic: 25: Words and grammar

Some words are so predictable that they do not have to be listed in a dictionary. Such words illustrate the fact that a word need not be a lexical item. They are merely grammatical.

may not be listed, depending on whether its meaning is predictable or not.

E.g. the word cats it's meaning is predictable from cat so we don't need to add cats in dictionary

The notions 'more basic' and 'grammatically conditioned variant'

(1) This pianist performs in the local hall every week.

(2) Mary told us that this pianist performed in the local hall

(3) The performance last week was particularly impressive.

(4) *This pianist perform in the local hall every week.

(5) *These pianists performs in the local hall every week.

(7) The performer last week was particularly impressive.

(8) The concert last week was particularly impressive.

(9) These pianists perform in the local hall every week.

We need a new term for the more abstract kind of word of which the word forms *performs*, *performed* and *perform* are all inflectional variants. Let us call this more abstract word a *lexeme*. We can now say that *performs*, *performed* and *perform* are all inflected forms of the lexeme *perform*, and we can describe the grammatical function of *performed* by calling it the past tense form of the verb *perform*. Being abstract in this sense, a lexeme is not strictly speaking something that can be uttered or pronounced. Only the word forms that belong to it can be. We refer to lexemes in English by means of their bare, unaffixed forms. Two words can be pronounced the same but spelled differently in English, and vice versa. It follows that the same word form can belong to two quite different lexemes, as does *rows* in:

- There were four rows of seats.
- One person rowed the boat.
- One lexeme may be represented by more than one word form, and one word form may represent more than one lexeme.

What links a word form with a lexeme in a given context is the grammatical word that the word form expresses there.

Topic: 26: Regular and irregular inflection

The words that have to be listed in a dictionary and words that do not. One does not have to list *performs* and *performed* alongside *perform*, or *pianists* alongside *pianist*, because they are merely *grammatically conditioned variants* of one basic word – of one lexeme, in fact. But it is not correct to say that dictionaries never have anything to say about inflectional morphology. There are two reasons why a word form such as *pianists* does not have to be listed, and these reasons are independent.

The first: once an English word is a noun denoting a kind of thing that can be counted like *pianist* or *cat* not *astonishment* or *rice*. We can be confident that it will have a plural form with no idiosyncrasies of meaning; it will mean simply ‘more than one X’, whatever X may be

The second: unless otherwise specified, we can be confident that the plural form of any countable noun will be formed by adding to the singular form the suffix *-s* (or rather, the allomorph of this suffix); in other words, suffixing *-s* is the regular method of forming plurals.

Any native speaker of English, after a moment’s thought, should be able to think of at least two or three nouns that form their plural in some other way than by adding *-s*:

child > children,

tooth > teeth,

man > men.

A dictionary entry for tooth will look like this: tooth noun (plural teeth). One of a set of hard white structures set in the jaw and used for biting and chewing.

- George Orwell envisages the eradication of morphological irregularity so

man > mans not men

In reality as opposed to fiction, this sort of regularisation is a well-known feature of the speech of young children and of non-native learners. If all this does not inhibit us from recognizing them as allomorphs of one morpheme, what about the different plural suffixes exhibited by nouns such as pianists, oxen, formulae and cacti?

Can we not classify -

Should we not also recognise a further allomorph that we might call ‘vowel change’, to accommodate men and teeth, which lack a suffix? Admittedly, these allomorphs are quite unlike each other in pronunciation – but if allomorphs are allowed to differ somewhat, why cannot we allow them to differ considerably? At what point, if any, does phonological divergence become too great? How the term ‘morpheme’ is used is important: whether in a more concrete sense, oriented towards pronunciation (in terms of which -s, -en, -ae and -i represent different morphemes), Or a more abstract sense, oriented towards meaning or grammatical function (-s, -en, -ae and -i are all allomorphs of one morpheme) a good way to avoid any confusion is to use terms such as ‘root’, ‘suffix’ and ‘prefix’, wherever possible, rather than ‘morpheme’. Because the role of inflectional morphology in English is much smaller than in languages such as German or Russian (although greater than in Chinese), what needs to be said about each word class is relatively limited.

Topic: 27: Forms of nouns

Most countable nouns in English have two word forms: a singular and a plural. Inflectionally, for any noun lexeme X, there are just two grammatical words, ‘singular of X’ and ‘plural of X’, contrasting in number.

- Lexeme cat > cats
- Root and suffix -s

The suffix -s is the regular suffix for forming plurals. Irregular suffixes expressing plurality include -i, -ae and -a found Latin or Greek;

- The suffix -(r)en in oxen, children and brethren;
- The Hebrew -im in cherubim and kibbutzim.

Some countable nouns that express their plural with no suffix at all but with a change of the vowel of the root like the (teeth, men) – or, more precisely, an allomorph of the root with a different vowel from the singular. Also some whose plurals display not even a vowel change: for example, sheep, fish, deer, trout.

How can we tell whether they are singular or plural?

- The answer is: according to the syntactic context.
- A deer was visible through the trees.
- Two deer were visible through the trees.

The class of nouns unchanged in the plural (called ‘zero-plural’ or carrying a ‘zero suffix’) could conceivably be just as random as the class of those with vowel change (tooth, man, etc). Nouns have plural forms that refer to entities that are countable like cats and pianists, but not *astonishments or *rices – except perhaps in contexts interpreted as denoting countable entities, such as astonishing events or varieties of rice. Not all nouns referring to countable entities have both singular and plural forms. A few nouns such as scissors and pants which exist only in an -s-syntactic contexts. This idiosyncratic lack of a morphological singular form (except in compounds such as scissor factory) creates a problem in contexts where the syntax seems to require such a form, as when the noun is preceded by the indefinite article a or an.

- We can say neither *a scissor nor *a scissors, and likewise neither *a pant nor *a pants.

For these lexemes, there is a conventional circumlocution or periphrastic form: pair of pants and pair of scissors. The unusual nouns scissors and pants provide an opportunity to deal with a possible doubt concerning whether the singular–plural contrast in nouns really deserves to be called inflectional. If inflection is a matter of grammatically conditioned variation, as is thought generally, it is easy to agree that the contrast between performs in (This pianist performs ...) and perform in (These pianists perform ...) is inflectional, because it is a contrast imposed by the grammatical context. But what about the noun phrases themselves? The choice between singular and plural there is determined not by grammar but by meaning, one may think – by what the speaker wants to say. If so, does this contrast really deserve to be called grammatically conditioned? The singular–plural distinction is the only grammatical distinction that is expressed morphologically in English nouns.

Topic: 28: Forms of pronouns and determiners

In _____ classes, namely _____ membership can be added to as new words come into use. By contrast, in English a new pronoun or a new preposition is not added. Determiners deserve a mention as some like nouns, display a singular–plural contrast as Pronouns. We have already encountered the distinction between this and these, as in *this pianist* and *these pianists*. These are the singular _____’s. Other determiners include the, a (but only one other determiner exhibits a singular–plural contrast: that, with singular and plural forms *that* and *those*. The determiners *that* and *this* demonstrate that number contrasts can have a grammatical effect inside noun phrase as well as between subject noun phrases and their accompanying verbs.

- *This* table is...
- *That* table is...

In many languages, the distinction that English expresses by word order in:

- ‘John loves Mary’
- ‘Mary loves John’

is expressed by inflectional means on the words corresponding to *Mary* and *John*. In English, the same technique is used. If one replaces John and Mary with the appropriate pronouns in these two examples, the outcome is as in following (1 & 2):

(1) He loves her.

(2) She loves him.

- *He* and *him* are sometimes said to contrast in **case**.

This kind of inflection has only a marginal role in English, being limited to pronouns; but, if we treat (say) *he* as a lexeme, we must recognise it as having two forms: *he* and *him*. Relationship between nominative and accusative forms is consistently suppletive, as in *I/me*, *she/her*, *we/us*, and *they/them*, except that for *you* the two forms are identical (*you*). This is consistent with the fact that pronouns are very common, and suppletion affects only very common words such as *go*. If *he* and *him* are flexemes of the lexeme *he*, and *we* and *us* are forms of *we*, what are *we* to say about corresponding words with a plural? *They*, *your* and *their*? Syntactically and semantically, these words fulfil just the same role as noun phrases with the apostrophe-s: *his* *belonging to him* just as *that man's bicycle* means ‘the bicycle belonging to that man’. One possibility is to say that these are pronoun forms belonging to a third case, the genitive or possessive, which stand in for apostrophe-s forms in noun phrases that consistently of a personal pronoun. Another is to classify these words as determiners, because they perform a determiner-like role and cannot be combined with other determiners (we cannot say *his his bicycle*). For present purposes, we need merely note how *his*, *our* and the rest behave, while leaving their exact grammatical classification undecided.

Topic: 29: Forms of Verbs

Already discussed some forms of English verbs such as *performs*, *performed* and *perform*. In English, a verb lexeme has at most five distinct forms, as illustrated here with *give*:

Verb Form	Example Sentence

a. third person singular present tense: gives	e.g. Mary <i>gives</i> a lecture every year.
b. past tense: gave	e.g. Mary <i>gave</i> a lecture last week.
c. progressive participle: giving	e.g. Mary <i>is giving</i> a lecture today.
d. perfect or passive participle: given	e.g. Mary <i>has given</i> a lecture today. The lecture <i>is always given</i> by Mary.
e. basic form (used everywhere else): give	e.g. Mary <i>may give</i> a lecture. Mary wants <i>to give</i> a lecture. Mary and John <i>give</i> a lecture every year.

The contrast between present at (a) and past at (b) is a contrast of tense. The other dimensions of contrast manifested in (a) are person (third person versus the rest) and number (singular versus plural, just as for nouns and pronouns). However, because only one word form (*gives*) exhibits these contrasts, they play a much smaller inflectional role in modern English verbs than in Old English verbs. For the form labelled ‘perfect or passive participle’, two examples are given, because perfect and passive contexts can be distinguished clearly. However, it is a peculiarity of English verb morphology that the corresponding forms are always the same. Another way of putting this is that, for any verb *V*, the grammatical words ‘perfect participle of *V*’ and ‘passive participle of *V*’ are expressed by the same word form. A verb lexeme has at most five forms. In fact, most verbs have only four forms, the past tense and the perfect (or passive) participle forms are the same.

Perform

a. third person singular present tense: performs

b. past tense: performed

c. progressive participle: performing

d. perfect or passive participle: performed

e. basic form (used everywhere else): perform

When two grammatical words that are distinct for some lexemes are systematically identical for others, as here, these forms are said to be *syncretised*, or to exhibit *syncretism*. The same syncretism also occurs with some irregular verbs, such as dig _____) and all those

in that they do not use the *-ed* suffix. The study of these irregularities belongs to grammar rather than to word formation. Other verbs or verb-like words whose behaviour belongs to grammar rather than word-formation are the auxiliaries, such as be and have, and modals, such as can, must, may. Instead of the usual verbal maximum of five forms, modals distinguish even just one (e.g. must), while be distinguishes eight (am, is, are, was, were, being, been, be).

Topic: 30: Forms of adjectives

Many English adjectives exhibit three forms, for example *green* here:

- Grass is *green*.
- The grass is *greener* now than in winter.
- The grass is _____ in early summer.

The grammatical words that green, greener and greenest express are the positive, comparative and superlative of green, contrasting on the dimension of comparison. All these exhibit suffixation with *-er* and *-est*, except for better and best, which are suppletive. Comparative and superlative forms of adjectives belong to inflectional rather than to derivational morphology. In some grammatical contexts in which comparative or superlative adjectives are unavoidable, anything else being ill-formed:

a. This field is greener than that one.

b. *This field is green than that one.

c. *This field is fertile than that one.

a. The greenest fields of all are here.

b. *The green fields of all are here.

c. *The superior fields of all are here.

Should every adjective lexeme possess a comparative and a superlative form (or, at any rate, every adjective denoting a property that can be present to a greater or lesser degree).

It is striking that many adjectives lack these forms:

- *Curiouser !
- *This field is fater than that one.
- *
- we use periphrastic forms with more or most
- More and more curious!
- This field is more fertile than that one.
- The most fertile fields of all are here.

Broadly speaking, the suffixes -er and -est appear on adjectives whose basic form has one syllable, or two provided that the second syllable ends in a vowel (e.g. tidy, yellow), while longer adjectives usually require the periphrasis.

More and most

Lesson 06

Words & Its Relatives: Derivation: I

Topic: 31: Relationships between lexemes

Perform: perform, performs and performed are grammatically conditioned variants of one lexeme perform, but *performance* was not one of these variants. The reason was that, whereas there are grammatical factors that determine the choice between perform, performs and performed (in appropriate contexts), there is no grammatical factor that requires specifically the presence of -ance on performance.

- The contexts determine where -s, -ed or -ing appear
- No contexts where, if a noun appears, it must carry the suffix -ance.

- The suffix -ance is not one of the small class of suffixes (so-called ‘inflectional’ suffixes) whose use is tightly determined by grammar.

It changes the class of words from verb to noun

What sort of suffix is it, then?

A short answer: not being inflectional, it must be derivational, since the term ‘derivation’ is used for all aspects of word-structure involving affixation that is not inflectional.

Let’s see how derivation works in English. Since performance is not a variant of the lexeme perform, it must belong to some other lexeme, which may itself have more than one form.

What lexeme could this be? performance, there is a performances(plural form). Just as cat and cats are the two forms (singular and plural) of the lexeme cat, It makes sense to regard performance and performances as the two forms of a lexeme performance. This tells us something about the relationship between perform and performance: it is a relationship not between word forms but rather between lexemes. In typographical convention, it a relationship between perform and performance. Thus derivational morphology is concerned with one kind of relationship between lexemes. Many ways in which lexemes can be related. Not concerned here with relationships solely of meaning AUBERGINE and EGGPLANT). Or of sound (such as the homonymy of ROW’ and ROW). Rather, we are concerned mainly with relationships involving affixation, and the grammatical and semantic tasks that such affixation can perform. The term base for the partially complete word form to which an affix is attached so as to create either an inflected word form or a new lexeme.

Equivalently, the base for an affixation process is what remains if the affix is removed. Some bases are roots, whether bound (e.g. wive-, the base for wives) or free (e.g. cat, the base for cats). Others, however, already contain a root and one or more affixes, such as helpful in its capacity as the base for helpfulness.

Topic: 32: Word classes and conversion

How word change classes?

- *The word class to which a lexeme belongs is mainly determined by its meaning.*
- *That*

Verbs are ‘doing words’, while nouns are ‘thing words’ and adjectives ‘describing words’. The trouble with these meaning-

together lexemes whose grammatical behaviour is quite different, and distinguish between ones whose grammatical behaviour is similar. Lexeme perform, which looks like a prototypical ‘doing word’, denoting something that actors and musicians do. The lexeme perfo

surely. Does that mean that perform and performance belong to the same word class?

- Perform has four forms: perform, perform, performing and performed
- Performance has two: performance and performances

This classification can be made, solely on the basis of their syntactic and inflectional behaviour, with no appeal to meaning. Meaning may be positively misleading, since a performance is not obviously a ‘thing’.

Compare now the lexemes perform and resemble

- Resembling, one may think, hardly counts as an activity.
- My great-uncle William resembles a giraffe is not to report some action of his, but rather to describe him.
- ‘describing words’ – adjectives?
- The meaning-motivated conclusion falls foul of syntactic and inflectional evidence.
- These adjectives have degrees (taller, tallest) or phrasal substitutes (more interesting, most interesting);
- Resemble has a set of forms.

So to identify verbs as ‘doing words’ risks misleading us into neglect of the syntactic and inflectional parallels that justify classifying not only perform but also resemble as a verb.

- Cannot a lexeme have both noun forms and verb forms?
- A root that can carry verbal suffixes such as -ed and -ing as well as the noun plural suffix - belong to two lexemes, not one.
- The more interesting question, then, is: do such roots exist? The answer is certainly yes.

For example, hope and fear

- Ambivalent words are wish, desire, father
 - a. She hoped that it would rain.
 - b. She feared that it would rain.
 - a. her hope that it would rain
 - b. her fear that it would rain

Some linguists: hope and fear, as nouns, are really ‘zero-derived’, carrying a phonologically empty and therefore unpronounceable ‘zero suffix’: Hope -*o*, fear -*o*. Others prefer a lexeme belonging to one class can simply be ‘converted’ to another, without any overt change in shape. Sometimes, despite the risks already mentioned of relying on meaning as a criterion, the basic meaning seems clearly appropriate to one word class rather than another; for example (father) that is more basic.

Topic: 33: Adverbs derived from adjectives

Let's take adjective dioecious, meaning 'having male and female flowers on different plants'. Certainly, dioecious must be listed in any reasonably complete dictionary of English. The corresponding adverb dioeciously would not have to be listed, because both its existence and its meaning can be taken for granted once the existence of dioecious is acknowledged. This neatly illustrates the distinction between lexemes and lexical items: dioeciously is a distinct lexeme from dioecious, since it belongs to a different word class, but it is not a distinct lexical item. This also illustrates a widespread though not universal characteristic of derivational processes: unlike inflection, they can change the word class of the bases to which they apply.

According to introductory treatments of English grammar all adverbs end in -ly. If that were true, it would be an unusual word class, all of its members being derived. In fact, simple or monomorphemic adverbs, though few in number, very common words (often, seldom never, soon). Some other adverbs are morphologically complex without containing -ly (nowhere, everywhere, today, yesterday,). Also, there are common adverbs that are formed by conversion: fast (as in The car was driven fast) and hard (as in They worked hard), derived from the adjective fast (as in a fast car) and hard (as in hard work).

Topic: 34: Nouns derived from nouns

- Not all derivational processes change word class.
- English has derivational processes that yield nouns with meanings such as 'small X', 'female X', 'inhabitant of X', 'state of being an X' and 'devotee of or expert on X'.

Here are some examples all of them must count as lexical items	unpredictable meanings
5) 'small X': -let, -ette, -ie	e.g. droplet, booklet, cigarette, doggie
(6) 'female X': -ess, -ine	e.g. waitress, princess, heroine
(7) 'inhabitant of X': -er, -(i)an	e.g. Londoner, New Yorker, Texan, Glaswegian

(8) 'state of being an X': -ship, -hood	kingship, priesthood, ladyship, motherhood,
(9) 'devotee of or expert on X': -ist, -ian	e.g. contortionist,, Marxist, logician, historian

Many of them have unpredictable meanings (a cigarette is not merely a small cigar, and a booklet is not merely a small book; brotherhood means not 'the state of being a brother' but rather 'secret or semi-secret society'). Why is there a word actress (albeit less used now than formerly), but there has never been a word 'writress' to designate a woman writer?

Nouns derived from members of other word classes

Nouns derived from adjectives and from verbs are extremely numerous, and it should be easy for you to think of many other examples on the lines of those given here.

Here are some suffixes used to derive nouns from adjectives:

- -ity, e.g. purity, equality, ferocity, sensitivity
- -ness, e.g. goodness, tallness, fierceness, sensitiveness
- -ism, e.g. radicalism, conservatism

Some of these nouns are formed from bases other than the free form of the corresponding adjective, e.g. ferocity from feroc- (not ferocious), conservatism from conservat- (not conservative). The ferocity pattern is fairly general for adjectives in -ious (compare rapacity, capacity alongside rapacious and capacious) but not absolutely general (for example, to delicious and specious there correspond dioeciouness and speciousness, not 'delicity or specity'). This gappiness is a reason for counting all nouns in -ity as lexical items.

Even more numerous are suffixes for deriving nouns from verbs. Here are just a few:

-ance, -ence, e.g. performance, ignorance, reference, convergence

-ment, e.g. announcement, commitment, development, engagement

-ing, e.g. painting, singing, building, ignoring

-((a)t)ion, e.g. denunciation, commission, organization, confusion

-al, e.g. refusal, arrival, referral, committal

-er, e.g. painter, singer, organiser, grinder

- The suffixes all have much the same function forming abstract nouns meaning ‘activity or result of Xing’,
- They are certainly not freely interchangeable: e.g. we have performnce but no ‘performant or ‘performance’.
- Affixation is most common way in which lexemes are derived in English, but is not the only way.

Some non-affixal ways of deriving abstract nouns (other than conversion) are:

(19) change in the position of the stress,	e.g. nouns permit, transfer alongside verbs permit and transfer
(20) change in the final consonant	e.g. nouns belief , proof defence alongside verbs believe prove, defend
(21) change in a vowel	, e.g. nouns song ,seat alongside verbs sing, sit .By contrast with some languages, however, the derivational use that

English makes of vowel change is minimal. Languages that exploit it much more consistently are members of the Semitic family, such as Arabic and Hebrew.

Topic: 35: Adjectives derived from adjectives

In adjective prefixes predominate.

The only suffix of note is -ish, meaning ‘somewhat X’, as in greenish, smallish and remotish ‘rather remote’. By contrast, the prefix un- meaning ‘not’ is extremely widespread: unhappy, unsure, unreliable undiscovered. Being common, most dictionaries do not attempt to list all un- adjectives. Another negative prefix is in- and im-, as intangible, illegal, irresponsible and impossible. The pairs of more or less synonymous adjectives, one negates with un- and the other with in- or one of its allomorphs:

- eatable/uneatable
- edible/inedible

- readable/unreadable
- legible/illegible
- lawful/unlawful
- legal/illegal

Some of the processes that derive adjectives from verbs overlap the divide between derivation and inflection. We met the suffixes -ed, -en and -ing, and vowel change, in passive and progressive participle forms of verbs.

The forms in the following can also be adjectives:

- a. a not very *interesting* book
- b. The party-goers sounded very *drunk*.
- c. The car seemed more *damaged* than the lamp-post.

Further suffixes that commonly form adjectives from verbs, with their basic meanings are:

-able ‘able to be Xed’:	breakable, readable, reliable, watchable
-ent, -ant ‘tending to X’:	repellent, expectant, conversant
-ive ‘tending to X’:	repulsive, explosive, speculative

- with-able variant, or allomorph, -ible the -ible words have clearly identifiable verbal meanings such as ‘eat’, ‘read’ and ‘touch’, and are bound rather than free.
- Some of the bound verb roots appear in a number of derived lexemes, such as the *aud-* root that occurs in: audible, audience, audible and auditory.
- Suffixes that form adjectives from nouns are more numerous.

Here are some:

-ful	, e.g. joyful, hopeful, helpful, meaningful
-less	e.g. joyless, hopeless, helpless, meaningless
-al	, e.g. original, normal, personal, national
-ish	e.g. boyish, loutish, waspish, selfish

Generally seen, adjectives in -ful and -less tend to come in pairs, although the correspondence is not exact: we have slothful, slothless but ‘penniless’ but not pennifull. This confirms again that, even when the meaning of a potential word may be easily guessable: a ‘slothless’ person would be hardworking, and a ‘penniful’ person would be well off, the existence of the word is not guaranteed.

Topic: 36: Verbs derived from verbs

Let’s discuss some unusual affixes that are prefixes. Most prominent are re- and the negative or ‘reversive’ prefixes un-, de- and dis-, as in the following examples:

paint, enter	repaint, re-enter
tie, tangle	untie, untangle
compose, sensitise	decompose, desensitise
entangle, believe	disentangle, disbelieve

- The prefix re- has relationship between morphemes and meaning.
- Semantically, the examples are mostly straightforward, although those with de- are less so: to decompose is not to undo the creative work of a musical composer!

Also worth mentioning here is the relationship between the verbs in the left and right

Intransitive verb: don't take object Lie (past lay) rise (past rose) fall (past fell) sit (past sat)	Transitive verbs: 'object 'noun phrase, usually indicating the thing or person that is the goal of the action of the verb lay (past laid) raise (past raised)fell (past felled) set (past set)
The book lay on the table.	Jill laid the book on the table.

The transitive verbs in (35) are all causative, that is they mean 'cause to X', where X stands for the meaning of the corresponding intransitive. Causative verb-pairs nearly all involve conversion, as in rather than either affixation or the kind of vowel change

a. Jill boiled the water.

b. The water boiled.

- The examples in (35) represent a residue of a vowel-change pattern that was more widespread at an earlier stage of the language.
- Verbs derived from other word classes
- Verbs derived from nouns and from adjectives are numerous.
- verbs from nouns are:
- de-, debug, deforest, delouse
- -ise, organise, patronise, terrorise
- -(i)fy, beautify, petrify

There are also some common verbs that are derived by replacing the final voiceless consonant of a noun with a voiced one, perhaps with some vowel change too:

- Nouns Verbs
- Bath bathe
- Breath breathe
- Wreath wreathe

A meaning for de- at (38) is clearly identifiable, namely ‘remove X from’ (compare its function in deriving verbs from verbs, e.g. *desensitise*).

The suffixes -ise and -ify can derive verbs from adjectival bases too, as in *nationlise*, *tenderise*, *intensify*, *purify*

The suffix -ate shows the same sort of ambivalence. Words such as *generate*, *rotate*, *replicate*, and *locate* clearly contain a root and a suffix, because the same roots crop up elsewhere (e.g. in *general*, *rotor*, *replica*, *local*).

one prefix to be mentioned: en- (with its allomorph em-), which forms verbs meaning ‘cause to become X’ or ‘cause to possess or enter X’ from a few adjectives and nouns: *enfeeble* *enslave*, *empower*, *enrage*, *enthrone*, *entomb*.

the prefix en- is combined with a suffix -en: *embolden* *enliven*. This suffix usually occurs without the prefix, however, and does so quite widely (e.g. *tighten*, *loosen*, *stiffen*, *weaken*, *widen*, *redde*n, *deepen* *toughen*

These verbs have either an intransitive meaning, ‘become X’, or a transitive one, ‘cause to become X’. The adjectives that can constitute bases for such verbs share an unusual characteristic, some verbs in -en that are imaginable, yet do not occur: *greenen, narrowen, strengthen, tallen. With strong we get round this restriction by adding -en instead to the corresponding noun, strength (which ends in a fricative sound), so as to yield strengthen.

Lesson 07

Words & Its Relatives: Derivation: II

Topic: 37: Compounds versus phrases

- **Compound words** containing roots,
- **Phrasal words** are words.
- Some types of compound are much commoner than others.
- Roots in English are mostly free rather than bound.
- How can we tell, then, whether a pair of such roots constitutes a compound word or a phrase that is a unit of sentence structure rather than a complex word?
- A definite answer is not always possible, but there are enough clear cases to show that the distinction between compounds and phrases is valid.

- a green "house,
with its literal meaning,
- a "greenhouse,

Phrases like green house stressed on last word	Like greenhouse stress on the first part
(1) black bóard 'board that is black'	blackboard 'board for writing on'
(2) silk wórm 'worm made of silk (e.g. a soft toy)'	silkworm 'caterpillar that spins silk'
(3) hair nét 'net made of hair'	hairnet 'net for covering hair'
(4) white hóuse 'house that is white'	(the) Whíte House 'residence of the US President'
(5) toy fáctory 'factory that is a toy'	tóy factory 'factory where toys are made'
pattern of semantic contrast between expressions stressed in different places is quite common	

It is characteristic of phrases in English to be stressed on the last word, unless some contrast is being stated or implied. Apart from stress, a second criterion traditionally used for distinguishing compounds from phrases is semantic. A compound tends to have a meaning that is more or less idiosyncratic or unpredictable. This criterion must be treated with caution, however, because, being semantically unpredictable does not correlate exactly with being a word. It is true that words are more likely to be lexical items than phrases are, so treating semantic idiosyncrasy as an indicator of compound status will not often be misleading.

Topic: 38: Compound verbs

Verbs form

Nevertheless, a variety of types exist which may be distinguished according to their structure.

(1) verb–verb (VV): stir-fry, freeze-dry

(2) noun–verb (NV): hand-wash, air-condition, steam-clean

(3) adjective–verb (AV): dry-clean, whitewash

(4) preposition–verb (PV): underestimate, outrun, overcook

- Only the PV type is really common, however, and some compounds with under-, over- and out- do not need to be classed as lexical items.
- out- can create a transitive verb meaning ‘outdo in Xing’ from any verb denoting a competitive or potentially competitive activity: outsail, outsing, outswim
- New words with over- created freely overpolish overcriticise
- All these compounds have a verb as the rightmost element, and also that, with most of them, the activity denoted by the compound as whole is a variety of the activity denoted by that rightmost element.
- Let us call these compounds right-headed, the rightmost element being the head.
- Most English compounds are right-headed, but not all.

On the analogy of (1-4) here are some examples of right-headed compound adjectives:

(5) noun–adjective (NA): sky-high, coal-black, oil-rich

(6) adjective–adjective (AA): grey-green, squeaky-clean, red-hot

(7) preposition–adjective (PA): under full, overactive

As with verbs, it is the type with the preposition *over* as its first element that seems most productive, in that new adjectives of this type, with the meaning ‘too X’, are readily acceptable: for example, overindignant, oversmooth. In overactive the head of the compound is the adjective active derived from the verb act. In structure, this adjective is not a mere string of morphemes (over + act + -ive), but rather a nested structure: [over[act-ive]].

Adjectives with a VA structure, corresponding to the VV verbs at

(2), would resemble a hypothetical ‘float-light’ ‘light enough to float’ or ‘sing-happy’ ‘happy enough to sing’.

One actual example is fail-safe ‘designed to return to a safe condition if it fails or goes wrong’. Other such compounds scarcely exist. This reflects the relative reluctance of verbs to participate in compounding generally in English.

- All the compounds in (5)–(7) are right-headed.
- There are also a few compound adjectives that are not right-headed.

Topic: 39: Compound Nouns and headed and headless compounds

It is with nouns that compounding really comes into its own as a word forming process in English. That is not surprising. Cultural and technical change produces more novel artefacts than novel activities or novel properties. These changes therefore generate new vocabulary needs that are more readily answered by new nouns than by new verbs or adjectives.

Examples can be found with each of the other main word classes supplying the left-hand element:

(13) verb–noun (VN): swearword, drophammer, playtime

(14) noun–noun (NN): hairnet, mosquito net, butterfly net, hair restorer

(15) adjective–noun (AN): blackboard, greenstone, faintheart

(16) preposition–noun (PN): in-group, outpost, overcoat

All of these have the main stress on the left – a characteristic important for distinguishing compound nouns from noun phrases. The fact that hair restorer, butterfly net and mosquito net are spelled with a space does not affect the fact that, from the grammatical point of view, they each constitute one complex word. Most of these are also right-headed. Arriving at the precise meanings of these compounds depends

on our knowledge of the world rather than on purely linguistic knowledge. The difference in precision with which we can interpret hair restorer on the one hand and hairnet etc. on the other hinges on the fact that restorer in hair restorer is derived from a verb (restore). Verbs, unlike most nouns and adjectives, impose expectations and requirements on the noun phrases that accompany them in the sentence.

For example, an X-restorer, whatever X is, something or someone that restores X.

- Here are some more compounds whose second element is derived from a verb:
- sign-writer, slum clearance, crime prevention, wish-fulfilment
- Some terminology, for convenience

Call a NN compound like hairnet or mosquito net, in which the right-hand noun is not derived from a verb and whose interpretation is therefore not precisely predictable on a purely linguistic basis, a primary or root compound.

- Headed and headless compounds
- The AN compounds ‘faint heart’ alongside blackboard and greenstone.
- a greenstone is a kind of stone and a blackboard is a kind of board, a faint heart is not a kind of heart but a kind of person –heart the head of the compound.

Rather, faint heart is headless, in the sense that its status as a noun is not determined by either of its two components. Headless AN compounds are loudmouth and redshank (a kind of bird that has red legs), and headless NN compounds are stickleback (a kind of fish with spines on its back) and sabre tooth. So, although heart is a noun, it is not appropriate to call heart the head of the compound. Rather, faint heart is headless, in the sense that its status as a noun is not determined by either of its two components. Headless compounds no internal ‘Centre’, called exocentric –that is, having a ‘centre’ outside themselves, figuratively speaking. Headed compounds would be regarded as having an internal ‘centre’; and, sure enough, they are sometimes called endocentric.

Topic: 40: Blends and acronyms

The example compound. Sporadically, we encounter a kind of compound where at least one component is reproduced only partially known as *blends*.

- A stranglechase
- chortle blended from chuckle and snort.

Partial blends, where only one component is truncated, are talkathon (from talk plus marathon).

- cheeseburger and similar blends such as beefburger and vegeburger may have been encouraged by a feeling that hamburger is a compound whose first element is ham – scarcely appropriate semantically, since the meat in a hamburger is beef.

The most extreme kind of truncation that a component of a blend can undergo is reduction to just one sound (or letter), usually the first. Blends made up of initial letters are known as acronyms, of which well-known examples are NATO (for North Atlantic Treaty Organisation).

- SCSI (scuzzy, small computer systems interface), and AIDS (from acquired immune deficiency syndrome).

Intermediate between an acronym and a blend is *sonar* (from sound navigation and ranging). The use of capital letters in the spelling of some of these words reflects the fact that speakers are aware of their acronym status. It does not follow that any string of capital letters represents an acronym. If the conventional way of reading the string is by pronouncing the name of each letter in turn, as with USA and RP, then it is not an acronym but an abbreviation. Acronyms are also pluralized like a noun phrase. Often a writer will add an 's' following an apostrophe, as in "PC's". However, some writers would prefer "DVDs" and "URLs".

Possessive plurals that also include apostrophes for mere pluralization and periods appear especially complex: for example, the C.D.'s labels (the labels of the compact discs). It is clear from these examples that blending and acronym are in active use for the creation of new vocabulary. However, they differ from derivational affixation and normal compounding in being more or less self-conscious, and are concentrated in areas where the demand for new noun vocabulary is greatest, such as (currently) information technology.

Topic: 41: Compounds containing bound combining forms

Most of the compounds that we have looked at so far involve roots that are free forms. But the vocabulary of English, especially in scientific and technical areas, includes a huge repertoire of compounds that are made up of bound roots, known as combining forms.

Here are just a few:

- anthropology, sociology, cardiogram, electrocardiogram, retrograde, retrospect, plantigrade

The meaning of the whole is clearly determinable from that of the parts: for example, anthrop(o)- 'human' plus -(o)logy 'science or study' yields a word that means 'science or study of human beings', and planti- 'sole (of foot)' and -grade 'walking' yields a word meaning 'walking on the soles of the feet'. This semantic predictability is crucial to the coining of new technical terms using these elements. Apart from containing bound roots, anthropology differs in two other ways from most compound nouns.

Firstly, it has a central linking vowel -o- that cannot conclusively be assigned to either root. In this respect it resembles many combining-form compounds. **Secondly**, although it is a noun, its stress is not on the first element – unless the linking -o- belongs there. In this respect it resembles e.g. monogamy, philosophy and aristocracy. We have already countered bound roots that could function as the base for derivational affixation, such as aud- in audible, audition etc. Not surprisingly, some combining forms can function in this way too (in other words, the dividing line between combining forms and other bound roots is not sharp): for example, soci- and electr(o)- from (23) also occur, indeed much more commonly, in social and electric. Given that combining forms, and the compounds that contain them, are so untypical of compounds in general, it is natural to ask how English has come to acquire them.

In fact, they come mostly from Greek or Latin, through deliberate borrowings to supply new needs for technical vocabulary that arose partly from the revival of learning in Western Europe in the fifteenth and sixteenth centuries known as the Renaissance. Some also came from the industrial revolution of the eighteenth century and its scientific spin-offs.

Topic: 42: Phrasal words

In some of the compounds that we have looked at so far, relationships are expressed that are the same as ones expressed in syntax: for example, the verb–object relationship between hair and restore in hair restorer. The relationship is expressed in this compound is quite different from how it is expressed in syntax, in that the two words appear in the opposite order: we say This substance restores hair, not *This substance hair-restores. There is a clear difference between compound word structure and sentence structure here.

An example of a phrasal word is the noun jack-in-the-box.

Structurally the appearance of a noun phrase exactly parallel to the phrases people in the street or (a) book on the shelf. But there are also complex items that function as words, yet whose internal structure is that of a clause or phrase rather than of a compound. There is no standard term for these items, so may be introduced with the term phrasal words. Though structurally a phrase, then, it behaves as a word. Contrast this with another item which is at least as characteristic in meaning and which has a superficially similar structure: brother-in-law.

Can phrases other than noun phrases constitute phrasal words?

- The answer is yes.

Adjectival examples are dyed-in-the-wool (as in a dyed-in the-wool Republican) or couldn't-care-less (as in a couldn't-care-less attitude). Syntactically, dyed-in-the-wool looks like an adjective phrase consisting of an adjective modified by a prepositional phrase, just like suitable for the party or devoted to his children, such a phrase cannot entirely precede the noun it modifies (we say a man devoted to his children not *a devoted to his children man or *suitable for the party music); therefore the behaviour of dyed-in-the-wool is that of a word rather than a phrase..

As for couldn't-careless, its structure is that of a verb phrase, but again its behaviour is that of an adjective:

- Your attitude is even more couldn't-care-less than hers!.

This seems an appropriate point to mention a small and rather old fashioned class of lexical items exemplified by governor general, attorney general, court martial and lord lieutenant.

How do they form their plural: like attorney generals, or like attorneys general?

If you prefer the former, then these items may seem at first like further phrasal words – except for the fact that they differ from normal English noun phrases in having an adjective following the noun rather than preceding it. It seems better, therefore, to treat them as examples of something that we have not so far encountered: endocentric words which, untypically, have their head on the left rather than on the right. On

the other hand, if you prefer the latter sort of plural (attorneys general), they seem more akin to brother(s)-in-law: not words but lexicalised phrases. If, finally, neither kind of plural sounds quite right to you, that is not surprising, because however these items are analysed, their structure is unusual.

Lesson 08

Morphological Structures

Topic: 43: Morphological structure: Introduction

The meaning of a word is not obtained just by adding the individual meaning of each morpheme. The way they are combined provides aspects of the word's semantics in a compositional way.

- This is what is called a structure.

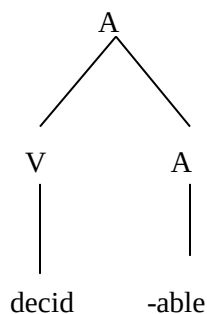
Consider the following:

- undecidable three units
- Prefix Un-, a verbal root, decid(e), a suffix, -able
- Let us derive the meaning step by step
- Undecidable roughly means: 'cannot be decided'

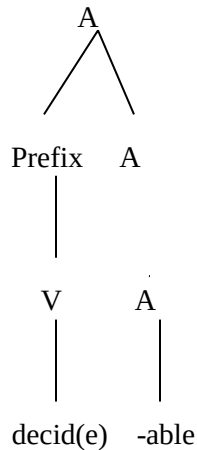
Why the meaning of the word is 'cannot be decided' and not 'can be not decided'? In the first case we refer to an entity that, in any case and no matter how hard one tries, cannot be decided; In the second case, we refer to something that can be decided, but which does not need to be. The way in which a speaker uses the word undecidable is the first and never the second.

How can we account for this?

The way the units are combined with each other. We need to represent how the units are organized inside the word, the structure of the word. The position of the un- morpheme is a decisive factor. In the first case it modifies the suffix -able and in the second it relates to the verb.



In second step the complex word combines with the negative prefix un-



Once we have decomposed the meaning of the word and associated it with each one of its units, we can establish the correspondence in the following:

- The prefix un- combine with adjective, decidable not verbal root decid(e). This account for the meaning of the whole word.
- Negation affect the combination of verb and –able not just the verb decid(e).

The formal properties, the structure shows several things:

A) The grammatical properties of the whole word. The whole structure is an adjective as in undecidable.

B) The units that are responsible for the fact that the word belongs to that particular category are called HEADS

C) It is the combination of the units that produces possible words, and explains that the prefix has a certain meaning as un- has a negative, and not reversative meaning in the word undecidable.

Topic: 44: Morphological properties

Any structure is different from a linear series of combined units. All relationships between units are identical, while in the Linear series there is always a unit which stands in a privileged position inside the structure that imposes its properties on the whole word called Head . In the word un-decid(e)-able, the head is the suffix –able, as it transmits a number of properties to the whole word.

a)The grammatical category.

The word undecidable is an adjective, and the affix –able is responsible for this categorization.

Words ending in –able are normality adjectives.

b) The semantics.

The word undecidable denotes a quality, the quality of being unable to be firmly established or refuted. This meaning contribution is associated with affix –able, as all words ending in –able (or - ible) denote the quality of being able to undergo a particular process (demonstrable, deducible, inexpressible...)

The head inside a morphological structure is identified because of the properties of the whole word are the same properties that the morpheme itself contain. The properties by the head are crucial in determining the behaviour of the word. The grammatical category is crucial in syntactic context, for determining what kind of other words it can combine with, the meaning of the word determines its semantic behaviour.

A third class ‘formal’ or morphological’, in this sense its members do not affect the syntactic distribution of the word or its semantic behaviour. They specify some morphological properties of the word: its gender, its conjugation class or whether its inflection is regular or irregular.

- Consider the English verb *undersell*

It consists of two units *under-* and *-sell* the unit *-sell* is the head. It imposes its grammatical category and meaning. This unit is also responsible for irregular inflection. In syntactic analysis a head is the only unit in a structure which can stand alone. In the phrase ‘The big boy’ the head is ‘boy’ and whose absence will make an ill-formed sentence.

- *The big failed the examination.

The head in morphology is frequently a bound morpheme which cannot stand alone without having been combined with other morpheme.

- As in *government –ment* cannot stand alone.

Thus syntactic criterion is not followed by morphology.

Topic: 45: A-Morphous Morphology

The existence of morphological structure has certain controversies. A number of proposals claim that words do not have an internal structure. Let’s look at this influential theory that argues for this view. This theory proposed by Anderson (1992) argues that words do not have internal structure.

A number of arguments against that analysis that proposes internal structure in words. A structure is defined as a set of units organized in a particular way. A prerequisite for having a structure is identifying a set of units that can be organized. This commits a theory that argues for word-internal structure.

Anderson's first argument against this view of word-internal structure is:

The relevant units, morphemes, do not exist on the basis of the problems that the classical notion of morpheme encounter. Anderson's view of morphology is REALIZATION, a system where there are no one-to-one correspondences between the meaning and the form of the word. In a system like this the meaning of a complex word cannot be established by identifying smaller units which contribute their meaning to the whole as in *cranberry* morpheme having two units' *cran-* and *-berry*. The meaning cannot be explained by combining the meaning of the two. A-morphous morphology proposes that word meaning cannot be reconstructed in that way because units we have segmented are not real.

The word cannot be divided into smaller units, and, by implication it is impossible to identify any internal structure

- A-morphous provides a second argument that show the intuitions of a speaker about meaning- form correlation are not reliable.
- The argument has to do with PHONOAESTHETICS

e.g. sequences of sounds which evoke a particular meaning.

- Anderson notices the words such as slide, slip, slither, slink and slippery share, in the intuition of most speakers, a part of their meaning and phonological sequence, /sl/, but it would not be justified to propose that *sl-* is a morpheme in English.

Topic: 46: Exocentricity

The most important unit of a structure is the head that imposes its properties. Exocentricity is a situation in which none of the units that can be segmented inside the word is responsible for the properties that the word exhibits. So the lack of a head makes it doubtful whether there is an underlying structure in that object. Exocentricity has been studied especially in the domain of compounding, so we will concentrate on these cases here.

- *To grandstand*
- *A pick pocket*

The case is exocentric because the unit that seems to impose its grammatical category (verb-*stand*) on the whole object does not impose the rest of its properties on the word: *Grandstood*

- *grandstanded*

This case is to be contrasted with *understand*, derived from stand by the prefix under-. The compound in (to grandstand) is exocentric in the sense that none of its elements can explain the semantics of the word: A pick pockets

Williams (1981) proposed that these objects are created through exocentric rules, that is transform the properties of the object, but do not add any new unit that introduces these properties. However, accepting exocentric rules amounts in practice to accepting that some words do not have a complete morphological structure. As we can see, Exocentricity is used in practice to label all the situations where the properties that an object has are not the immediate reflection of the properties of one of its units. For this reason, it seems appropriate to classify that the different kinds of Exocentricity depends on the kind of property that shows the mismatch. We will discuss, in turn, category features, semantic features, and morphological features such as whether a verb is irregular or regular, or to which gender it belongs

Topic: 47: Bracketing paradoxes

A crucial property of a structure is that both the formal and the semantic properties of the word need to be represented in the structure. For this reason, the situations where the semantic and the formal properties of an object do not coincide are problematic for the idea that words have internal structure.

These situations are known as BRACKETING PARADOXES (Williams 1981, Beard 1991). There are three kinds of bracketing paradox. One kind of bracketing paradox is the situation in which the formal properties of one of the units requires it to combine with a particular base, but the meaning of the word suggests that it combines with a unit smaller than the base.

- Take the word *international*. The formal properties of the word require the prefix *inter-* to combine with the adjective *national*, as the word **internation* does not exist in English.

a. [inter [[nation]N al]A]A

b. [[inter[nation]N]N al]A

This structure gives the wrong semantics for the word, ‘*relation* between at least two national things.’ The word means more precisely ‘among nations’. Thus, as far as the meaning of the word goes, the prefix does not combine with the adjective *national*, but with the noun *nation* from which *national* is formed (as in the structure in (b)).

b. [[inter[nation]N]N al]A

However, (b) is impossible on formal grounds, because the prefix cannot combine directly with the noun *nation*, as **internation* does not exist, and this structure contains inside it this impossible word. The two possible structures make wrong predictions at some level. Another kind of paradox comes when a unit combines with a word but semantically modifies not only the word, but a bigger structure in which the word is included.

Consider the expression *Vulgar Latinist*.

- This does not refer to *Latinist who happens to be Vulgar*, but to someone *who studies Vulgar Latin*.

This requires the suffix *-ist* to combine with whole phrase Vulgar Latin, and not only with the noun Latin, because that would give us the wrong semantics. One of the properties of affixes is that they cannot combine with phrases, but only with morphological objects. Last kind of paradox take into account phonological factors. In the case unhappy, there is a paradox in the sense that the meaning of the adjective (more unhappy) require the suffix *-er* to combine with the trisyllabic adjective unhappy. It is known that generally trisyllabic adjective take the adverb more in the comparative.

Avoiding this phonological infraction implies proposing that *-er* combines with happy and then *un-* is combines with happier. This gives the wrong semantics as it mean; *not more happy*’ which the speaker does not interpret. There is no analysis for bracketing in the general case which cover the three paradoxes.

Topic: 48: Parasyntesis

The phenomenon known as Parasyntesis constitutes a problem for morphological structures because it casts doubt on one of its crucial properties, namely binary branching: two morphemes can be combined never triplet or bigger sets. Non-binary branching was rejected before on the grounds that when more than two units are combined at a time the relationship which is established between them is not clearly stated. Parasyntesis is the situation where two different affixes - normally a prefix and a suffix - seem to be added simultaneously to the same base. One example of this is the Spanish verb *engrandecer* 'to enlarge', from the adjective *grande* 'big.'

Neither the set formed by the base and the suffix (a hypothetical verb *grandecer) nor that which includes the prefix and the base (a hypothetical adjective *engrande) constitutes a word in Spanish. In consequence, if it is assumed that only existing words can form the basis for further derivation, one is forced to conclude that the verb *engrandecer* has been formed by adding two units at the same time, the prefix *en-* and the suffix *-ece*, to the base *grande*. The same structure, with a zero suffix, has also been proposed in English for words such as *enlarge* ; in this way, the analysis avoids saying that the prefix *en-* is responsible for the category change, since in English prefixes generally do not have this ability.

It is crucial in Parasyntesis that the combination with only one of the two affixes does not produce an existing word in the language. In other words, English discontented is not a case of Parasyntesis, because the root and the prefix form an existent word (discontent); equally, the root and the suffix form one (contented).

Different proposals have been made to avoid the ternary structure in (26).

a. [en [gran(e)]A ece]v

b. [en [large] ø]v

One of the most widely accepted is to consider the prefix and the suffix to be a discontinuous morpheme, a circumfix. In this way, the structure would be binary branching, as only two units, [en . . . ece] and [grande], are combined. The analysis of Parasyntesis as a circumfix has an empirical problem. The circumfix that should be proposed in the case of *engrandecer* has a second part which is identical to a Spanish suffix

- Palid –ece (fade)
- and a first part that, again, is identical to a Spanish prefix
- En-frente (in front)

The fact that these units exist independently of each other suggests that it is not accurate to propose that a circumfix is responsible for the verb *engrandecer*. The weak side of this analysis is that has to postulate an intermediate phase with a possible but non-existent word hypothesis, *grandece*. Some linguists accept but others deny.

Lesson 09

Derivation and Semantics

Topic: 49: Introduction: Derivation & Semantics

The Utne Reader quoted the following Bucharest sign in its March–April 1996 issue:

- The lift is being fixed for the next day.
- During that time we regret that you will be unbearable.

Why is this sign so funny?

First of all, unbearable already exists in English with the meaning ‘difficult to tolerate’, and this is the meaning we first think of when we read the sign. But this isn’t the whole story. If we replace unbearable with a similar but unambiguous word, like transportable, the sentence is no longer humorous, but it still sounds less than native. It’s because words like unbearable, untransportable, and uneatable describe inherent qualities of people or things, qualities that don’t change simply because an elevator is out of order. To take another example, the fact that someone is allergic to chocolate doesn’t make the chocolate cupcakes in the window uneatable. In this simple sign, there is a complex interaction between affixation and semantics going on, and that is the sort of thing that we explore in this chapter.

We begin by introducing a fundamental problem in lexical semantics, the study of word meaning: the meanings of individual lexemes can be highly diverse. We then examine in some detail the semantics of derived lexemes to see what generalizations we can draw.

Topic: 50: Polysemy Problem

The most fundamental aspect of a word’s meaning is that it refers to some entity or relation (real or imaginary) in the world. We can refer to this entity or relation as the word’s semantic type. Formal approaches to grammar have provided us with terminology that allows us to make even more fine-tuned distinctions between words. The word reptile refers to all individuals in the world that are reptiles.

Verbs like respect or love refer to relationships between individuals.

- We can differentiate:

- bear from teddy bear foliage from leaf
- literature from book on the basis of the mass/count distinction.

Verbs are given labels such as ergative, accusative, transitive, or intransitive. The main problem of lexical semantics is that the meanings of individual lexemes are highly diverse. We call this the problem of polysemy.

As an example, take the verb lose.

- They lost their passports;
- Jake lost his job;
- Sarah lost her husband to cancer;
- I lost my temper;
- we both lost ten pounds.

But all of the meanings of lose reflected here are related – they are all instances of the same lexeme. Because lose has more than one related meaning, we say that it is polysemous.

There are many types of polysemy.		
(1) mass/count distinction	(1) a. I love watermelon. (mass)	b. I sold three watermelons. (count)
(2) Figure–ground reversal	a. Hugh broke the window.	b. The kids climbed through the window.
(3) Container–contained alternation	a. A hot glass put under cold water will shatter.	b. Franny downed the glass in two seconds flat.
(4) Place–people alternation	a. The president and his family live in the WhiteHouse.	b. The White House announced yesterday that the peace talks willcontinue.

The sentences in (2–4) contain pairs of words with very different – even contradictory – interpretations. Yet they represent single lexemes. Examples like these show that the same phonetic string can convey different, but related meanings depending on the linguistic and pragmatic context. This introduction to polysemy leads into the discussion which focuses on the semantics of derived lexemes.

Topic: 51: The Semantics of Derived Lexemes

When somebody makes up a word, they are inventing it for use under a particular circumstance. Sometimes the circumstance can be very peculiar. Take the sentence:

- Joe was Houdini'd and died.

The second factor that can affect a word's meaning is its history. We might think of every lexeme not just as a word and its meaning, but as the word and every time it has ever been used: every time we hear the word, we revise its lexical entry in some way.

That this indeed goes on is particularly evident from first-language acquisition research. Children in earlier stages of language acquisition may under extend a word by using it to refer to only a subset of its actual referents, or overextend a word by using it to refer to objects or individuals that are typically covered by the word, as well as to others that are “perceptually similar” (Clark 1993: 33).

For example, a child might under extend the word dog by using it to refer to more typical examples of the species, but not to varieties like Chihuahua or Pekingese (Kay and Anglin 1982), or overextend tree by using it to refer to potted plants, trees, and even balsam fir wreaths. Such under- and overextensions are generally short-lived, which indicates that children revise lexical entries as they are exposed to more and more tokens of a word.

It is not unreasonable to think that the meaning of a word is a compilation of every single use of that word. Every word has a history. It has your own personal history – how you have heard the word. It has the history of the word as it has been used by other people.

Over time, the meanings of words can become more complex and diverse, making the task of the morphologist looking for semantic patterns of word formation more complicated than it would be if the semantics of word formation were purely compositional. A syntactic construction may have pragmatics to deal with, but it doesn't have history. One question you might want to ask is what kinds of meanings arise via lexeme-formation rules.

- Are derived forms like lexemes, with potentially very complicated meanings?
- Or are they like syntactic collocations, with simple meanings?

Topic: 52: The semantics of affixation

Let's take the English suffix -ism. This affix has some very highly lexicalized meanings, one of which is 'doctrinal system of principles'. We find this meaning in words like the following, and many others having to do with religion, philosophy, science, politics, or the arts.

Religion	Philosophy	science, politics, or the arts

Catholicism	Platonism	romanticism	McCarthyism
Judaism	Marxism	realism	socialism
Buddhism	idealism	surrealism	fascism

The suffix -ism has an even more specific and lexicalized meaning:

- ‘a peculiarity of speech’. We talk about colloquialisms, spoonerisms, and Obamaisms (Obamaism can either be a system of beliefs or a peculiarity of speech).

So -ism is an example of a suffix with two very highly lexicalized meanings, both of which might be considered to be more characteristic of words than of affixes.

The German suffix -ei is like -ism in having at least two very highly lexicalized meanings. The first, illustrated by the words in (8), attaches to a noun and makes another noun meaning ‘the place in which X works’.

a. Bäcker ‘baker’ → Bäckerei ‘bakery’

b. Drucker ‘printer’ → Druckerei ‘printing office, print shop’

c. Sattler ‘saddler’ → Sattlerei ‘saddlery, saddler’s workshop’

d. Tischler ‘joiner’ → Tischlerei ‘joinery’

The second, seen in (9), attaches to a verb stem and creates a noun referring to the ‘act of doing X’:

a. plaudern ‘chat (v)’ → Plauderei ‘chat (n)’

b. zittern ‘tremble’ → Zitterei ‘trembling’

c. prügeln ‘clobber’ → Prügelei ‘brawl, fight’

d. quengeln ‘whine’ → Quengelei ‘whining’

English has a cognate suffix of French origin, -(e)ry, as in bakery, tannery, winery; bribery, flattery, foolery.

Topic: 53: The semantics of zero-derivation

We now address a type of word formation which is much more abstract: zero-derivation. Zero-derivation results in lexemes whose interpretation is context-dependent. The peculiarity of zero-derived verbs is that they often have a wide range of meanings. To give you just one example, the verb to sand denotes two very different actions.

While sand is well established as a verb, zero-derivation is a productive derivational process in English (cf. 4.2.2), as shown by the following nonce forms presented by Clark and Clark. All of these sentences are actual quotations:

a. Ruth Buzzi house guested with Bill Dodge (Herb Caen, SF Chronicle)

b. He wristed the ball over the net (tennis commentator)

c. When you’re starting to Sunday School members, then I think you’re going too far (a Californian legislator)

d. Will you cigarette me? (Mae West)

e. We all Wayned and Cagneyed (NY Times magazine)

Location	(N is at a place) blanket, saddle, roof (put something at N) kennel, ground, cellar
Duration	(spend the duration of N) summer, holiday, vacation, weekend

Agent	(N acts) jockey, referee, umpire, pilot
Goal	(make into N) fool, orphan, baby, cripple, pile, loop, powder
Instrument	(use N) ship, nail, glue, shampoo, fork
Miscellaneous	lunch, hay, whale, dog

- while shipping originally took place via ship, today we ship things by truck or air. We might redefine this category as ‘do what you do with N’.

The miscellaneous category includes some interesting words. We use the verb whale to mean ‘catch whales’ or fish to mean ‘catch fish’ but are hard pressed to come up with many other verbs of this type. We don’t use a verb deer to mean ‘hunt deer’ or butterfly for ‘catch butterflies’. The zero-derived verbs with the most extensive semantic possibilities are probably those that are derived from personal names. To understand them, you need to know something about the person and often about a particular event.

12), you have to know the circumstances of Houdini’s death:	
(12) Joe was Houdini’d and died.	
(13) a. My sister Houdini’d her way out of the locked closet.	
b. I would love to Houdini those ESP experiments.	

The semantic obscurity of verbs derived from personal names results in speakers forgetting very rapidly that there even was a zero-derivation. To restate this analysis, the wide array of meanings of zero-derived verbs results from two properties. **The first** is that the rule by which simple, specifying only that we take a noun and form a verb. **The second** is that conversational convention dictates only that the verb have something to do with the noun. We can reasonably compare this analysis down even further and say that the proper analysis of zero-derived verbs is that they are simply verbs.

The fact that they denote an activity connected with the noun is derivable on purely conversational grounds. While any English speaker can see the connection between nurse (noun) and nurse (verb) or bottle (noun) and bottle (verb), most aren’t aware that boycott and lynch are of the same ilk. Robert Lees, in his classic book on English nominals (1960), derived the meanings of denominal verbs from sentences containing them. The verb summer would be derived from the phrase spend the summer and the verb kennel from keep in a kennel. While other derivational formations don’t have as dramatic an array of potential meanings as zero-derived verbs, we still find variety.

Topic: 54: English agent nouns in -er

Another derivation that works in the same way as zero-derived verbs but is a little simpler. Marchand gives examples of English agent nouns in -er. He points out that they fall into four basic categories, which can be broken down even further into two separate sets.

The four basic categories are listed below:

Persons:	baker, dancer, gambler, driver
Animals:	pointer, retriever, warbler, trotter
Material objects:	blotter, eraser, fertilizer, shutter
Immaterial objects:	reminder, clincher, thriller, eye-opener

These nouns can be further divided into their habitual and non-habitual uses.

Habitually:	He is a gambler. we usually mean that he gambles regularly.
non-habitually:	All ticket-holders may enter

There are two possible analyses of agent nouns, both of which are reasonable.

- We won't try to choose between them here.
- One is the strategy to assign the derivation a sparse semantic rule.

So an agent noun is 'someone or something connected with what the base denotes', or alternatively, 'somebody or something whose function or characteristic is to perform a particular act'.

- For now we assume the latter.

It permits the categories person, animal, material object, and immaterial object, as well as a habitual or non-habitual interpretation. The other method which linguists might use to account for possible meanings of X-er agent nouns involves prototypes, also called archetypes. The idea is that not all members of a given category are equal. Reasonably the prototypical agentive is a person who habitually performs a particular type of action. So the prototypical agentive is a word like baker, dancer, gambler, or driver, in the habitual sense. These reflect the core meaning of this particular formation.

Other forms, like retriever, blotter, or clincher, involve relaxation of the core meaning. We use them to distinguish one specimen from other members of its class.

- A pointer
- A retriever

When we claim that persons are the prototypical members of the category of agentive nouns, we can also argue that some other members of the category – material objects like screwdrivers, for example – aren't agents at all. Instead, they are instruments, because they don't have will. In short, the second method of analyzing agentives is to establish a central case, the prototype, and to work out from that to get the others.

Lesson 10

Inflections

Topic: 55: Introduction Inflection

What is the plural of euro, the name of the European currency? The answer seems obvious: euros – one euro, two euros,

- -s is the productive plural marker in English and is normally used with new coinages, e.g. modem or byte: modems, bytes.
- In Ireland, the only largely English-speaking country in the euro zone,
- four euro and not four euros.
- Why?
- The answer lies in the multilingual nature of the euro zone.

Name of the currency itself bespeaks such multilingualism: its devisers had to come up with a name that could be both written and pronounced in all of the languages in the zone.

- It is pronounced differently in each of the languages.
- A 10-euro bill has the solution to this multilingual problem that was found. First, unlike the US

10-dollar bill, where ten dollars is written out in full.

- The word for ‘ten’ is written in numerals: 10.

The reason is simple: 10 can be written identically while still being pronounced differently in each of the languages of the zone. **The second problem:** the plural form of euro, which follows the numeral, must also be written uniformly on the bill and, though several major European languages besides English use the plural marker -s(French, German, and Spanish do), not (Italian, etc. never uses an -s plural). The bill reads simply 10 EURO/EYPW and the spoken languages have all followed the same practice of not using any plural marker either, making the plural of euro euro in Ireland. The linguistic point of this story is that even when several languages share the same word, as they do with euro, the inflectional systems of the languages differ, and this difference normally affects the forms of the words. Inflection varies from one language to another more than any other systematic aspect of language. Let’s see how it varies across languages widely but still remain quite relentlessly systematic within themselves.

Topic: 56: What is Inflection?

The inflection is from Latin root *flect-* means ‘*bend*’ but this bending is now become *altering the shape of the word* to fit in a particular sentence. Every sentence is a syntactic frame with positions for a series of words.

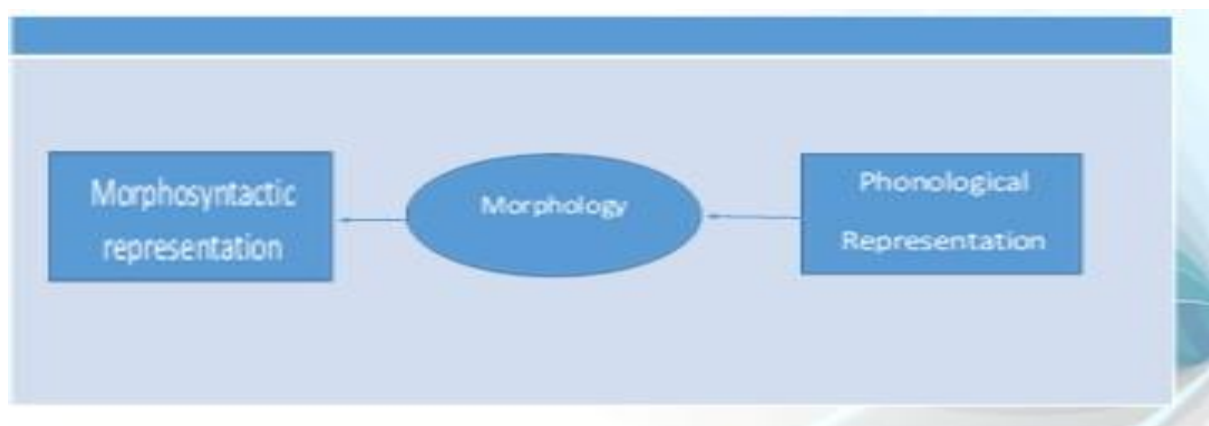
- This_____is clever.

Noun

In order to fill one of those positions, you take a lexeme from the lexicon and bend it to fit. In this way, inflectional morphology is determined by syntax.

- What kinds of things do lexemes express through inflection?

In general we speak of inflection expressing morphosyntactic information: syntactic information that is expressed morphologically. This includes the abstract syntactic categories of, *tense*, *aspect*, *number*, and *case*. Since the syntax provides the morphology with morphosyntactic features, the job of the morphology must be to get from there to the actual phonological realization:



- The answers to that question.

- Exponence: refers to the realization of morphosyntactic features via inflection.
- In *seas*, the morpheme[z] is the exponent of the morphosyntactic feature plural,
- In *sailed*, [d] is the exponent of past tense or past participle

In both cases there is a one-to-one relationship between form and meanings, since one morpheme realizes one morphosyntactic feature, a situation that Matthews calls simple exponence. Lastly, inflection for *case*, *number*, and *gender* in many Indo-European languages involves cumulative exponence. like Latin -o: in *am-o*: 'I love',

- Context-free and context sensitive inflection.

Context-free inflection when there is a simple directional mapping between a morphosyntactic feature and a particular phonological string.

- as /-ɪŋ/, refers to context-free inflection:
- All present participles in English bear the same suffix.

Inflection for past tense in English is context sensitive in the sense that the feature [past] is realized as many things depending on the lexeme it attaches to, with /-d/ suffixation being the default case

- Inherent vs. assigned inflection

Nouns and pronouns are marked as having a particular gender in the speaker's mental lexicon. For them, gender is inherent. For any other lexical category that reflects the gender of nouns and pronouns, such as adjective and verb, gender cannot be inherent.

- It must be assigned.
- Government vs. concord
- Another word for concord is agreement.

Concord or agreement occurs when one element in a sentence takes on the morphosyntactic features of another element. One example of concord is noun–adjective agreement in the Romance languages or German. Adjectives take on the number and gender of the noun they modify. Government is more or less what it sounds like: one word dictates the form of another. Case assignment by verbs is usually thought of in this way. When a noun is required to appear in objective case, for example, it cannot be said that it agrees with (reflects the case of) the verb. This is because verbs don't have case. The same holds for prepositions.

Inflectional categories

While most languages have morphological inflection of some sort, the actual inflectional categories can differ quite widely across languages. In this section, we briefly survey both the most common categories and some of the ways languages may differ. It is convenient to make a first broad cut

into nominal and verbal categories, though the nominal categories often appear on adjectives and verbs through concord. The most common nominal categories are number, gender and case.

Topic: 57: Inflection vs. Derivation

Within a lexeme-based theory: derivation gives new lexemes, and inflection gives you the forms of a lexeme that are determined by syntactic environment. What exactly does this mean, and is there really a need for such a distinction? The first question about the distinction is whether there is any formal basis for distinguishing the two types of morphology. Can we tell them apart because they do different things to words?

Distinction between inflectional and derivational morphemes

- Inflection does not change the core lexical meaning or the lexical category of the word to which it applies. Derivation does the former and may do the latter.
- Inflection is the realization of morphosyntactic features, i.e., those that are relevant to the syntax, such as case and number. Derivation is not.
- Inflectional morphology is more productive than derivational morphology.
- Derivational morphology tends to occur closer to the root or stem than inflectional morphology.
- Derived lexemes are more likely to be stored in the lexicon than inflected forms.

Despite the generalizations made above, the morphological form that inflection and derivation may take is very similar. Cross-linguistically, both can be expressed through prefixal, suffixal, or non-segmental means. The difference between inflection and derivation is therefore not so much a difference in form as a difference in function – what they do and what they tell us.

Topic: 58: Inventory of Inflectional Morphology Types

What are the ways in which speakers inflect, or “bend,” lexemes to make them fit into a certain syntactic slot? We present a list of inflectional morphology types here. Not comprehensive, it will give a broad understanding of how inflection may be expressed.

Affixation and stem alternations

The most common means by which inflectional categories are expressed. For that reason we do not address affixation on its own here. We present it with stem alternations, another means of expressing inflection in the world’s languages. We wish to emphasize that just as affixation does not necessarily occur with stem alternations, stem alternations do not necessarily occur with affixation. We present them together merely for organizational purposes.

Latin is one language where in many cases, affixation interacts obligatorily with stem alternations in the expression of inflectional categories.

- *rēxisti*- ‘you ruled’. The perfective stem *rēx*- can be contrasted with the present stem *reg*-.

Apophony

In some cases affixation is supplemented or replaced by apophony, or vowel changes within a root, as shown below for English and the Bernese dialect of Swiss German: (14) a. sing, sang, sung b. drive, drove, driven

- suuffe [su:f e] ‘drink (inf)’ gsoffe [°qs c f: e] ‘drunk (past part)’
- schwimme [ʃvɪm: e] ‘swim (inf)’ gschwomme [°qʃvɔ̃m: e] ‘swum’
- pfyffe [pfi:f e] ‘whistle (inf)’ pfiffe [pfif: e] ‘whistled (past part)’

Other terms for apophony are internal change and, particularly when referring to English and other Germanic languages, ablaut. All three of these terms are sometimes applied to the vowel changes that apply to roots in Semitic languages. In the context of Germanic linguistics, ablaut is often reserved for apophony in verb paradigms, as seen. A second term, umlaut, is used to describe the apophony found in singular–plural noun pairs like goose~geese or foot~feet. Umlaut is a phonologically conditioned alternation in which a vowel assimilates in part to a succeeding vowel. The term is used even when the succeeding vowel has been lost. For instance, goose, geese and foot, feet resulted from vowel harmony with a high vowel in the plural suffix, which has since disappeared.

Topic: 59: Inventory of Inflectional Morphology Types II

In the Semitic language family, inflection often involves internal variations in vocalic and syllabic patterns, while the consonantal frame stays fairly stable. We call this root-and-pattern morphology: in certain Arabic nouns. The inclusion of the loanword ‘bank’ is to show that this particular way of forming the plural (referred to as the broken plural) is productive. In the examples below all of the plural forms begin with the syllable pattern CVCVV+:

Root	Singular	plural	gloss
Jndb	jundub’	janaadib	‘locust’
slt’n	sult-aan	salaat’iin’	‘sultan’
nfs	nafs	nufuus	‘soul’

bnk	bank	bunuuk	'bank'
-----	------	--------	--------

Root-and-pattern morphology is yet another way of “*bending*” a form to fit a particular syntactic context.

Reduplication

A but from a phonological perspective.

Examples Indonesian plural here.

- kuda-kuda ‘horses’
- rumah-rumah ‘houses’
- singkatan-singkatan ‘abbreviations’

Indonesian plural reduplication is not obligatory. Speakers of Indonesian have the option of using the unreduplicated form to refer to either singular or plural. So kuda not only means ‘horse’, but also ‘horses’; rumah can refer to one house or more than one; and so on. The reduplicated plural is most likely to be used when the number of the noun is not clear from the context. The reduplicated plural is most likely to be used when the number of the noun is not clear from the context.

Suppl

Suppletion is said to take place when the syntax requires a form of a lexeme that is not morphologically predictable. In English, the paradigm for the verb *be* is characterized by suppletion.

- *Am, are, is, was, were*, and *be* have completely different phonological shapes, and they are not predictable on the basis of the paradigms of other English verbs.
- We also find suppletion with pronouns.

Examples of suppletion in other languages

French aller ‘to go’/vais ‘go (1sg)’

être ‘to be’/suis ‘am (1sg)’

Spanish ir ‘to go’/fue ‘went (1sg)’

ser ‘to be’/fue ‘was (1sg)’

Finnish hyvä ‘good (nom. sg)’/parempi ‘better’, paras ‘best’

Greek [enas] ‘a, one (m.nom.sg)’/[mja] ‘a, one (f.nom.sg)’

Swedish ett ‘one’, två ‘two’/första ‘first’, andra ‘second’

We can look to historical linguistics for an explanation for how suppletive forms arise. The paradigm of the verb ‘to go’ in French, for example, comes from three different Latin sources. The idiosyncrasies of languages today can often be explained by looking at the languages of yesterday. In certain cases, such as with catch~caught or think~thought and other verbs like them in English, it is most convenient to speak of partial suppletion. In these cases, the initial phoneme or phonemes of the word remain the same, but there is both internal change and change to the end of the word (loss of segments and addition of a past tense indicator [t]).

Topic: 60: Syncretism

We speak of syncretism when a single inflected form corresponds to more than one set of morphosyntactic features. Syncretism is common cross-linguistically, and it raises a number of questions relevant to morphological theory. In the Bulgarian imperfect and aorist paradigms, the second person singular and third person singular forms are identical. (The aorist is a past tense.) We show this in (19) for the verbs *krad* ‘steal’ and *igráj* ‘play’ (these are the citation forms of the lexemes):

In the follow ing , -š is a preterite suffix and -e is 3sg agreement

Imperfect	2sg <i>krad-é-š-e</i>	igrá-e-š-e
	3sg <i>krad-é-š-e</i>	igrá-e-š-e
Aorist	2sg <i>krád-e</i>	igrá
	3sg <i>krád-e</i>	igrá

We see here that a single inflectional form such as *kráde* may express more than one set of morphosyntactic features: 2sg aorist or 3sg aorist. Romanian also displays widespread syncretism. The term *typology* refers to a classification based on the comparative study of types, and morphological typology was the first systematic method used by linguists in the nineteenth century to compare the structures of different languages.

While other sorts of typology flourish today, especially syntactic typology, morphological typology has languished since it was criticized by the first American structuralists, especially Edward Sapir in his *Language* (1921). Still, the traditional terms are used often enough to warrant mention, and the distinctions, while they may not be valid for entire languages, are still useful for describing individual morphological phenomena.

The basic typology has to do with a scale running from analytic to synthetic languages, which encodes the degree to which the individual meaningful elements in a language are expressed separately. At the analytic end we have the *isolating languages*, of which Vietnamese is the prototypical example, because the only morphology it has is compounding.

It has no derivational or inflectional processes of any kind. The next type is inflective, of which the more analytic subtype is agglutinating.

- An *agglutinating language* like Turkish or Hungarian has affixes, but they are strung out quite separately, each expressing a single notion, and easily identified.

		‘house’	‘house’
Singular	Nominative	ház	folió
	Accusative	házat	foliót
Plural	Nominative	házak	foliók
	Accusative	házakat	foliókat

The accusative case marker is -at (the vowel is deleted after a stem-final vowel), while the plural marker is -ak (with the vowel again deleting after a stem-final vowel). When a word is both accusative and plural, both affixes appear one after the other. Compare a *fusional language*, distinct suffixes: -s, -m, -ī, and -ōs (here, the stem vowel deletes before the suffix vowel), so we say that the two morphosyntactic features in each of the cells of the table (e.g. Nominative singular) are fused. Latin is actually much more complicated, since these two nouns represent only one of five main types (declensions), each of which has a distinct set of forms.

		‘lord’	‘garden’
Singular	Nominative	dominus	hortus
	Accusative	dominum	hortum

Plural	Nominative	dominī	hortī
	Accusative	dominōs	hortōs

- **Polysynthetic languages.** The languages of this type cited most often come from North America. One example of a polysynthetic language is Nuuchahnulth (called Nootka in earlier literature), a language spoken on Vancouver Island in British Columbia.

Even English can have one-word sentences. Go! is a perfectly well formed such utterance. But compared to English, polysynthetic languages are able to express much more complex notions using a single word, including subject, verb, object, and other information. The English gloss of this single *Nuuchahnulth* word, for example, contains a complex subject noun phrase, a progressive verb, and a prepositional phrase expressing location.

Lesson 11

Lexical Source of English Word Formation

Topic: 61: Lexical source of English word formation

Not the summary of the whole history of the English language. Just those aspects of its history over the past thousand years or so that help to account for some of the peculiarities of word formation in contemporary English.

English

A West Germanic language, related closely to the other West Germanic languages: Dutch, German, Frisian and Afrikaans and less closely to the North Germanic languages: (Norwegian, Danish, Swedish, Icelandic and Faeroese). England was ruled for a long period after 1066 by a monarch and a nobility whose native language was a variety of French; This ruling group gradually switched to English for everyday purposes; French a language of law, administration, and culture every educated person learnt it. It is not surprising, then, that the vocabulary of English contains a high proportion of words borrowed from French than other Germanic languages. French is one of the so-called Romance languages, descended from Latin, along with Portuguese, Spanish, Italian and Romanian. Latin has had a more direct influence too. Official language of the western half of the Roman Empire and the vehicle of a huge and varied written literature, second only to Greek. The liturgical language of all West European Christians until the Protestant Reformation 16 century, and of Catholics till 1970s

Many words were adopted into English from Latin directly, rather than by way of French. The Romans preferring instead to create new Latin terms to translate Greek ones. The possibility of direct Greek influence on English did not arise, however, until 16 c Western Europeans began to learn about Greek culture for themselves in the fifteenth century. In word formation, influence of Greek has been in the invention of scientific and technical words. The variety of the sources that have contributed to the

vocabulary of English came in whole or in part, from the same ancestral morpheme in the extinct Proto-Indo-European language from which Greek and the Romance and Germanic languages

Proto-Indo-European

Proto-Indo-European from which Greek and the Romance and Germanic Languages are descended.

An example Proto-Indo-European root: heart

In Latin as cord-, In French as coeur from which was formed a derivative courage, In Latin cordial cordial appears in cardiac, from Greek word kardiakos

Another Indo-European root : bear

In Latin as fer-, in Greek as pher,

The former as the bound root in verbs such as, confer and latter in the name Christopher

English has also acquired the root via French suffer (or in Modern French souffrir)

A striking feature of these words is that the inherited Germanic forms, and in the forms borrowed from Latin, French or Greek the cognitive roots are bound. In borrowing words, English speakers borrowed not only the roots and affixes but also the pattern of word formation that they conform to --a pattern which does not allow roots to appear naked, so to speak, unaccompanied by some derivational or inflectional affix. Some borrowed roots are free, and a few inherited ones are bound. Most of the roots that are bound in all contexts (that is, most of the roots that have no free allomorphs) do not belong to the vocabulary that English has inherited from its Proto-Germanic ancestor.

Topic: 62: The rarity of borrowed inflectional morphology

If English borrows a foreign pattern of word formation, it should be expected to borrow inflectional affixes that conform to that pattern, as well as roots and derivational affixes. English does not use French or Latin inflectional affixes on verbs borrowed from those languages a language acquires lexemes through borrowing rather than individual word forms.

If English speakers import a new verb V from French, they will not import just its past tense form since we expect to be able to express in English not only the grammatical word ‘past tense of V’ but also the grammatical words ‘third person singular present of V’, ‘perfect participle of V’, and so on. It is much more convenient to equip the new French-sourced verb with word forms created in accordance with English verbal inflection – specifically, the most regular pattern of verbal inflection (suffixes -s, -ed and -ing). The only condition under which English speakers are likely to borrow foreign word forms along with the lexemes if the grammatical words are few in number(not hard to learn), and if their functions in English and the source language correspond closely. This condition is fulfilled with nouns.

Greek and Hebrew, which resemble English in distinguishing singular and plural forms in nouns:

Greek	phenomenon phenomena schema schemata
Latin	cactus cacti formula formulae datum data
Hebrew	cherub cherubim kibbutz kibbutzim

The foreign plurals are all vulnerable, however. Phenomena and data seem solidly established, but for the others not: schemas, cactuses, formulas, cherubs and kibbutzes. Even data tends to be accommodated to English morphology, but by a different method:

- The usual plural suffix in both medieval and modern French is -s, just as in English.

Just a few French borrowings sometimes retain, in formal written English, an idiosyncratic plural suffix -x, e.g. tableaux, plateaux.). The effect of these borrowings is to divide the class of nouns with irregular plurals into two classes: nouns that belong to everyday vocabulary and whose irregular plural survived for frequent use (e.g. teeth, children, mice), and relatively rare or technical survived as a badge of learning or sophistication. What we do not find are irregular plurals that fall between these extremes, in nouns that are not particularly common but do not belong to technical or learned vocabulary either.

- But what about oxen

Topic: 63: The reduction in inflectional morphology

Modern English nouns have no more than two inflected word forms: singular and plural. In Old English, there was a contrast of case, like that found in modern English personal pronouns (nominative we versus accusative us etc.), Old English nouns could distinguish also a genitive case, and a dative case whose meanings included that of modern to in Mary gave the book to John. These two numbers and four cases yielded a pattern of eight grammatical words for each noun lexeme

	Nominative	Accusative	Genitive	Dative

Singular	nama ‘name’ stān ‘stone	Naman	Naman	Naman
		stān	stānes	stāne
Meaning	Name			
Plural	naman	Naman	Namena	namum
	stānas ‘	stānas	stāna	stānum

Neither name nor stone had eight distinct word forms, one for each grammatical word; instead, they display different patterns of syncretism. All Old English nouns had more than two forms that are available in modern English. If nouns distinguished four cases in Old English, it is reasonable to guess that pronouns should have done so too; and that guess is correct. Pronouns had five cases, including an instrumental. Adjectives had five forms. Old English verbs displayed a similar inflectional luxuriance. Most modern English verbs have four distinct forms (e.g. perform, performs, performed, performing), while some common verbs have five (e.g. speak, speaks, spoke, spoken, speaking).

		Indicative	Subjunctive
	Person	Present	Present
Singular	1st (‘I’)	helpe	helpe
	2nd (‘you’)	helpest	helpe
	3rd (‘(s)he’)	helpeD	helpe
Plural	1st (‘we’)	helpaD	helpen
	2nd (‘you’)	helpaD	helpen
	3rd (‘they’)	helpaD	
		Indicative	Subjunctive

	Person	Past	Past
Singular	1st ('I')	healp	hulpe
	2nd ('you')	hulpe	hulpe
	3rd ('(s)he')	healp	hulpe
Plural	1st ('we')	hulpon	hulpen
	2nd ('you')	hulpon	hulpen
	3rd ('they')	hulpon	hulpen

Besides the imperative forms ('help!'), or the verbal adjective helpende, which, just like other adjectives in Old English, had forms that distinguished three genders, two numbers and four cases.

- Why did English lose this wealth of inflection?

The temporary eclipse of English by French as the language of culture and administration after 1066. Partly also it is due to dialect mixture. **The examples** of 'Old English given here come from the dominant dialect of written literature, that of south-western England. But this was not the dialect of London, which became increasingly influential during the so-called 'Middle English' period (from about 1150 to 1500), and established itself as the main variety used in printing. English inflectional morphology was already by 1600 almost the same as in 2000, so that modern readers of Shakespeare encounter only a few obsolete inflected forms such as 'helpest' and he helpeth, that preserve two Old English suffixes

Topic: 64: Characteristics of Germanic and non-Germanic derivation

The inherited Germanic root heart is free while the cognate roots cord- and card-, borrowed from Latin and Greek, are bound, and the same applies to inherited bear by contrast with borrowed *-fer* and *-pher*. If this kind of contrast is general, then it has implications for inherited and borrowed affixes too. We will expect that native Germanic affixes should attach to free bases, while the affixes that attach to bound bases should generally be borrowed. And this turns out to be correct.

The derivational affixes that we have considered so far, classified according to their origin:

Germanic

select almost exclusively free bases

Romance or Greek

readily permit or even prefer bound ones

-ish	-(a(t))ion	-ine
-ed	-(i)an	-ise
-en	-(i)fy	-ism
-er	-al	-ist
-hood	-ance, -ence	-ment
-ie (as in doggie)	-ar	dedis-
-let	-ent, -ant	
-ship	-ess	
-y (as in misty)	-ette	

-let and -ette,: both mean roughly ‘small’, and -ette also sometimes means ‘female’.

. booklet, piglet, droplet and starlet, all with clearly identifiable free bases.

. cigarette, and it may also include (depending on your country of origin) suffragette, laundrette, kitchenette, maisonette and drum-majorette.

Similar conclusions emerge from comparing some abstract-noun forming suffixes in the two columns: -ship and -hood in the Germanic column,

-(a(t))ion, -ance/-ence, and -ism in the Romance and Greek column.

- **certainly possible to find words whose bases are free (e.g. consideration, admittance, etc. many of the bases selected by these affixes are bound,**
- **being either bound allomorphs of roots that are elsewhere free (e.g. consumption, preference, Catholicism)**
- **or else roots that lack free allomorphs entirely (e.g. condition, patience, solipsism).**

Nouns in -ship and -hood always seem to have freebases: friendship, kingship, governorship; childhood, adulthood, priesthood.

The historical basis for a : the root of an English word is more likely to be free than bound, yet a large number of bound roots exist in modern English also, do to massive borrowing from French and Latin.

An example is the suffix -ment, as in development, punishment, commitment, attainment – though it is sometimes found with a bound base, as in the nouns compliment and supplement. Another example is the prefix de-, as in deregister, and decompose. This tolerance for free bases is surely connected with the fact that, in the terminology de- is formally and semantically rather regular, and can readily be used in neologisms (e.g. de-grass in The courtyard was grassed only last year, but now they are going to de-grass it and lay paving stones. For an affix restricted to bound bases, such a neologizing capacity would be scarcely conceivable in a language where, as in English, most bases are free.

Topic: 65: Fashions in morphology

The title highlights a respect in which morphology differs from syntax. It makes sense to ask whether a certain word formation process (a particular affix, let's say) is in or out of fashion. Syntax is stable in a way that morphology is not. The fact that many morphological processes are haphazardly 'gappy' whereas few if any syntactic constructions are 'gappy' in this way.

In morphology, gaps get filled, or else gappy processes lose their regularity and survive only in a few lexically listed lexemes, like the process of forming abstract nouns by suffixing -th to adjectives, while other processes become increasingly regular to replace them. A systematic study of morphological fashions belongs to a historical study of English word formation rather than to an introductory survey such as this.

Two fashions are important that manifested themselves in the last half of the twentieth century, against trends of the last couple of centuries. The first is a fashion for certain Latin- and Greek-derived prefixes; the second is a fashion for a certain kind of headless compound. Conscious borrowings from Latin and (to a lesser extent) Greek were fashionable in certain literary styles of the sixteenth and seventeenth centuries, because of a perceived need to enrich the English vocabulary. But such borrowings, often obscure and even incomprehensible to ordinary readers, were also attacked as 'inkhorn terms' – mere products of the some scholars' desire to show off his knowledge of Latin.

- The result is that the Latin- and Greek-derived element in the vocabulary of English has, since the eighteenth century, been pruned rather than increased.

Prefixes

Latin			Greek		
Super	Superstar, rich,	superman, super-	Hyper	hyperactivity, hypermarket	

		Macro		
Sub		Micro		
		Mega, Nano	Giga, megastore, megalith, megabucks,	

- A fashion for a certain kind of headless compound or exocentric compound
- For example : redhead, lazybones, pickpocket, longneck, etc.

Topic: 66: History and structure

- Characteristics of a language that are due purely to historical accident are the characteristics that, in principle, are least likely to interest a general linguist.
- The Norman Conquest in 1066 is just such an accident, so its consequences for the vocabulary of English from French may seem to deserve a place only in histories of the English language, not in books about its morphological structure.
- But there is more to it than that.
- If it had not been for the Norman Conquest and its aftermath, English morphology would not have acquired the at first sight rather bewildering mix of characteristics discussed so far.
- What's more, one cannot dismiss characteristics acquired through the Latin lexical intake as 'unproductive' and therefore not truly part of modern English morphology;
- Some Latin-derived processes, such as suffixation of -ion and -ence, are in limited domains just as formally regular as processes such as adverb formation with -ly.
- If the history of the community of English speakers in the British Isles had been otherwise, the English language would be considerably different today not just in its repertoire of lexical items but in how its words are structured.

Morphological Productivity and the Mental Lexicon

Topic: 67: What is Morphological Productivity?

- A given morphological pattern is more productive than another is to say that there is a higher probability of a potential word in the first pattern being accepted in the language than there is of a potential word in the second pattern.
- English nouns make their plural in a number of different ways, as can be seen in the following set of words (Bauer 2001): cats, dogs, horses, oxen, deer, mice, hippopotami, cherubim
- What does a speaker of English do when confronted with novel words, such as: ?
- argaz ‘crate of specific style’
- smick ‘type of cracker biscuit’
- brox ‘piece of computer hardware’
- ceratopus ‘type of dinosaur’
- cheppie ‘type of antelope
- Chances are that, despite the existence of other plural formations, an English speaker will mark all of these words as plural by suffixation of /z/: argazes, smicks, broxes, ceratopuses, and cheppies.
- The only likely exception is ceratopus, ceratopi, on the basis of Latin as alumni, foci, and nuclei.
- The word argaz ‘crate’, being a Hebrew noun.
- It would therefore be possible to pluralize it as argazim, on the pattern of cherubim, seraphim, and kibbutzim.
- brox as broxen
- broxes and Vaxes
- One term potential word.
- A potential word could be a word but isn’t.
- One example is Mugglehood ‘the state of being a non-wizard’, which we created from Muggle ‘non-wizard’, a word found in J. K. Rowling’s Harry Potter books, and the English suffix -hood.
- In 2004, only three instances of Mugglehood, but also numerous instances of Muggledom and Muggleness.

- Searched for it again on January 3, 2010, showed 292 hits,
- Perhaps someday it will be included in the dictionary.
- The suffix -th creates nouns from adjectives (e.g., deep → depth, wide → width).
- a meaning that is similar to -ness. Length means the same thing that longness would mean;
- decidedth, mean the same thing as decidedness.
- But only-ness can be called productive.
- Studying productivity, we study phenomena and distinctions like these.
- One question we need to ask about productivity is whether it is part of linguistic competence.
- Competence is Chomsky's (1965) term for the knowledge that speakers and hearers have of their language.
- Some people would say that productivity is not part of linguistic competence either, because, in order for something to be considered part of competence, it must be structural and 'all or none'.
- Productivity is a probabilistic notion, and some linguists believe that if something is probabilistic, it is not structural and hence is not part of the grammar.
- Under this view, productivity would have to be treated as a phenomenon that is related to a speaker's competence, but not part of it.

Topic: 68: Productivity and Structure Negative Prefixes in English

- Productivity can tell us something about language structure: the more productive a morphological derivational process is, the more likely it is to have a compositional one whose meaning is transparently predictable from the meaning of its input.
- The converse is also true: the less productive a derivational process, the more likely it is to result in a non-compositional, semantically idiosyncratic, non-transparent output.
- To illustrate this point, consider negative prefixes in English.
- Zimmer (1964) looked at three of them, *non-*, *un-*, and *in-*, and discovered that the most productive of the set, *non-*, also has the most semantically transparent derivatives.
- This is shown by the contrast between the two:
 - non-Christian unchristian
 - non-human inhuman
 - What's the difference?

- Non-Christian means ‘not Christian’. unchristian can mean same and ‘not behaving in a Christian manner’ or even ‘uncivilized and barbaric’.
- Likewise, non-human simply means ‘not human’, while inhuman refers to the absence of human qualities like pity or kindness.
- A person can be both human and inhuman, but not human and nonhuman.
- The non-words in general simply negate their bases, the in- and un- words have the meaning ‘completely opposite to X’, where X is the meaning of their bases, in the way that east and west or long and short are opposed.
- To put it in a more technical way, non- is a logical or contrary negator.
- Using logical notation, we could represent non-Christian as in, where $\neg$ means ‘not’:
- $\neg$ Christian
- Un- and in- are contradictory negators, whose addition to a word X results in a new word meaning ‘opposite of X’.
- Zimmer’s observation extends to other derivational affixes.
- The suffix -ness is more productive in English than -ity (Aronoff 1976).
- Consider the pair collectivity~collectiveness.
- While both may mean ‘the quality or condition of being collective’, only collectivity has the additional meaning ‘the people considered as a body or whole’.
- Overall, when we compare many such pairs, the -ness derivative has more transparent semantics.
- Sometimes the -ity derivative sounds or looks odd, while the -ness derivative is pretty much always acceptable. Compare conduciveness with conductivity.
- Most English speakers would say that the former is more acceptable than the latter.
- We can even go beyond morphology to make the observation that syntax, which is always productive, is by definition compositional.

Topic: 69: Degrees of Productivity

- With two observations:
- a. Though many things are possible in morphology, some are more possible than others.
- b. Though there are infinitely many potential words in a language, some are more likely to become actual words than others.
- Morphological processes do not fall into two neat categories, productive and unproductive.

- They are best seen as being spread out along a scale, with some more productive than others.
- There are a number of types of constraints that limit productivity and that contribute to or even determine to what degree a particular formation is productive.

constraint on morphological productivity is phonological

(i) those that depend on the segmental make-up of the base;	Modern Hebrew pet-name suffix -le, borrowed from Yiddish. It can only attach to bases ending in a vowel (Glinert 1989: 437): ába ‘father’ > ábale ‘daddy’
(ii) those that depend on the suprasegmental make-up of the base, including stress placement;	-al suffixation in English. This suffix may attach only to verbs that are stressed on the final syllable, as seen in the examples in acquittal, arrival, denial, dismissal, rebuttal, referral, revival
(iii) those that depend on the number of syllables in the base.	‘-ish’ may attach only to monosyllabic adjectives, as in ‘greenish’

- Morphological constraints
- Productivity can also be limited by morphological constraints.
- a morphological process may require the base to have a certain structure, to belong to a certain morphologically or syntactically defined class, such as a gender, or to end in a particular affix.

	First type: certain English suffixes, such as -ish, attach only to unsuffixed bases.	
	Second type, affixes that attach only to bases that have a particular etymological origin. Modern Greek, where the suffix –adoros may only attach to words of Romance origin. Thus kombinadoros	
	A basic type of syntactic constraint on word formation requires the base to belong to a certain lexical category. Some affixes attach only to nouns.	

Others attach only to verbs, or to adjectives.

English re-, for example, attaches to verbs.

- verbs.
- Semantic constraints
- Productivity can also be constrained by semantic factors.
- Barker (1998) studies the English suffix -ee found in words like advisee, addressee, enlistee, or employee using a large set of naturally occurring examples such as the following:

a. There was the Asian influenza casualty ... who was replaced gallantly by an influenza recoveree, Mr Robert Harben.

b. [experiment involving shining lights into the subject's eyes] The adaptee then cannot tell the difference between yellow and white, i.e., is yellow-blind.

c. These young musicians were chosen from over 200 auditionees

d. The ground rules were simple: to find ways to relax that required absolutely no effort on the part of the relaxee.

e. The paella didn't turn out very well, but fortunately my dinees were quite understanding.

Barker finds that -ee suffixation is constrained by three semantic factors.

First, the referent of the newly derived noun must be sentient.

Second, the referent of the -ee noun is typically characterized by a relative lack of volitional control: the dinees have no control over the paella they are served, and the adaptee does not manipulate the lights.

Finally, the -ee noun and the stem must be episodically linked. Barker (1998: 712) gives the example of the noun lessee, from the verb lease: "every leasing qualifies some individual as a lessee, and for every lessee, there must be a leasing event which qualifies them as a lessee. "He continues by pointing out that it is not the case that all deverbal nouns must be episodically linked to their stems

- To summarize, if a language has two ways of doing something, one of which is less productive, the less productive rule or process has a linguistic purpose.
- The less productive rule may result in a word that stands out, useful in newspapers, magazines, and advertisements, or it may result in a form that sounds more technical or learned.

Topic: 70: Testing Productivity

- Using a tool to come to a better understanding of how speakers and hearers use and process language.
- Morphological productivity offers many opportunities for researchers with this kind of philosophy.
- Linguists, in recent years, have used many different tools to come to a better understanding of how we produce complex words such as: standard dictionaries, rhyming dictionaries, and large corpora, technologically sophisticated techniques (positron emission tomography (PET)).

- **English suffixes**

- Aronoff and Schvaneveldt (1978) conducted an experiment to verify that productivity figures in individuals' linguistic competence and to judge its consistency across speakers and words.
- The experiment focused on -ness, a native English suffix, and -ity, of Latin-Romance origin, which often attach to the same morphological and semantic classes of words.
- We see this in triplets like the following, where all three members can be found in a dictionary:
- immense immenseness immensity
- scarce scarceness scarcity
- exclusive exclusiveness exclusivity
- porous porousness porosity
- The two suffixes differ, however, in that -ness is more productive overall, especially with certain types of stems like those of the shape X-ive.
- Speakers were presented with three sets of words:

(i) actual words, like activity or assertiveness, where actual means listed in Webster's Collegiate Dictionary;

(ii) possible words, like effervescivity or affirmativeness, where X-ive occurs in the dictionary, but not the -ness or -ity suffixed form; and

- (iii) non-words.
- Here neither X-ive nor X is listed in the dictionary.

- Examples of non-words are remotiveness and lugativity.
- Over the course of the experiment, 141 subjects were asked to judge 40 words, 100 possible words, and 40 non-words, which were presented in randomized lists.
- For each subject, half of the words were suffixed in -ity and half in -ness.
- The 40 actual words were the same across all subjects, but possible words and non-words were counterbalanced so that half of the subjects got, for example, effervesciveness and elaborativity, and the other half elaborativeness and effervescivity.
- This was done to ensure that judgments would be based on the fecility of the suffix and not some peculiarity of the stem it attached to.
- The final variable in the study involved the instructions given to the subjects, who were divided into three groups of 47 each.
- One group was asked to judge whether the items were in their vocabulary;
- The second group was asked whether the items were English words;
- The third group was asked whether the words were meaningful.
- As it happened, the instructions had little effect on the subjects' judgments.
- It turned out that English speakers preferred the actual words in -ity (they were balanced for frequency).
- When it came to the non-words, they didn't care, and on the potential words, they preferred the -ness words.

This shows that speakers can tell the difference between a more productive and less productive rule.

- Any given study can advance our understanding of morphological productivity without being the last word on the subject.
- There is a great deal of value in studies that ask new questions or try out new ways of testing hypotheses.
- Design errors may or may not influence a given set of results.
- When they do, they can often be corrected in follow-up studies, which are just as essential to advancing our knowledge.

Topic: 71: The Mental Lexicon

- Studies of the sort just discussed have goals beyond linguistic description and aim rather to describe what is actually going on inside the head of a language user when they use complex words.

- Psycholinguistics studies this phenomenon, though when the research involves measuring neurological activity, as much modern research does, then it may also be termed neurolinguistics or cognitive neuroscience.
- The goal of psycholinguistic and neurolinguistic research on morphology is to understand exactly how the mental lexicon works, the lexicon that all language users store in their brains.
- Pinker (1999) is a very lively and informal presentation of the major issues in the field.
- Aitchison (2003) is a more general introduction.
- It is important to remember in to distinguish the lexicon itself from how it is used by the speaker or hearer.
- Most studies of the mental lexicon are concerned with word recognition, a classic problem in psychology: How do people associate the speech or writing signal with the entries in their mental lexicon?
- Until very recently most word recognition studies dealt mostly with written words, not speech.
- Psychologists were interested in how people recognize words, both their sound and their meaning, when they read.
- Only in the last twenty years or so have psycholinguists turned their attention to spoken word recognition, which, for linguists at least, is a much more interesting and certainly more basic problem than reading.
- After all, as linguists have known for over a century, the primary form of language is spoken (or signed) and written language is secondary.
- The basic and universal task of word recognition is thus spoken word recognition.
- From an experimental point of view, it has always been much easier to deal with written text than with speech, which is why it is only recently, with the advent of tools for regularizing the speech signal, that those psycholinguists have begun to turn to spoken language recognition.
- The other side of accessing the mental lexicon, and in some sense the opposite or reverse of word recognition, is word production.
- When a person speaks, how does that person know what words to use, the words that will eventually come out of his or her mouth?
- This is a much more difficult problem than word recognition and has received much less attention from psycholinguists (the classic presentation is Levelt 1989).
- Most psycholinguists assume that there is only one mental lexicon and that both word recognition and word production access this same lexicon, though in different ways.

- From an experimental perspective, though, the mental lexicon is not something that we can study directly.
- We study it rather only through the intermediaries of production and recognition, both of which must access this dictionary hidden deeply within our heads.

Topic: 72: Psycholinguistics, and Neurolinguistics

- To understand what sorts of issues might arise in the study of the mental lexicon, let's look at some of the aspects of Kujamaat Jóola. Consider vowel harmony first.
- The vowels of Kujamaat Jóola fall into two sets, tense and lax, and all the vowels in a word must be either tense or lax.
- When a suffix is added to a word, the vowel of the suffix may be either tense or lax, depending on the vowels of the stem.
- If a suffix contains a tense vowel, then the vowels of the stem must also become tense.
- Overall, tense vowels win: as long as a complex word contains one tense vowel, all its vowels must become tense.
- There are several ways in which those stems and suffixes that alternate between tense and lax could be represented in a person's mental lexicon.
- Since morphemes with only tense vowels never alternate, it is to assume that these morphemes have only one form in the mental lexicon.
- But other morphemes sometimes have lax vowels and sometimes tense vowels.
- We could say that each of these morphemes has two alternative forms (allomorphs) in the mental lexicon and that the form with lax vowels is selected just in case there are no tense vowels in the word.
- We could say that there is a mental operation that makes lax vowels tense if there are tense vowels in the same word.
- A psycholinguist or neurolinguist would ask how these different systems might be encoded in a person's brain and would want to devise experimental tests and measures for determining whether a particular mechanism is actually at work
- We certainly know that speakers and signers do not hesitate when speaking or signing, no matter how complex the morphology of their language.
- Semantics alone thus provides some information about which forms must be stored not tell us how they are stored.
- Most linguists: they are stored as complex structures, but no purely linguistic evidence to tell us whether that is so.

- Similar questions can be asked about all the inflectional forms of a single verb or noun.
- Are they all stored?
- And if so, are they stored together or separately and are they stored with or without structure?
- As it happens, a fairly broad consensus has emerged among psycholinguists in the last decade that “morphological considerations need to be introduced into any model of the mental lexicon
- In other words, regardless of whether any particular words are stored or not and regardless of whether related words are stored together or separately, they do have morphological structure.
- Whether related words are stored together or separately, we just don’t know.
- Even if everyone agrees that all words in the mental lexicon have morphological structure, many other issues remain undecided.

One of the most central is that of the number of different mechanisms speakers use to access their mental lexicons, either in recognition or in production.

- The most obvious answer is that there is only one mechanism, but in fact, of the theories of accessing the mental lexicon that are out there in the literature, we can identify two broad types, called the single-route and dual-route models.
- A very accessible account of both model types is provided by Pinker (1999).
- It is well known that the more frequent or familiar a word is, the faster a speaker can recognize it (Aitchison 2003).
- An English speaker will recognize clean more quickly than glean, even though the two differ phonologically in only one distinctive feature, because the first is about thirty times more frequent than the second.
- Most of the 500 most frequent words in English are simple, but even if a word is morphologically complex and perfectly regular, it may still be common.
- Unfortunately, almost all the psycholinguistic and neurolinguistics research that has been done to date is based on English or other European languages whose inflectional and derivational morphology is fairly simple.

Lesson 13

Morphology and Syntax

Topic: 73: Morphology and Syntax

- Already seen, inflection and syntax are intimately related to one another.
- Inflection is the realization of morphosyntactic features through morphological means.

- Morphosyntax deals with the relationship and interactions between morphology and syntax.
- A variety of topics in morphosyntax, includes morphologists' and syntacticians' definitions of inflection, structural constraints on morphological inflection, inflection and universal grammar, and grammatical-function-changing morphology.
- Since our target audience consists of students whose only exposure to syntax comes from an introductory course in general linguistics, we avoid bringing in advanced syntactic analyses.

According to the traditional view, the relation between morphology and syntax is the following: while morphology builds up word forms—typically by combining roots with other roots and with affixes, but also by applying other operations to them.

- Syntax takes fully inflected words as input and combines them into phrases and sentences.
- The division of labour between morphology and syntax is thus perfect: morphology only operates below the word level whereas syntax only operates above the word level.
- These two components of grammar are ordered in strict sequence, such that the syntax takes over after the morphology has done its work.

Topic: 74: Morphological vs. Syntactic Inflection

- We begin by distinguishing between two applications of the word inflection, one found chiefly in the morphological literature and the other in syntactic literature.
- For a morphologist, the presence of inflectional morphology in a language depends on the existence of multiple forms of a lexeme.
- If a lexeme has only one form, then there can be no morphological inflection.
- In syntax, there is no such requirement.
- Chinese lexemes have only one form, abstracting away from phonologically determined alternations (mostly changes in tone).
- While Chinese has a few clitics or particles, including one that expresses past tense, these are generally not considered affixes.
- The same is true of Vietnamese, though the two languages are unrelated.
- For the morphologist, therefore, these two languages have no inflection.
- From a syntactician's point of view, whether or not Chinese and Vietnamese have inflection is an entirely different matter.
- Even if a language does not express a particular notion such as number or case, it is typically assumed to be present in the syntax.
- Likewise, a syntactician may argue that a verb always agrees with its subject in an abstract sense.

- This abstract agreement is considered just as real in Chinese, where the verb form never depends on its subject,
- In Russian, where the form of the verb changes depending on the person (first, second, third) and number (singular or plural) of the subject.
- In sum, morphological inflection is realized overtly, where syntactic inflection may or may not be.
- The existence of zero allomorphs does not contradict this definition of morphological inflection.
- Zero allomorphs is contrasted with non-zero allomorphs.
- In other words, in some instances the inflectional feature encoded by the zero allomorph is realized overtly.
- Another difference between the morphological and syntactic usage of the term inflection is that morphologists speak of inflection only when dealing with bound forms.
- The reason for this is clear inflection informally is as “bending” of a lexeme.
- English has a syntactic category of modals, or modal auxiliaries.
- used to accompany other verbs to indicate that the action or state described by the sentence is something other than simple fact.

If we substitute other modals for might, the degree of uncertainty changes somewhat with each substitution:

Alicia might go to the birthday party isn't a simple fact about Alicia's going to a party; there is an element of uncertainty.

Alicia may go to the birthday party;

Alicia can go to the birthday party;

Alicia could go to the birthday party;

Alicia must go to the birthday party;

Alicia should go to the birthday party;

Alicia would go to the birthday party;

Alicia will go to the birthday party.

- Should we consider these modals to be morphological inflection?
- No.
- In order to be classified as morphological inflection, a syntactic category must be expressed through bound forms.
- In the case of the English modals, we are not dealing with bound forms, but rather with separate words.
- Again, syntacticians differ from morphologists on this point.
- Most would treat auxiliaries as part of the inflectional system of a language.
- Morphologists are not denying the validity of this treatment, only distinguishing the full word modals (i.e., syntactic inflection) from morphological inflection.

Differences between morphological and syntactic inflection

Morphological inflection is realized overtly.

Syntactic inflection may or may not be overt.

• In order to be classified as morphological inflection, morphosyntactic properties must be expressed by bound forms or other morphophonological means (e.g., ablaut, suppletion).

Syntactic inflection may be realized by free forms (e.g., auxiliaries).

Topic: 75: Structural Constraints on Morphological Inflection

- Cross-linguistically, constraints on the realization of inflectional morphology.

In Russian, for example, verbs show gender agreement with their subject (feminine, masculine, or neuter) only in the past tense.

- This fact about Russian has a historical explanation: the past tense form was originally an adjective, and Russian adjectives agree in gender with the nouns they modify.
- In Modern Hebrew, verbs agree in gender with their subjects only in the present tense.
- Historically, Hebrew present forms were originally participles.
- Gender agreement is not optional in Hebrew and Russian.
- Instead, its morphological realization is context-dependent.

- It is surprisingly easy to find languages where verb inflection is obligatory in some contexts but impossible in others.
- a few such
- All depend on syntactic context, rather than on tense, which is expressed as an inflectional part of the verb itself in Russian and Hebrew.
- In the Kujamaat Jóola the subject agreement is expressed obligatorily except in the past subordinate and positive imperative forms of the verb.
- In some related languages, like Balanta verbs agree with their subject only in certain syntactic contexts.
- In Balanta, verbs may be marked for subject agreement, but generally only in the absence of a subject noun phrase.
- When a subject noun phrase is present, a subject prefix on the verb does not express agreement. Instead, it indicates that the subject is focused:

(1) a. **b e- ɲaɲ be-nte**

cl2-people come

‘The people came’

b. **b e-ɲa –aɲ be- do-oló b e-beeθa ma**

cl2-people cl2-few cl2-see 3sg.obj

‘A few people saw him.’

In Arabic, the basic generalization is that subject–verb number agreement appears on the verb when the word order is SV (subject–verb) (a) but not when it is VS (4b) (data from Ouhalla 1994: 43):

a. l-tullaab-u wasal-uu

the-students-nom arrived-3pl

‘The students have arrived’

b. wasal-a l-tullaab-u
arrived-3sg the-students-nom

‘The students have arrived’

In (b) third plural subject agreement is blocked, and instead we get default third person singular agreement.

- Arabic (Semitic): Verbs agree with their subjects for number when the word order is SV but not when it is VS. When the word order is VS, the verb bears default 3sg agreement.
- • Balanta (Atlantic): Verbs agree with their subjects only when there is no subject NP or the subject NP is focused (emphasized).

Topic: 76: Inflection and Universal Grammar

- Universal Grammar is the theory developed by Noam Chomsky that states that all languages are identical at some level of analysis.
- Tremendous influence on the field of linguistics, and most linguists agree with Chomsky that language has an innate component.
- A key phrase in the definition of Universal Grammar that we have provided is “at some level of analysis.”
- What is the level of analysis at which languages are identical?
- At which levels do languages differ?
- More specifically, are inflectional categories universal?
- In one sense, inflectional categories are universal.
- A mistake to say that the realizations of inflectional categories are stable cross-linguistically.

Examples of inflectional categories

Nominal Verbal

- case (e.g., nominative, accusative, genitive, dative, ablative, ergative, absolutive)
- tense (e.g., past, present, future)

- **aspect (e.g., perfective, imperfective)**

- **mood (e.g., indicative, subjunctive, optative)**

- **voice (e.g., active, passive)**

- **subject and object agreement**

- **number (e.g., singular, plural, dual, trial, paucal)**

- **gender (e.g., masculine, feminine, neuter; animate, inanimate)**

- Let's consider gender. Gender is highly problematic from a universalist point of view.
- The number of noun classes in the languages of the Atlantic family vary widely.
- Kujamaat Jóola 19,
- Gombe Fula 25, Serer 16,
- Wolof 10
- Manjaku 14,
- Balanta seven.
- Some Atlantic languages have multiple dialects, and dialects do not necessarily share the same number of noun classes.
- To other language families of the world, we find variation not only in the number of genders but also in the entire organization of the gender system.
- Despite differences between gender systems some similarities do emerge.
- German has three genders called masculine, feminine, and neuter: Masculine Feminine Neuter
- Mann 'man' Frau 'woman' Parlament 'parliament'
- Masculine, feminine, and neuter are obligatory inflectional categories of German.
- This means that every noun in the language, including borrowings like Parlament 'parliament', must belong to a gender.
- A noun cannot be genderless.
- Furthermore, gender is obligatory in that a noun cannot simply carry it around: its gender category must be expressed through agreement.

- In many languages gender is an obligatory inflectional category.
- Every noun must have a gender, and that gender must be expressed in the morphology.
- The gender categories themselves and the number of them are completely different.
- Many languages, have certain types of inflectional categories appear over and over again.
- For example, nouns are regularly inflected for case, number, and gender.
- For verbs, the picture is similarly limited.
- Verbs might inflect for tense, aspect, mood, voice, or agreement, but you generally do not find languages where verbs inflect for other categories.
- Although the inflectional categories themselves may not be universal, universal principles govern what is inflectionally possible and what is not.

Topic: 77: Grammatical Function Change

- The governor broke the law.
- The law was broken by the governor.
- The law was broken
- There are times in life when the passive is convenient.
- In English we can also say:
- Solomon made the governor break the law
- Sentences like the one are called causative because they usually express the meaning ‘cause to do something’, or sometimes ‘allow, persuade, help to do something’.
- In English we can also say:
- The governor broke the law for Smith
- This sentence introduced another participant, Smith, the person for whom the governor broke the law.
- We could discuss the morphology of break in the English sentences but its forms are fairly limited: broke, broken, break.
- None of these forms is limited to expressing a passive, causative, or ‘for X’ interpretation.
- (The -en of broken is sometimes considered a passive morpheme, but it is not limited to passive sentences.
- We could also :

- The governor has broken the law(which is active).
- If we look at other languages, we often find that the passive, causative, and other types of grammatical-function-changing phenomena are associated with particular morphology.
- Grammatical function change refers to “alternations in the grammatical encoding of referential expressions,” to use the definition presented by Baker (1988: 1).
- We see that the agent can be encoded as a subject or object, depending on the form of the verb used: broke or was broken.
- Passive, causative, and other phenomena are grammatical-function changing phenomena because they can be seen as triggering the encoding change.
- There are various types of grammatical-function-changing phenomena that are found cross linguistically.

Topic: 78: Productivity in Grammatical Function Change

- Passive, causative, and other phenomena are grammatical-function changing phenomena shows productivity of syntax.
- We do not analyze them, beyond presenting basic definitions, because to do so would require us to go too deeply into syntax.
- Grammatical-function-changing phenomena involve morphology–syntax interactions at their most intimate.
- Passive
- Construction in which a grammatical subject of the verb is subjected to or affected by the action of the verb.
- The verb’s agent may be expressed as an adjunct (in English, a by phrase). Example: His memoirs were written by a ghost writer.
- Antipassive Construction in which the object of the verb is expressed in an oblique case or becomes null. The following is an example of the antipassive voice:
- "Mary-erg eats pie-abs." → "Mary-abs eats."
- "He-erg is speaking the truth-abs." → "He-abs is speaking."
- Causative Construction in which the subject causes an event.
- The causative may be an auxiliary verb or an inflectional morpheme attached to a verb, but the term is also applied to verbs that inherently express a relationship whereby the subject causes a given event (to kill is to cause to die).Sam makes us smile.

- Applicative Construction in which the addition of a morpheme allows a verb to take an additional object.
- This object is often understood as benefiting from or being adversely affected by the action of the verb.

English does not have a dedicated applicative prefix or suffix. However, prepositions can be compounded with verbs for an applicative effect. For example, from

- **Jack ran faster than the giant,**

the intransitive verb ran can be made transitive, and the oblique noun giant the object:

- **Jack outran the giant.**

The applicative verb can be made passive, something which is not possible with ran:

- **The giant was outrun by Jack**

- Noun incorporation Morphological construction in which a noun stem is compounded with a verb stem and yields a derived verb stem. Example: Siberian Koryak qoya- ‘reindeer’ combines with the verb stem -nm- ‘to kill’, yielding the derived form qoyanm- ‘to reindeer-slaughter.
- Although incorporation does not occur regularly, English uses it sometimes: breastfeed, and direct object incorporation, as in babysit.

Lesson 14

Morphology and Psycholinguistics

Topic: 79: Morphology and mind

- how morphological knowledge (knowledge of complex words and of morphological rules) is represented in the human mind and how it is used in language processing.
- This is an important topic because the mental representation of morphological knowledge is a battle ground for different theories about the nature of linguistic rules.
- This is an important topic because the mental representation of morphological knowledge is a battle ground for different theories about the nature of linguistic rules.

- A related topic is the balance between storage and computation. Knowledge in a particular domain of human cognition always comprises both storage of information and the ability to compute new information.
- For instance, when we want to use the word books, we have two ways to do this: either we retrieve this plural form of book from our lexical memory, or we create it on line, by adding the plural suffix -s to the stem book. What determines the choice between these two routes?
- Empirical research of such questions may thus help us to get a better understanding of the nature of human cognitive capacities.
- Morphology may therefore be qualified as a window on the human mind. The empirical domains of psycholinguistic research will be reviewed: the mental lexicon ('the dictionary in your head'), the acquisition of morphological knowledge, and the role of this knowledge in language perception and production.
- In the previous chapter, on morphological change, we saw that language change also provides information about the mental representation of morphological knowledge.
- There are a number of ways in which we can find out how morphological knowledge is represented in the mind and used in language processing: by studying naturalistic data (corpora of language use, speech errors, effects of language impairment, etc.) and experimental data (lexical decision tasks, production tasks, etc.).
- Morphology is a battlefield for competing models of linguistic knowledge, and for discussion on the nature of linguistic rules.
- The frequency of words in actual language use correlates with their activation level in the mental lexicon.
- Statistical and probabilistic data are therefore relevant for adequate models of morphological knowledge.

Topic: 80: The Use of mental lexicon

- Lexical knowledge of a native speaker does not have the format of a dictionary.
- First, the number of lexical entries in a good dictionary is much higher than that in our individual mental list of words.
- There are many words that most speakers do not know.
- The passive vocabulary, the number of words that you understand.
- The active vocabulary, the set of words one uses in language production is much smaller.
- A dictionary is conservative by nature, and hence it will contain words from the past that nobody uses any more.

- Mental lexicon will always be ahead of the dictionary, and contains a substantial number of words that are not listed in dictionaries.
- New words (neologisms) are coined continuously, and dictionaries always lag behind.
- A second difference between a dictionary and the mental lexicon is that words in the mental lexicon bear a number of relationships to each other.
- Words with similar meanings or similar phonological forms appear to be related in the mental lexicon, as can be concluded from speech errors.
- Someone talking about the sympathy of a musical conductor used the word symphony instead of sympathy.
- In a dictionary, semantic relations between words are usually not expressed directly.
- The mental lexicon as a multidimensional web of words, with all kinds of connections between those words: semantic similarities, phonological similarities, and morphological relationships.
- In a dictionary, words have one type of relationship only, that of alphabetical ordering.
- This ordering is based on the degree of orthographical similarity between words.
- A third difference between a dictionary and a mental lexicon is that the latter also stores information about the frequency with which you come across a word.
- Linguists may compute the frequency of words on the basis of large corpora of actual language use..
- Frequency counts indicate how many tokens are found for each word type in a particular corpus.
- Function words such as determiners have a relatively high frequency,
- Within the set of English adjectives, the word nice is used far more frequently than the word opaque.
- A high frequency word is more easily recognized.
- Frequency of exposure to a word correlates with the activation level of that word in the mental lexicon.

A word with a higher activation level is activated faster in language processing.

- A related effect is the cumulative frequency effect.
- The frequent singular form of a noun will also have a frequency effect on its plural form
- The token frequencies of related inflectional word forms contribute to the activation level of each of them.

- This is called the base frequency effect.
- Another example is the cumulative root frequency effect: the summed frequencies of all words with the same root also appear to affect response latencies of individual complex words containing that root.

Topic: 81: Acquisition of morphology

- How does a child acquire the morphological system, the set of morphological rules of his or her mother tongue?
- Morphological rules have to be discovered on the basis of words that are formed according to these rules.
- The first stage: the acquisition of individual complex words.
- The child uses morphologically complex words correctly without making use of morphological knowledge, by retrieving them from memory.
- Next, he may discover certain recurring properties e.g. plural nouns in English, the ending -s (with the allomorphs [s], [z], and [ɪz]) to the stem.
- Rules with exceptions go a further refinement.
- Three stages of morphological acquisition (Berko, 1958).
- Firstly, children learn a number of past tense forms of English verbs by rote producing correct past tense forms of both regular and irregular verbs (asked, went).
- Secondly, children acquiring English have discovered the rule for past tense forms, but will also apply the rule to the class of irregular verbs.
- They will produce the correct form asked, but the incorrect form goed instead of the correct went, a case of overgeneralization.
- third stage, both the rule and the set of exceptions have been acquired, correct *asked* and *went*.
- So the learning process has the shape of a U-curve: the number of correct past tense verbs in stage I decreases in stage II, and increases again in stage III.
- This learning curve has been found for many languages.
- In Romance languages, with a number of different conjugations for verbs, it is the default conjugation that is overgeneralized.
- In languages with case marking, children will use case-marked word forms from very early on because it is words in their surface forms (not stems) that form the basis of acquisition.
- For nouns, the nominative form is usually the default case form, used in all syntactic contexts.

- Children can discover the role of case marking in syntactic contexts at a very early age.
- Russian children already use accusative nouns after transitive verbs before the age of 2 (Clark and Berman 2004).
- In the domain of word-formation children discover the building principles for complex words quite soon.
- Able to coin new words themselves at a very early age.

Children coin words for concepts for which there are already established words.

- For adults, established words have priority above new coinings, in accordance with the principle of conventionality that gives priority to established words, and thus blocks the creation of new words that are synonymous to existing ones.
- Since children discover morphological processes on the basis of the words they are exposed to, they must be able to compare and analyse words, and assign them morphological structure if possible.
- Children appear to analyse words consciously, and ask questions like: Does cornflakes have corn in it? (Clark 1993: 40).

Topic: 82: Sources of evidence

- How morphological information is represented in the mind?
- Sources of evidence: experimental and naturalistic data.
- Psychologists use experiments, and in the realm of morphology lexical decision tasks are the most widely used way of probing into the mental representation of morphology.
- Lexical decision tasks are often used in combination with priming, the prior presentation of another word.
- This means that the subject who has to make a lexical decision first receives some other information.
- An example of priming is the following: exposure to one stimulus influences a response to a subsequent stimulus, a case of identity priming.
- A higher level of activation by previous access to the same word.
- Other ways of priming are phonological priming (the prime word is phonologically similar to the target word), and semantic priming (with a semantically similar word as the prime).

If these primes reduce response latencies for the target word, one can conclude that words in the mental lexicon are connected to words that are either phonologically or semantically similar.

- In morphological priming, a word that is morphologically related to the target word is used as a prime. For instance, calculate will function as a prime for calculation.
- Phonological and semantic priming, not an independent phenomenon.
- Naturalistic data evidence for theories of morphology in the mind.
- Data concerning the actual behaviour of language users.
- Children coining new words or word forms may be interpreted as evidence for the children's having acquired morphological rules.
- Speech errors
- *Take the freezes out of the steaker* (take the steaks out of the freezer).
- The roots freeze and steak exchanged erroneously.
- Exchange of roots bears on the issue of the lexical representation of morphological structure.
- The word freezer a complex word is stored as such in the mental lexicon, given its specific conventional meaning shows, its internal morphological structure is still accessible.
- Complex words can be stored in lexical memory with their morphological structure still present.
- The language behaviour of people with a language deficit, such as aphasia, may throw light on the mental representation of morphology.
- Aphasics that suffer from agrammatism, usually referred to as Brocaaphasics, may not be able to produce correct plural forms of nouns, except for nouns with a high frequency plural form such as the word eyes(Aitchison 2004).
- Three German aphasic patients with agrammatism it was found that their syntactic abilities were severely impaired, whereas their ability to use inflectional morphology was still intact.
- This fact can be interpreted as evidence for the position that syntactic rules belong to a different module of the grammar than the rules of inflectional morphology.

Topic: 83: Models of morphological knowledge

- The observed ability of native speakers to coin new words or word forms is the basic argument for assuming that morphological knowledge encompasses more than storage of the complex words that language users are exposed to.
- There must be mechanisms in the mind that enable us to extend the set of complex words in a language.
- Languages with a rich system of inflectional morphology such as Turkish, it would even be quite absurd to assume that all these forms are memorized as such.

- The memory load for such languages can be reduced considerably by making use of rules.
- A possible model is that language users acquire morphological rules by constructing abstract rules or templates on the basis of their lexical knowledge.
- There are also morphological patterns that are not productive, but can still be called patterns e.g. English past tense verb forms.
- Regular past tense forms are created by suffixing -ed to the stem.
- On the other hand there are a number of stem-changing verbs where the past tense form and the past participle are marked by having a different vowel than the present tense form e.g. an -ing/k -ang/k-ung/k pattern for verbs: ring, sing, spring, drink, shrink, sink, stink
- The class of verbs with this kind of vowel alternation will normally not be extended.
- It is clear that native speakers of English are able to recognize the abstract pattern involved.
- Some speakers tend to also inflect the verb to bring as brang and brung.
- If people are asked to make the past tense forms of non-existing verbs with a similar phonological make-up as the words spling, they tend to come up with either splang or splung.
- Some verbs once regular have changed into irregular ones: dive (with the past tense form dove not dived in American English).
- These facts have led a number of linguists to defend a dual system model of morphological knowledge.
- The past tense and participial forms are not stored in lexical memory, but always created by rule.
- The irregular forms are stored in memory.
- These stored irregular forms are linked to each other in an associative way, and thus the language user will be able to discover similarity patterns such as -ing/k -ang/k -ung/k.
- This explains why the pattern may be extended incidentally to similar verbs such as to bring.
- The dual system theory has also been defended based on a type of evidence : neurological evidence. Jaeger et al. (1996) have argued that regular and irregular verbs in English are processed by different neural systems. The processing location determined by PET-scans.
- A second type of rule-less approach is the model proposed by Bybee(1988, 1995, 2001).
- In her model, individual inflectional forms of words are listed in the lexicon.
- This also applies to past tense verb forms, both regular and irregular ones.
- The regularities in the formal relationship between present tense and past tense forms are captured by abstract schemas that can be constructed on the basis of sets of words.

- A third single system approach without rules is the analogy-based model.
- To be sure, analogy has been recognized as a source of new complex forms in all kinds of morphological models.
- In the case, a new words formed on the basis of a related existing word, without an abstract morphological template being involved.
- The coinage of the English word seascape is based on the word landscape land : sea = landscape :
- This survey of different models of morphological knowledge, and the debate about these models illustrates that morphological phenomena play an important role in the debate on the nature of linguistic knowledge.
- Morphology is an important window on the mental representation of natural language and the human mind.

Topic: 84: Morphological processing

- The insight that complex words are often stored as such in the lexicon raises the question of how they are processed.
- There are two ways in which a complex word can be processed, the direct route, and the indirect route.
- When perception of complex words is involved, the direct route means that we do not first parse the complex word, but go directly to its representation in the lexicon, in order to access its meaning.
- The indirect route means that a complex word is parsed into constituent morphemes, and that its meaning is computed after we have gained access to its constituent morphemes and their meanings.
- There are data that might be interpreted as showing that language users try to parse words into morphological sub-constituents.
- For instance, when speakers of Dutch are confronted with non-words in a lexical decision task, it takes more time to make the correct decision when that word begins with a syllable that could have been a prefix, compared to words where this does not apply.
- Dutch words aderibag, afgeblar, afbegepakt
- 1 no potential prefix
- 2, begins with two prefixes (af-, ge-),
- 3. begins with three prefixes (af-, be-, ge-).

- Correlating with this difference, the last nonword has longest response latency, the language users try to strip prefixes from words that are not in their lexicon.
- The existence of two routes has led to the assumption of morphological race models.
- In such models, both routes will be used for word recognition, and compete with each other.
- If the word is not stored in lexical memory, the indirect route is the only one that will lead to recognition, on the basis of parsing.
- If the word is stored, the direct route will be faster than the indirect route if the stored complex word has a high frequency, that is, a high level of activation.
- For a complex word with a low frequency, on the other hand, parsing is the obvious route.

Both routes are followed, and one of them will turn out to be the fastest, depending on the level of activation of the complex word.

- The parsing of complex words is also affected by relative frequency: the frequency of the complex word compared to that of its base word.
- If the complex word has a lower frequency than the base word, this will make parsing of that complex word easier and more efficient than direct retrieval.
- If the inverse situation obtains, direct access to the complex word will be the most efficient route.
- dazzlement will be parsed than its base dazzle,
- government will be accessed as a whole than that of its verbal base govern
- Note that the meaning of government is not completely compositional, unlike that of dazzlement.
- The more complex words with a certain affix will be parsed, the more productive the corresponding word-formation process
- Parsing the affix will increase its activation level
- The morphological race model can also be used as a model of word production.
- It helps us to understand why it is irregular forms of high frequency that are maintained in a language, whereas low-frequency irregular forms disappear more easily.
- Went is a high-frequency form, this route will be faster than the indirect route where the past tense form has to be created by rule, resulting in the incorrect goed.

Morphology and Language Change

Topic: 85: The nature of language change: Internal

Geoffrey Chaucer's famous Canterbury Tales

(1) When that Aprill with his shoures soote

“When that April with his showers sweet”

The droghte of March hath perced to the roote

“The drought of March has pierced to the root”

And bathed every veyne in swich licour

“And bathed every vine in such liquid”

Of which vertu engendred is the flour;

“Of which virtue engendered is the flower”

- Middle English as used in these lines is different from Modern English.
- If we assume that it is still the same language, English, we have to conclude that languages can change.

But why do they?

- The statement that languages change is in fact metaphorical in nature.
- It presupposes that we conceive of a language as an organism that grows, changes, and sometimes dies.
- Languages have a mode of existence outside their users.
- A language primarily exists in the minds of its speakers.
- That is why we say that a language has died when its last speaker has died.
- So it is speakers that change their language while using it in language perception and production.
- Why do language users change their language?

- In fact, 'change' is not always the appropriate word for what is going on.
- It would be better to speak of construction or innovation of language.
- What does change when new words or new meanings of words get established is the lexical norm of that language, not the system behind it.
- The main reason for changing the lexical norm of a language is that language users need expressions for new concepts, or new things.
- One way of meeting this need is extending the meaning of existing words.
- mouse
- hard disk
- anti-virus-program
- Lexical gaps can also be filled through borrowings.
- Dutch, for example, has borrowed many words from English recently e.g. printer
- The change of the lexical norm consists of two steps.
- The first step is that an individual language user coins a new lexical unit (a word or a phrase), borrows one from another language,
- Or gives an existing word a new interpretation.
- Supra-individual lexical norm: change accepted by many users
- Lexical innovation: users need to express new concepts.
- The study of language change has a psychological and a social dimension.
- A second source of linguistic innovation besides changing the lexical norm is reanalysis.
- Language users cannot grasp the system behind a language in a direct fashion.
- The only evidence they have are outputs of the system, concrete cases of language use.
- This opens up the possibility that a language user reconstructs the system underlying the perceived outputs in a slightly different way than previous users.
- Children do the same by using *goed* and *bringed*.
- In sum, language users may be involved in language change at different stages of their lives.

Topic: 86: The nature of language change: External

- Language change may also be effected by external causes, in particular language contact.

- This is called external change.
- Language contact may lead to borrowing of words from other languages.
- Most diplomacy.
- English was influenced much earlier and more strongly by French due to the Norman conquest of 1066.
- Borrowing had its impact on the morphology of Germanic languages. In Dutch, a number of affixes of French origin are used productively in combination with non-native stems, and sometimes even with native stems (as in *flauw-iteit* “silly joke” derived from the native adjective *flauw* “silly”).
- Borrowing of inflectional forms may also take place.
- English has Greek and Latin plurals, such as *criteria*, *schemata*, and *data*.
- These learned plurals came to be replaced with regular plurals such as *schemas* and *museums*.
- Lack or rarity of morphology is considered by some linguists to be typical for pidgin and creole languages.
- From recent research: both pidgins and creoles do have morphology, both inflection and word-formation (derivation, compounding, reduplication).
- Language contact as a source of loss of inflectional morphology can be observed in the development of Afrikaans: mother language Dutch.
- Afrikaans has an extremely simple kind of inflectional morphology, with a present tense form that is identical to the stem, without person and number marking.
- Morphological change is the erosion of inflectional systems in the Germanic and Romance languages, referred to as deflection.
- English has the poorest inflectional system of all.
- Most Germanic languages have lost their case system and the three-gender system.
- Language change is obviously not restricted to the morphological system of languages.
- Quotation above from Chaucer also shows phonological, syntactic, and semantic changes.
- Phonological change is exemplified by the loss of the final vowel of *roote* that has become *root* in Modern English.
- The word order noun–adjective in *shoures soote* corresponds with the Modern English word order adjective–noun: *sweet showers*.

- An example of semantic change is that the present-day meaning of liquor is no longer “liquid”, but denotes alcoholic liquids only.

Topic: 87: Historical sources of morphology

- Morphological systems not only erode, they may also arise.
- An example is the emergence of nominal compounding in Germanic languages that arose from noun phrases of the type:
- [N-GEN N]NP
- Harry’s book
- The pre-nominal noun with genitive case functions as a modifier of the head noun.
- Such noun phrases were reinterpreted as compound nouns.
- Univerbation, the reinterpretation of phrases as words, does not necessarily lead to language change.
- In linguistics, univerbation is the diachronic process of combining a fixed expression of several words into a new single word.
- Some English examples include *always* from all [the] way (the *s* was added later),
- *onto* from *on to*,
- *albeit* from *all be it*,
- and colloquial *gonna* from going to.
- These changes imply the addition of lexemes to the lexicon, but do not affect the morphological system
- Although this kind of univerbation did not lead to new morphological rules, : when a word is multi-morphemic, this does not necessarily imply that it has been created by a morphological process.
- The frequency plays a role in grammaticalization: defined as ‘the process by which a frequently used sequence of words or morphemes become automated as a single processing unit’ (Bybee 2003: 603).
- [G]rammaticalization is that subset of linguistic changes through which lexical item in certain uses becomes a grammatical item, or through which a grammatical item becomes more grammatical’ (Hopper and Closs Traugott 1993: 2).
- An example of the change from lexical to grammatical item is the development of verbs into auxiliaries.

- In English the verb *to have* not only functions as a main verb, with the meaning “to possess”, but also as an auxiliary in perfect tense forms.
- The verb *can* has lost its status as lexical verb completely, and functions as a modal auxiliary only.
- In these examples grammaticalization does not create morphology.
- This does happen if a (lexical or grammatical) morpheme becomes an affix.
- The phenomenon of degrammaticalization has raised the question whether grammaticalization is subject to the condition of unidirectionality.

Can a grammatical morpheme become less grammatical, or even lexical as well.

This depends on one’s definition of grammaticalization

- If unidirectionality is one of its defining properties, changes into the opposite direction simply require another descriptive term, for instance degrammaticalization.
- The reinterpretation of the English genitive case morpheme -s as a clitic is a restricted form of degrammaticalization.
- It is a restricted case because the suffix does not end up as a full lexical word.
- In Early Modern English it has also been interpreted as the weak form of the possessive pronoun *his*, that is, as a full word.
- Shakespeare writes: the count his galleys “the count’s galleys”.
- This change is related to the fact that English lost its case system and this made a reanalysis of the genitive suffix -s necessary.
- Degrammaticalization is restricted in nature, Grammaticalization is a pervasive phenomenon in natural languages.
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Topic: 88: Changes in morphological rules

- Erosion of inflectional morphology has taken place in most Germanic and Romance languages.
- This means that some inflectional rules have disappeared from the grammar of those languages.
- Dutch, for instance, has lost its morphological cases.

- The case system has only survived in a number of fixed expressions, prepositional in origin.
- The prepositions in these phrases govern a particular case.

Some of these frozen expressions are considered as words, others are still spelt as phrases. The case endings are in italics.

- *te-gelijker-tijd* “at the same time, simultaneously
- “*met dien verstande* “with that understanding
- “*in dier voege* “in such a manner”

An interesting case of change in inflectional morphology is that of adjectival inflection in Afrikaans.

In Dutch, adjectives in attributive position are inflected according to the following rule: ‘add the suffix -e [ə] to the adjectival stem unless the NP in which it occurs carries the features [indefinite], [neuter], and [singular]’

een groot-*ø* paard “a big horse” een grot-*e* koe “a big cow”

het grot-*e* paard “the big horse” de grot-*e* koe “the big cow”

(de) grot-*e* paarden “(the) big horses” (de) grot-*e* koeien “(the) big cows”

- Quite a complex system for many non-native speakers.
- No wonder this system broke down in Afrikaans, a creolized variant of Dutch.
- In the new system, the adjectives are divided into two classes: either they always carry the inflectional schwa in attributive position, or they never do.
- The adjectives that do take the schwa are morphologically complex adjectives, and simplex adjectives that exhibit stem allomorphy.
- For instance, the adjective *sag* “soft” has the stem allomorph *sagt*, as shown in the inflected form *sagte*.
- Hence, it will have the form *sagte* in attributive position, whereas *sag* is used in predicative position.
- What we can learn from these facts is that relics of a previous stage of the language, “historical junk” in the words of Lass (1990), might be kept, and even re-used in a different way.
- That is, these words have lost their morphological structure.
- Another type of possible change in derivational rules is a change in the category of the base words to which they apply.

- Very productive rules tend to extend their domain of application to new categories.
- The productive diminutive suffix -tje of Dutch, for instance, is no longer restricted to the domain of nouns, but also attaches to adjectives, verbs, and adverbs:
- strijk “to stroke” strijk-je “small string orchestra” blond “blond” blond-je “blond girl” uit “out” uit-je “outing”
- The changes of the French suffix -age also illustrate category change (Fleischmann 1977).
- The suffix -age entered many European languages through borrowing of French words in -age. English has extended its use to native, Germanic nominal, and verbal stems,
- base N: foot-age, front-age, mile-age, shipp-age, wreck-age, yard-age
- base V: break-age, brew-age, cover-age, drain-age, leak-age, sew-age.
- The word shortage shows that -age has even been extended to adjectival bases. In Dutch we find both denominal and deverbal nouns in -age; as in English, it has also been attached to native stems, an indication of its productivity.

Topic: 89: Changes in word structure

- Complex words, once they are coined, may be subject to reanalysis and reduction.
- In reanalysis, words receive a different structural interpretation.
- A classical example is the reinterpretation of the word *hamburger*.
- This word, derived from the base noun Hamburg, and denoting a specific kind of food originating from that city, received the following structural reanalysis:
- [[hamburg]er] > [[ham][burger]]
- Reanalysis can only be observed when the reanalysed structure serves as a model for new words.
- Many new words in -burger have been coined, such as *beefburger*, *cheeseburger*, *fishburger*,.
- *turkeyfurter* on the model of *frankfurter*
- The piece burger itself has been reinterpreted as a noun, witness the brand name Burger King.
- So in fact we cannot conclude that a new suffix -burger has developed in English.
- Burger is a new noun denoting a particular kind of fast food, and can be used as the head of nominal compounds.
- Semantic reinterpretation may also lead to a new set of words.
- This has been the case for -gate, as used in Watergate (the name of an apartment building in Washington, DC, that was burglarized by order of President Nixon).

- The morpheme -gate received a new interpretation, “political scandal”, thus leading to many new words such as Monicagate and Irangate.
- This suffix also acquired the more general meaning of “scandal”.
- Reanalysis may also have the effect of a sequence of affixes becoming an affix.
- A morphological structure [[[x]A]B] can be reinterpreted as [[x]AB].
- An example from Dutch is the suffix -erij, originally a combination of the suffixes -er and -ij: base noun in -er noun in -ijbak “to bake” bakk-er “baker” bakk-er-ij “bakery” > bakk-erij.
- This phenomenon of two affixes becoming one is referred to as affix telescoping
- Prefixes may also lose their morphological status.
- This is quite clear in English borrowings from Latin such as abortion and adoption.
- prefixes ab- and ad- respectively
- The syllabifications these words are a.bor.tion and a.dop.tion respectively.
- The Dutch prefix ge-, as in geloof “to believe” is no longer productive as a verbal prefix.
- In Afrikaans, the verb geloof has become a simplex verb, glo, with deletion of the schwa

Topic: 90: Some notions about grammar

- The word grammar is derived from Greek, which means "art of letters", from γράμμα (gramma), "letter", itself from γράφειν (graphein), "to draw, to write".
- The same Greek root also appears in graphics, grapheme, and photograph.
- In linguistics, grammar is *the set of structural rules governing the composition of clauses, phrases, and words in any given natural language*.
- The term refers also to the study of such rules, and this field includes phonology, morphology, and syntax, often complemented by phonetics, semantics, and pragmatics.
- Speakers of a language have *a set of internalized rules for using that language* and these rules constitute that language's grammar.
- Native language – acquired not by conscious study or instruction, but by observing other speakers.
- Much of this work is done during early childhood; learning a language later in life usually involves a greater degree of explicit instruction.
- Grammar is the cognitive information underlying language use.

- The term "grammar" can also be used to describe the rules that govern the linguistic behaviour of a group of speakers.
- The term "English grammar" may have several meanings.
- The whole of English grammar, that is, to the grammars of all the speakers of the language.
- Alternatively, it may refer only to what is common to the grammars of all, or of the vast majority of English speakers (such as subject–verb–object word order in simple declarative sentences).
- Or it may refer to the rules of a particular, relatively well-defined variety of English (such as standard English for a particular region).
- A specific description, study or analysis of such rules may also be referred to as a grammar.
- A reference book describing the grammar of a language is called a "reference grammar" or simply "a grammar".
- A fully explicit grammar that exhaustively describes the grammatical constructions of a language is called a descriptive grammar.
- This kind of linguistic description contrasts with linguistic prescription, an attempt to discourage or suppress some grammatical constructions, while promoting others.
- John Dryden, objected to it leading other English speakers to avoid the construction and discourage its use.
- Outside linguistics, the term grammar is often used in a rather different sense.
- it may be used more broadly, including rules of spelling and punctuation, which linguists would not typically consider to form part of grammar, but rather as a part of orthography.
- In other respects, it may be used more narrowly, to refer to prescriptive grammar only and excluding those aspects of a language's grammar that are not subject to variation or debate.
- Jeremy Butterfield claimed that, for non-linguists, "Grammar is often a generic way of referring to any aspect of English that people object to."

Traditional Grammar: Categories & Functions

Topic: 91: Some authorities of grammatical tradition

The Babylonians: some early attempts at language description, The first systematic grammars, of Sanskrit, originated in Iron Age India,

- Yaska (6th century BC),
- Pāṇini (6-5th century BC)
- Pingala (c. 200 BC), Katyayana, Patanjali (2nd century BC).

Tolkāppiyam, the earliest Tamil grammar, is mostly dated to before the 5th century AD. In the West, grammar emerged as a discipline in Hellenism from the 3rd century BC forward with authors like Rhyanus and Aristarchus of Samothrace. The oldest known grammar handbook is the Art of Grammar, by the ancient Greek scholar Dionysius Thrax (c. 170–c. 90 BC), a student of Aristarchus of Samothrace who established a school on the Greek island of Rhodes. Dionysius Thrax's grammar book remained the primary grammar textbook for Greek schoolboys until as late as the twelfth century AD. The Romans based their grammatical writings on it and its basic format remains the basis for grammar guides in many languages even today.

Latin grammar developed by following Greek models from the 1st century BC, due to the work of authors such as Orbilius Pupillus, Remmius Palaemon, Marcus Valerius Probus, Verrius Flaccus, and Aemilius Asper. A grammar of Irish originated in the 7th century with the Auraicept na n-Éces. Arabic grammar emerged with Abu al-Aswad al-Du'ali in the 7th century. The first treatises on Hebrew grammar appeared in the High Middle Ages, in the context of Mishnah (exegesis of the Hebrew Bible).

The Karaite tradition originated in Abbasid Baghdad. The Diquduq (10th century) is one of the earliest grammatical commentaries on the Hebrew Bible. Ibn Barun in the 12th century compares the Hebrew language with Arabic in the Islamic grammatical tradition. Belonging to the trivium of the seven liberal arts, grammar was taught as a core discipline throughout the Middle Ages, following the influence of authors from Late Antiquity, such as Priscian.

Treatment of vernaculars began gradually during the High Middle Ages, with isolated works such as the First Grammatical Treatise, but became influential only in the Renaissance and Baroque periods. In 1486, Antonio de Nebrija published Las introducciones Latinas contrapuesto el romance al Latin, and the first Spanish grammar, Gramática de la lengua castellana, in 1492. During the 16th-century Italian Renaissance, the *Questione della lingua* was the discussion on the status and ideal form of the Italian language, initiated by Dante's *de vulgari*. The first grammar of Slovene language was written in 1583 by Adam Bohorič.

Grammars of non-European languages began to be compiled for the purposes of evangelization and Bible translation from the 16th century onward, such as *Grammatica o Arte de la Lengua General de los Indios de los Reynos del Perú* (1560). From the latter part of the 18th century, grammar came to be

understood as a subfield of the emerging discipline of modern linguistics. The Deutsche Grammatik of the Jacob Grimm was first published in the 1810s. The Comparative Grammar of Franz Bopp, the starting point of modern comparative linguistics, came out in 1833.

Topic: 92: Concept of Parts of speech

In traditional grammar, a part of speech is a category of words (or, more generally, of lexical items) which have similar grammatical properties. Words that are assigned to the same part of speech generally display similar behavior in terms of syntax—they play similar roles within the grammatical structure of sentences—and sometimes in terms of morphology, in that they undergo inflection for similar properties. Almost all languages have the word classes noun and verb, but beyond these two there are significant variations among different languages.

For example, Japanese has as many as three classes of adjectives, where English has one (not to be confused with the seven types of English adjectives, or the fact that English adjectives can modify both nouns and pronouns). Chinese, Korean, Japanese and Vietnamese have a class of nominal classifiers; and Many languages do not distinguish between adjectives and adverbs, or between adjectives and verbs. Because of such variation in the number of categories and their identifying properties, analysis of parts of speech must be done for each individual language. Nevertheless, the labels for each category are assigned on the basis of universal criteria.

In the Nirukta, written in the 5th or 6th century BC, the Sanskrit grammarian Yāska defined the following main categories of words:

नाम nāma – noun (including adjective)

आख्यात ākhyāta – verb

निपात nipāta – particle, invariant word (perhaps preposition)

The ancient work on the grammar of the Tamil language, Tolkāppiyam, argued to have been written around 2,500 years ago, classifies Tamil words.

- **peyar (பெயர்; noun),**
- **vinai (வினை; verb),**
- **idai (part of speech which modifies the relationships between verbs and nouns),**

- **uri** (word that further qualifies a noun or verb).

Western tradition

A century or two after the work of Nirukta, the Greek scholar Plato wrote in his Cratylus dialog that "... sentences are, I conceive, a combination of verbs [rhêma] and nouns [ónoma]".

Aristotle added another class, "conjunction" [sýndesmos], which included not only the words known today as conjunctions, but also other parts (the interpretations differ; in one interpretation it is pronouns, prepositions, and the article).

By the end of the 2nd century BC grammarians had expanded this classification scheme into eight categories, seen in the Art of Grammar, attributed to Dionysius Thrax:

- **Noun (ónoma):** a part of speech inflected for case, signifying a concrete or abstract entity
- **Verb (rhêma):** a part of speech without case inflection, but inflected for tense, person and number, signifying an activity or process performed or undergone
- **Participle (metokhê):** a part of speech sharing features of the verb and the noun
- **Article (árthron):** a declinable part of speech, taken to include the definite article, but also the basic relative pronoun

By the end of the 2nd century BC grammarians had expanded this classification scheme into eight categories, seen in the Art of Grammar, attributed to Dionysius Thrax:

- **Pronoun (antōnymíā):** a part of speech substitutable for a noun and marked for a person
- **Preposition (próthesis):** a part of speech placed before other words in composition and in syntax
- **Adverb (epírrhēma):** a part of speech without inflection, in modification of or in addition to a verb, adjective, clause, sentence, or other adverb
- **Conjunction (sýndesmos):** a part of speech binding together the discourse and filling gaps in its interpretation

The Latin grammarian Priscian (fl. 500 AD) modified the above eightfold system, excluding "article" (since the Latin language, unlike Greek, does not have articles), but adding "interjection". In the older English terminology: noun substantive, noun adjective and noun numeral. Later the adjective became a separate class, as often did the numerals, and the English word noun came to be applied to substantives only. Works of English grammar generally follow the pattern of the European tradition as described above, except that participles are now usually regarded as forms of verbs rather than as a separate part of speech, and numerals are often conflated with other parts of speech:

Topic: 93: Parts of Speech in English

- Works of English grammar generally follow the pattern of the European tradition, except that participles are now usually regarded as forms of verbs rather than as a separate part of speech.
- noun
- verb
- adjective
- adverb
- pronoun
- preposition
- conjunction
- interjection
- article or (more recently) determiner
- Linguists recognize that the above list of eight or nine word classes is drastically simplified.
- Modern linguists have proposed many different schemes whereby the words of English or other languages are placed into more specific categories and subcategories based on a more precise understanding of their grammatical functions.
- In English grammar, open class refers to the category of content words—that is, parts of speech (or word classes) that readily accept new members. Contrast with closed class.
- The open classes in English are nouns, lexical verbs, adjectives, and adverbs.
- The open or closed status of word classes varies between languages, even assuming that corresponding word classes exist.
- In Japanese, verbs and adjectives are closed classes, though these are quite large, with about 700 adjectives, and verbs have opened slightly in recent years. Japanese adjectives are closely related to verbs.

- By contrast, Japanese pronouns are open class—if they can even be considered a class—and nouns become used as pronouns with some frequency.

Topic: 94: Prescriptive grammar

- The term _____ or should not be used rather than describing the ways in which a language is actually used.
- Contrast with descriptive grammar.

Also called normative grammar and prescriptivism

- In a prescriptive grammar there is right and wrong language. Rules of “good” or “proper” usage, which dictate what is “good grammar” and what is “bad grammar” Example:
- (1)She doesn’t know him.
- (2)She don’t know him.
- Besides there are other differences of the following types.
- Case of pronoun: never use an object case after the verb ‘to be’.
- It is me.
- Instead say; It is I.
- Mary runs faster than me.
- Say: Mary runs faster than I.
- Similarly don’t say: who are you talking to?
- Say: whom are to talking to?
- Never end a sentence with a preposition.
- Whom are you talking to?
- Say: to whom are talking.
- Don't use ain't
- Don't use seen as the past tense of see (as in I seen him at the party last night).
- Don't use contractions
- Don't use sentence fragments
- Don't use dangling participles

- Coming to college, there were many tall trees.
- Many of these rules are essentially taken from Latin.
- Latin was the classical language known by all educated people and one regarded as model for all languages.
- However, there is no reason why English should follow the Latin rules.
- For example English has no case endings for noun (except in genitive case). So why to follow case?

Topic: 95: Problems of Prescriptive Grammar

- Although everyone knows or thinks they know what a word is and what a sentence is, both terms defy exact definition.
- The sentence as a linguistic concept has been defined in over 200 different ways, none of them completely adequate.
- Here are the most important attempts at defining the sentence:
- The traditional, or common sense definition states that a sentence is a group of words that expresses a thought.
- The phrase *an egg* expresses a thought but is it a sentence?
- A sentence like *I closed the door because it was cold* expresses two thoughts and yet it is one sentence.
- Another definition is that a sentence is a group of words expressing a topic (old information) and some comment (new information) about that topic: John left.
- The problem with the topic-comment definition is that many sentences have no clear topic and comment structure: It's raining.
- Many sentences have no clear topic and comment structure: It--is raining. (The word *it* here is the so-called dummy *it* used to fill the subject slot for impersonal verbs in English.

Some sentence types make no internal syntactic structure; there is no distinction between subject and predicate:

a) Emotive sentences such as Gee! Wow. Darn! Yes! No!

b) Imperatives: Go! Leave! Taxi! All aboard!

c) Elliptic sentences: Who took the car? John.

d) small talk phrases: Hello. Good-bye. Good morning.

- In polysynthetic languages the single word serve as a complete sentence much more frequently. In such languages, morphology rather than syntax usually expresses the distinction between subject and predicate.
- Structural ambiguity
- Flying plan can be dangerous.
- The king is eager to please.
- The king is easy to please.
- Words and phrases can be grouped according to their sentence building functions.
- Not all languages have the same parts of speech. Many languages have postpositions rather than prepositions
- A common assertion is that all languages have at least nouns and verbs.
- It is true that all languages have some means of conveying information as a concept or as an event, but what a noun or verb is differs from language to language.
- In the Salishan languages of the Puget Sound, a single word can be translated into English as village and a village exist or there is a village;
- in other words, morphemes denoting stationary concepts are often bound roots that require verbal affixes to stand as words.
- So parts of speech--even nouns and verbs-- turn out to be at best fuzzy categories across languages, not identical or even present in every language.

Topic: 96: Concept of Sentence in Prescriptive Grammar

- A sentence is a group of words that are put together to mean something.
- A sentence is the basic unit of language which expresses *a complete thought*.
- It does this by following the grammatical rules of syntax.
- For example: ' Sana is the most intelligent girl in the class'.
- A complete sentence has at least a subject and a main verb to state (declare) a complete thought.

Short example:

- Walkers walk.
- In English and many other languages, the first word of a written sentence has a capital letter.
- At the end of the sentence there is a full stop or full point (American: 'period').

A Sentence typically contains a subject and predicate, conveying a statement, question, exclamation, or command, and consists of a main clause and sometimes one or more subordinate clauses.

- A phrase or clause is part of a sentence.
- This is an example of a sentence:
- The dog is happy.
- In this sentence,
- 'The dog' is the subject, and 'is' is the verb.
- This is an example of a phrase:
- The happy dog

There is no verb, so we do not know anything about what the happy dog is doing. Therefore, it is not a sentence.

- A simple sentence has only one clause.
- The cat is sleeping.
- Notional definition: a sentence expresses a complete thought.
- What does a complete thought mean?
- A cabbage
- A man
- If it rains, I shan't come
- Pair of subject predicate
- It's raining.
- What is *it*?
- *The sentence is a linguistic unit that has the grammatical structure we assign to it.*

What is Syntax?

Topic: 97: Hierarchy of Sentence

- Words, phrases, clauses, and sentences constitute what is called the GRAMMATICAL HIERARCHY of a sentence.

Hierarchy of Sentence			
Sentences	consist of one or more than one	sentence	Stop!
Clauses	consist of one or more than one phrase		stop
Phrases	consist of one are more than one word		stop
Words			stop
	are separated by space		

- Sentences are at the top of the hierarchy, so they are the largest unit.
- Words are at the lowest level, though again, some grammars go below the word to consider morphology.
- At the clause level and at the phrase level, two points should be noted:
 - Although clauses are higher than phrases in the hierarchy, clauses can occur within phrases, as:
 - The man *who lives beside us* is ill
 - Here we have a relative clause *who lives beside us* within the NP *the man who lives beside us*.
 - Clauses can occur within clauses, and phrases can occur within phrases.

Topic: 98: Descriptive Grammar

The term descriptive grammar refers to an objective, nonjudgmental description of the grammatical constructions in a language. It's an examination of how a language is actually being used, in writing and in speech. Specialists in descriptive grammar examine the principles and patterns that underlie the use of words, phrases, clauses, and sentences. Kirk Hazen notes, "Descriptive grammars do not give advice: They detail the ways in which native speakers use their language. A descriptive grammar is a survey

of a language.

For any living language, a descriptive grammar from one century will differ from a descriptive grammar of the next century because the language will have changed." ("An Introduction to Language." John Wiley, 2015). "Descriptive grammar," Edwin L. Battistella notes in "Bad Language," "is the basis for dictionaries, which record changes in vocabulary and usage, and for the field of linguistics, which aims at describing languages and investigating the nature of language."

The term descriptive is a little bit misleading, as descriptive grammar does provide analysis and explanation of the language's grammar and not just description of it. Contrast the type with prescriptive grammar, which notes how something should or should not be used, what is right and wrong. Prescriptive grammarians attempt to enforce rules concerning "correct" or "incorrect" usage. According to Donald G. Ellis, "All languages adhere to syntactical rules of one sort or another, but the rigidity of these rules is greater in some languages. It is very important to distinguish between the syntactical rules that govern a language and the rules that a culture imposes on its language."

- People spoke long before there were linguists around to uncover the rules of speaking....
- Prescriptive grammars, on the other hand, are the stuff of high school English teachers.
- The sentence "I ain't going,"
- *Who did you arrest?
- Whom did you arrest?
- *What are you waiting for?
- For what are you waiting?
- *I saw several octopuses at the beach.
- I saw several octopi at the beach.

Descriptive grammar, consists of rules that all speakers of a speech community adhere to, barring speech errors. Linguists are usually more interested in such rules, as they are reflective of the properties of actual language, not ideal language.

Topic: 99: Sentence structure

- A simple sentence contains a subject and a verb, and it may also have an object and modifiers.
- It contains only one independent clause.
- Examples:
- She reads.
- She completed her literature review.
- He organized his sources by theme.
- They studied APA rules for many hours.

Compound Sentences

- A compound sentence contains at least two independent clauses.
- These two independent clauses can be combined with *a comma and a coordinating conjunction* or *with a semicolon*.
- Coordinators (Coordinating Conjunctions)

The seven coordinators by the phrase FAN BOYS:

- For And Nor But Or Yet So

For	Our team did not play well, for they did not practiced hard (the second clause is the reason for the first clause).
And	Our teachers are very diligent, and our management too is excellent. (The two clauses express equal, similar ideas).
Nor	Good students study well, nor do they waste their time. (nor means “and not” it joins two negative independent clauses. Notice that question word order is used after nor).
But	Other team players may exercise, but they may not exercise as Ali does. (The two clauses express equal, contrasting ideas).
Or	All students should take healthful diets, or they will risk getting some disease. (the two clauses express alternative possibilities).
Yet	Sportsmen give a little time to their studies, yet in the long run, they are better off than their peers. (The second clause is a surprising or unexpected contrast to the first clause).
So	He organized his sources by theme, so he did not face any problem. (the second clause is the reason of the first)

- Complex Sentences

- A complex sentence contains at least one independent clause and at least one dependent clause.
- Dependent clauses can refer to the subject (who, which) the sequence/time (since, while), or the causal elements (because, if) of the independent clause.
- Here are a few examples:
- Although she completed her literature review, she still needed to work on her method section.
- They studied APA rules for many hours as they were so interesting.
- Compound-Complex Sentences
- Sentence types can also be combined.
- A compound-complex sentence contains at least two independent clauses and at least one dependent clause.

Complex-compound sentences

- **She completed her literature review, but she still needs to work on her method section even though she finished her course last semester.**
- **Although he organized his sources by theme, he decided to arrange them chronologically, and he carefully followed the MEAL plan for organization.**
- **With pizza and coffee at hand, they studied APA rules for many hours, and they decided that writing in APA made sense because it was clear, concise, and objective.**

Using some complex-compound sentences in writing allows for more sentence variety. Pay close attention to comma usage in complex-compound sentences so that the reader is easily able to follow the intended meaning.

Topic: 100: Noun Phrases

Noun Phrases

Often a noun phrase is just a noun or a pronoun:

- **People like to have money.**
- **I am tired.**
- **It is getting late.**

or a determiner and a noun ...:

Our friends have bought a house in the village.

Those houses are very expensive.

... perhaps with an adjective:

Our closest friends have just bought a new house in the village.

Noun Phrases

Sometimes the noun phrase begins with a quantifier:

- **All those children go to school here.**
- **Both of my younger brothers are married**
- **Some people spend a lot of money.**

Numbers:

Quantifiers come before determiners, but numbers come after determiners:

- **My four children go to school here. (All my children go to school here.)**
- **Those two suitcases are mine. (Both those suitcases are mine)**

Noun Phrases

So the noun phrase is built up in this way:

- **Noun:** people; money
- **Determiner + noun:** the village, a house, our friends; those houses
- **Quantifier + noun:** some people; a lot of money
- **Determiner + adjective + noun:** our closest friends; a new house.
- **Quantifier + determiner + noun:** all those children;
- **Quantifier + determiner + adjective + noun:** both of my younger brothers

Noun Phrases

The noun phrase can be quite complicated:

- a loaf of nice fresh brown bread
- the eight-year-old boy who attempted to drive a car
- that attractive young woman in the blue dress sitting over there in the corner

Some words and phrases come after the noun. These are called postmodifiers. A noun phrase can be postmodified in several ways.

- with a prepositional phrase:

a man with a gun

the boy in the blue shirt

the house on the corner

- with an –ing phrase:

the man standing over there

the boy talking to Angela

- with a relative clause:

the man we met yesterday

the house that Jack built

the woman who discovered radium

an eight-year-old boy who attempted to drive a car

That clause: very common with reporting or summarising nouns like idea, fact, belief, suggestion:

He's still very fit, in spite of the fact that he's over eighty. She

got the idea that people didn't like her.

There was a suggestion that the children should be sent home.

- with a to-infinitive.

This is very common after indefinite pronouns and adverbs:

You should take something to read.

I need somewhere to sleep.

I've got no decent shoes to wear.

Noun Phrases

There may be more than one postmodifier:

**an eight-year old boy with a gun who tried to rob a sweet shop that
girl over there in a green dress drinking a coke**

Noun Phrases

There may be more than one postmodifier:

**an eight-year old boy with a gun who tried to rob a sweet shop that
girl over there in a green dress drinking a coke**

Topic: 101: Structure of verb phrase

Verbs have distinctive word

preceded within the verb phrase by auxiliary and modal verbs, and the negation word not. The verb phrase is usually preceded by a noun phrase called the subject. A sentence represents an action or an event or a state of affairs. In many cases, the verb is the word which names the kind of action or event or state of affairs. English verbs have two distinctive morphological characteristics.

i) the shape of the verb can encode the tense of the sentence;

(1) He eats meat.

(2) He ate meat.

present and past, is coded by the morphology of the verb.

ii) The second special morphological characteristic is that an English verb can agree with its subject if the verb is present and the subject is third person singular.

- She plays piano well.

The first grammatical form that can appear in a verb phrase in English is the auxiliary verb. In English, the twelve auxiliary verbs are have, be, do, and the nine modal verbs (can, could, may, might, must, shall, should, will, and would). Auxiliary verbs perform the functions of progressive, perfect, passive, operator, and modal within verb phrases.

Progressive Auxiliary Verb | Verb

The dog **was barking**.

Perfect Auxiliary Verb | Verb

The boys **have completed** their work.

Passive Auxiliary Verb | Verb

The letter **has been written**.

Operator Auxiliary Verb | Verb

please **do call** me

Modal Auxiliary Verb | Verb

Progressive, perfect, passive, and modal auxiliary verb may appear with other progressive, perfect, passive, and modal auxiliary verbs. The order in which auxiliary verbs can appear together is Modal-Perfect-Passive-Progressive. For example:

- **Perfect | Progressive | Verb**

had | been | sleeping

- **Perfect | Passive | Verb**

has | been | broken

- **Modal | Progressive | Verb**

could | be | swimming

- **Modal | Perfect | Passive | Verb**

might | have | been | swindled

- **Modal | Perfect | Passive | Progressive | Verb**

should | have | been | being | watched

S		V	Complement
subject		verb	

		In direct object	irect	Subjecti ve complement	Objecti ve complement	Adv erbial
ohn	J s earched		t he room			
he girl	T s i			a student		at a university
is brother	H rew g			happier		gra dually
t	I ained r					stea dily all day
e	H ad given h	th e girl	a n apple			
hey	T ake m	hi m			the chairman	ever y year

Topic: 102: Structure of Adjectives Phrase

What is an adjective? An adjective describes or modifies noun/s and pronoun/s in a sentence. Words like small, blue, sharp etc. It normally indicates quality, size, shape, duration, feelings, contents, and more about a noun or pronoun. We cannot say whether a word is an adjective by looking at it in isolation. The form does not necessarily indicate its syntactic function some suffix are indeed found only with adjective, eg:-ous, but many common adjective have no shape e.g nice, good, sharp etc.

The major functions of Adjective:

(1) Attributive

- This premodifies a noun phrase
- *A beautiful painting*

- His *main* argument

(2) Predicative

Predicative adjective can be

(a) subject complement

- Your son is *intelligent*.

(b) objective complement

- He made his family *proud*.

(3) Postposition

Follow the item they modify (usually regarded as reduced relative clause) especially with indefinite pronouns ending in -body -one, -thing, -where e.g. I want to try on something *larger* (ie 'which is larger')

Obligatory postposition adjective have different sense when used attributively

• The president <i>elect</i>	'Soon to take office'
• The city of London <i>proper</i>	'as strictly defined'

In several compounds –mostly legal or quasi-legal – the adjective is postposition.

Attorney *general*, body *politic*, court *martial*, heir *apparent*, notary *public*

Some a-adjectives and concerned, involved

- The house *ablaze* is next door to mine.
- The people *involved* were not found.

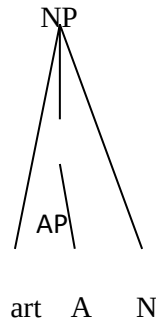
(4) Head of a noun phrase

Does not inflect for number or for genitive case and must take a definite determiner

- *The extremely old* need a great deal of attention.
- *The young in spirit* enjoy life.

- *The rich* will help only *the humble poor*.

Adjective phrase in the noun phrase



The white house

- We can add *quite* to this phrase
- The quite *white* house
- *the white quite house
- *the quite house
- Structure
- The head of an adjective phrase is an adjective which may be
- Simple: big
- Comparative: bigger
- Superlative: biggest
- The premodifiers of adjective or always adverbs: *extremely* hot
- Post modifiers can be either adverbs:
- *Very* tall *indeed*
- *Nice* enough
- Or preposition phrase
- M M M
- A PP
- Rather too hot (for Comfort)

Universal Grammar

Topic: 103: Structure of Adverb Phrase

An adverb is a word that is used to change, modify or qualify several types of words including an adjective, a verb, a clause, another adverb, or any other type of word or phrase, with the exception of determiners that directly modify nouns. A good way to understand adverbs is to think about them as the words that provide context. Specifically, adverbs provide a description of how, where, when, in what manner and to what extent something is done or happens. Normally, we can spot an adverb by the fact that it often ends in –ly, but there are lots of adverbs that don't end in this way e.g. fast, well, etc.

Moreover, adverbs can be used in many combinations with each other very nicely, extremely fast, etc.

- When? She always arrives *early*.
- How? He drives *carefully*.
- Where? They go *everywhere* together.
- In what way? She eats *slowly*.
- To what extent? It is *terribly* hot.

The phrase structure rule for a noun phrase says that an adjective phrase can be contained in a noun phrase. In contrast, an adverb phrase cannot, as is seen below.

(5) The [AP quite quick] swimmer

(6) *The [AdvP quite quickly] swimmer

Adverb phrases most characteristically roam free in the sentence and without very major effects on the meaning of the sentence.

Quite quickly I saw the swimmer.

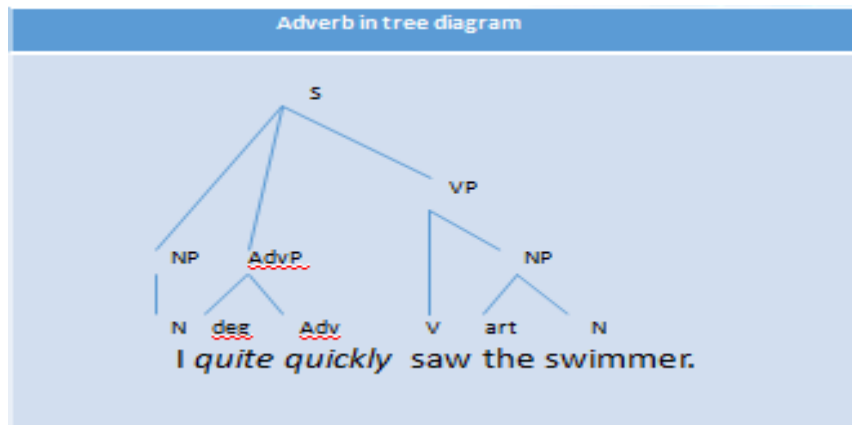
I quite quickly saw the swimmer.

I saw quite quickly the swimmer.

*I saw the quite quickly swimmer.

The adverb phrase cannot roam into the noun phrase.

I saw the swimmer quite quickly.



- Two syntactic of function adverb
- (1) Adverbial
- (2) Modifier of adjective phrase and adverb

Adverb as an adverbial

Adjuncts:	Integrated with the structure of the clause to some extent	They are waiting <i>outside</i> He spoke about it <i>briefly</i>
Disjuncts	Not integrated within the clause. Semantically express an evaluation of what is said.	<i>Frankly</i> , I am tired <i>Fortunately</i> , no one complained They are <i>probably</i> at home
Conjuncts	Have connective function. Indicate the connection what is said and what was said before.	If you open all the window, then I am leaving I have not looked into his qualification. He seems very intelligent, <i>though</i>

Adverb as modifier

Modifier of an adjective	That was a <i>very</i> funny film
--------------------------	-----------------------------------

Modifier of adverb	There I have seen so <i>very</i> many letters like that one
Modifier of prepositional phrase	The nail went <i>right</i> through the wall His parents are <i>dead</i> against the trip
Modifier of determiner, predeterminer, Postdeterminer	<i>Nearly</i> everybody came to our party (intensifying adv. premodifying pronoun) <i>Over</i> two hundred survivors were reported I paid <i>more than</i> ten pounds for it

- Modifier of noun phrase
- Intensifiers: *quite*, *rather* (esp BrE)
- Predeterminer: *such*
- Exclamatory: *what*
- It was *rather* a mess
- I never heard *such* wickedness
- He was *quite* some player
- *What* a fool he is!

Topic: 104: Minimal noun phrase

A single written sentence may stand for two different sentence structures, each having a different meaning. The structures differ because the words are grouped into different phrases.

Phrases can be discovered by replacement and movement of sequences of words.

- MEANING AND PHRASE STRUCTURE
- I was reading the letter to John

How does one sequence of words produce two different and alternative meanings? Two alternative organizations of the words, each of which delivers a different meaning. The ambiguity is based on two different structures for the same sequence of words. This is called a STRUCTURAL AMBIGUITY. To understand how this string of words can produce two different meanings we must say that they are invisibly organized into groups.

A 'box' of words is called a PHRASE

• I was reading the	letter to John	
• I was reading the	letter to	John

REPLACEMENT AND PHRASE STRUCTURE

A sentence has a syntactic structure which is hidden from view but which can be revealed by various tests.

TEST: REPLACEMENT BY IT

If a sequence of words can be replaced by the word *it* without significantly changing the meaning, then that sequence of words is a phrase.

A 'box' of words is called a PHRASE

I was reading the	the letter to John	
I was reading	it	
I was reading the	letter to	John
I was reading	it	

Movement test

I was reading	the letter to John		
The letter to John	was being read		
The letter	was being read	to John.	

So we are replacing like for like, with the word *it* being a minimal or stripped-down noun phrase standing for a more filled-out noun phrase. It is called a PRONOUN, though it should more properly be called a pro-

Topic: 105: Tree structure for a sentence

A sentence represents an EVENTUALITY.

An eventuality is an action or an event or a state of affairs: something that happens or something that is. The sentence represents an eventuality by separating out the type of eventuality from the abstract or concrete things which are involved in the eventuality. The type of the eventuality is prototypically represented by the verb, and the abstract or concrete things involved in the eventuality are prototypically represented by noun phrases.

(1) The man in the hat casually gave a slimy fish to Toby.

Here the eventuality type is an action of *giving*, represented by the verb *gave*. The things involved in the eventuality are represented by the three noun phrases, *the man in the hat* (the giver), *a slimy fish* (the thing given), and *Toby* (the thing given to). Apart from the type of eventuality and the things which participate in it, a part of the sentence can also express the circumstances of the eventuality. The adverb phrase *casually* expresses the circumstances here. The component parts of the sentence thus contribute distinct components of meaning to the overall meaning of the sentence.

Step 3: A verb phrase is usually inside a sentence, so draw a line up from VP to S.

S
|
VP
|
V

The man in the hat casually gave a slimy fish to Toby

Step 4: Now identify the noun phrases by the substitution test. A noun phrase might be replaceable by *him* or *her* or *it*.

The man in the hat casually gave a slimy fish to Toby

He casually gave it to him

Step 5: Underline the NPs and write NP above each of them.

S
|
NP VP NP NP
|
The man in the hat casually gave a slimy fish to Toby

The man in the hat casually gave a slimy fish to Toby

Step 6: Identify any articles in the sentence and write art above them.

S
|
NP VP NP NP
|
art art V art
The man in the hat casually gave a slimy fish to Toby

The man in the hat casually gave a slimy fish to Toby

Step 1: Look for the main verb and write V above it.

V

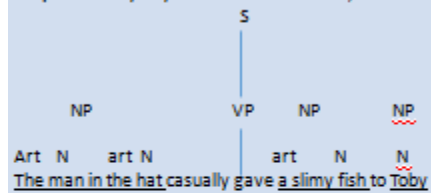
The man in the hat casually gave a slimy fish to Toby

Step 2: A verb is always inside a verb phrase, so draw a line up from V to VP.

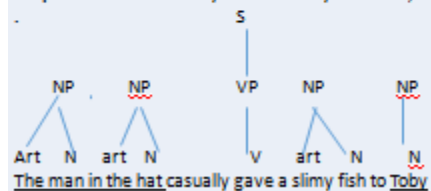
VP
|
V

The man in the hat casually gave a slimy fish to Toby

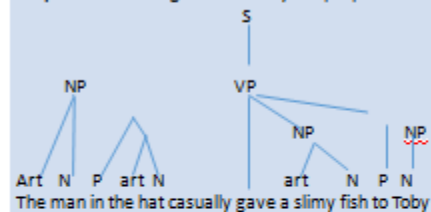
Step 7: Identify any nouns in the sentence, and write N above them. .



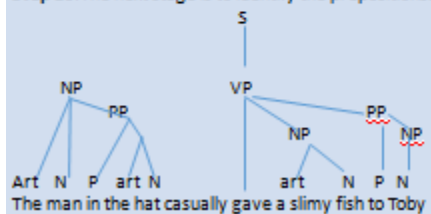
Step 8: An article is always immediately under NP, and N is always immediately under



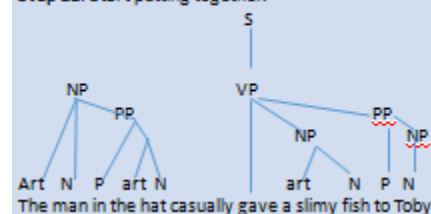
Step 9: The next stage is to identify the prepositions. Write P above the prepositions.



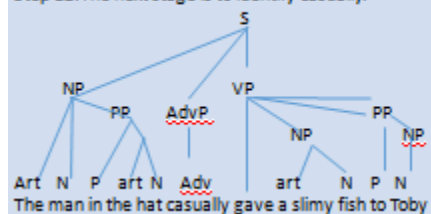
Step 10: The next stage is to identify the prepositions. So join PP.

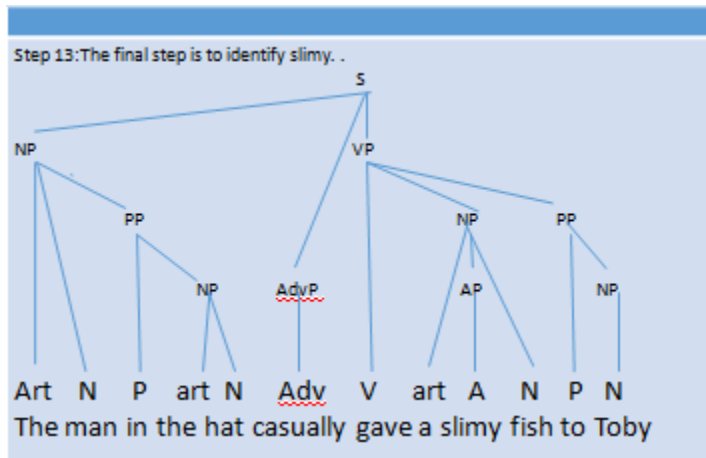


Step 11: Start putting together.

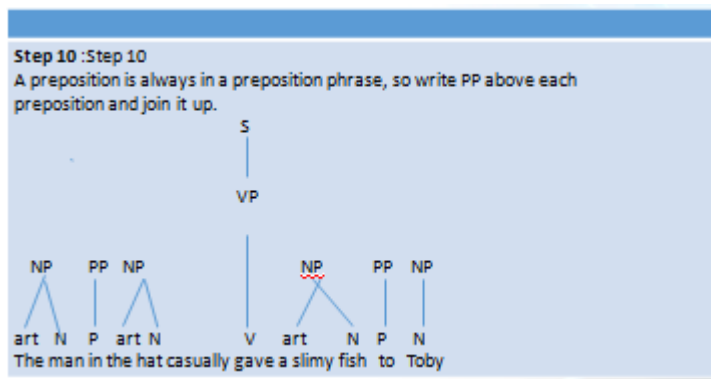


Step 12: The next stage is to identify casually.





- Tree diagram clearly identifies each component of the sentence.



Topic: 106: I C ANALYSIS

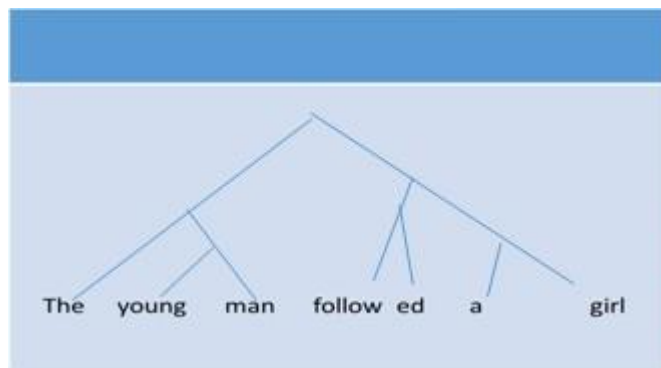
In the last few decades syntax has undoubtedly become the most intensively studied topic within linguistics. Many new ideas and new linguistic models have been put forward. Even the best descriptive accounts have been written in conservative terms. These models have partly been influenced by modern theoretical proposals. The structuralists (American of 1950s) discovered and defined morphemes, but needed to establish what other grammatical units were and how they were distributed. Their approach was to divide the sentence up into IMMEDIATE CONSTITUENTS or ICs. Their aim was to discover relevant linguistic elements. Not clear whether there are any definable units beyond morphemes, apart from sentence. The principle of IC analysis is to cut a sentence into two and then to cut those parts into two and to continue with the segmentation until the smallest indivisible unit, the morphemes are reached.

- As a general principle the division is binary:
- The young man followed a girl.
- The young man + followed a girl.
- Followed + a girl
- a + girl

- The young man = the + young man
- Young + man
- Follow + ed

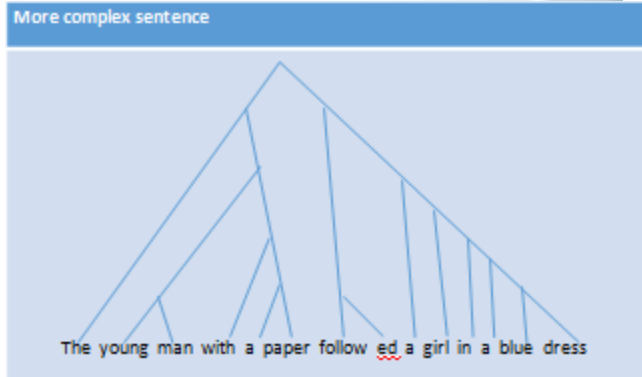
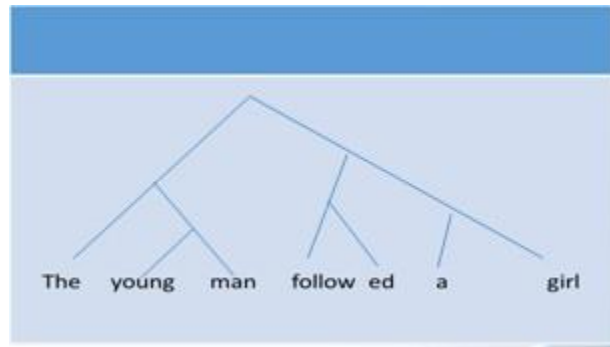
The || young ||| man | follow ||| ed || a ||| girl.

((The (((young) (man)))) ((followed) ((a) (girl)))).



Topic: 107: Method of display

- The best method of display is to use a tree diagram where branching show the division



- Where to make the first cut?
- The answer lies in the notion of *expansion*.
- A sequence of elements said to be an expansion of another, if it can be substituted for it.
- Substitution is the basic procedure in all structuralist analysis.
- The first ICs are the following:
- *The young man with a paper* and *followed the girl with blue dress*.
- Not
- *The young man with a paper* and *followed* and *the girl with blue dress*.
- Reasons:
- Firstly, *The young man with a paper* is an expansion of *John* as
- *John followed the girl with blue dress*
- Similarly
- *followed the girl with blue dress* can be an expansion of *arrived*
- *The young man with a paper* *arrived*.

- On the basis of our arguments we can conclude: the ICs of our longer sentence are *The young man with a paper and followed the girl with blue dress*.
- Expansion being technical indicates the substitution of one sequence of element with another. Look at the following:
- Children
- American children
- Three American children
- Three American children with a dog
- Those three American children with a dog
- Expansions can be of different types

Endocentric

Ai

element, a word, for which it can be substituted. This word is known as Head; like children in above example. Exocentric construction An exocentric construction cannot be substituted for any of its element and thus has no head. For example *with a paper*

- *Is not an expansion of with or a or paper*
- Not an expansion of any of its parts

An exocentric construction cannot be substituted for any of its element and thus has no head. Because of endocentricity that we can talk of noun phrase, verb phrase for these are endocentric constructions with nouns verbs etc., as their heads.

Topic: 108: Flaws in the ICs

- Distribution is a criterion(of structuralism)

Same construction occurs in different places and help us with identification.

- *To go*
- *To go* is fun
- wants *to go*
- It is *wants* and *to go*
- Not *wants to* and *go*

Similarly

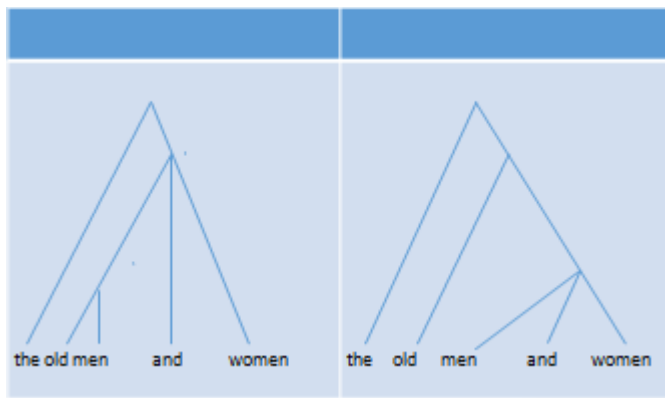
- The king of England is divided
- *The and king of England*
- Not *the king* and *of England*
- In view of
- *He became king of England*

Simply dividing a sentence into ICs does not provide much information. Nevertheless it can sometimes prove illuminating. It shows a certain type of ambiguity or difference of meaning which is related to the hierarchical structure of the IC analysis.

- Ambiguity is clear by paraphrasing the following
- The old men and the women
- Is it 'the old men and the women

Or

- the old men and the old women
- we would have two different analyses by the tree:



A complication

- With preference like the following:
- *Men and and women*
- Or *men* and *and women*
- Similarly
- The Egyptian cotton shirt

- First cut after Egyptian (a cotton shirt made in Egypt)
- Or after cotton (a shirt made of Egyptian ‘cotton’)

Dividing sentences out into their constituents part is unsatisfactory for several reasons,

Firstly, it does not indicate what kind of elements those constituent parts are; it does not even identify except implication. **Secondly**, it does not show clearly that noun phrases are built on nouns, verb phrases on verbs etc. **Thirdly**, IC analysis does not tell us how to form new sentences, i.e. to produce sentences that have not already been attested in some corpus of data.

Lesson 19

Sentence Presenting

Topic: 109: Phrase structure rules

What is needed is a model that:

- will not merely segment the constituents, but will also identify them in grammatical terms,
- will, in addition, change the emphasis of the linguistic analysis from mere description of sentences that have actually occurred to the specification of what sentences are possible in a language.

Such a model is generative; it is said to GENERATE all the grammatical sentences of a language. The model is associated with transformational generative grammar, since it deals with the constituency structure and not the transformational aspects of that grammar. IC analysis uses brackets, or bracketing. The bracketed elements can, however, be given grammatical identification by what is aptly known as LABELLED BRACKETING.

Let us consider the sentence:

The man followed a girl. *a* can be identified as determiners, *man* and *girl* as nouns, and *followed* as a verb further as *follow-* and *-ed*, or as the past tense of FOLLOW, but this is not wholly relevant and be ignored). Moreover, *the man* and *a girl* are also constituents of the same, but larger, type, noun phrases, and if our IC analysis is correct *followed a girl* is also a constituent, a verb phrase, while the whole is a sentence or clause.

Symbol

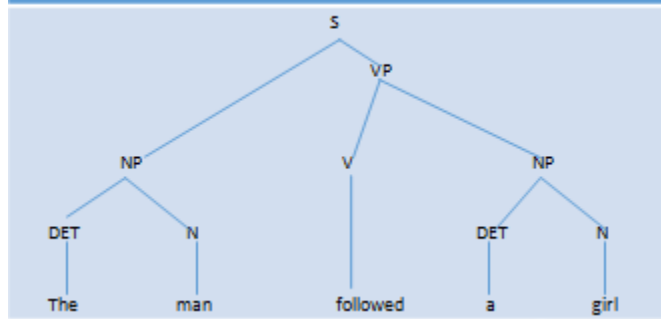
- DET for determiner

- **N for noun**
- **V for verb**
- **NP for noun phrase**
- **VP for verb phrase**
- **S for sentence**

A labeled bracketing of the sentence

[S [NP [DET the] N. man]] [VP [V followed] [NP [DET a] N. girl]]]

The representation of phrase structure is known as Phrase Marker or P-Marker



- Nodes: points that are joined by the lines
- Terminal nodes: nodes on the bottom line
- Dominate: one node is higher in tree than the other
- Immediately dominates: joined by a single line
- Dominance: a larger constituent may consist of one more of smaller kind.

The tree structure preserve the linear order of constituents just like IC analysis does.

The first noun phrase precedes the Verb phrase. The verb phrase precedes the second noun phrase, the determiner precedes the noun. Precedence like dominance is clearly shown in the tree diagram. Label bracketing and phrase structure trees provide much more information than IC analysis, Except implication does not state how new sentences can be generated

This can be done with the use of Phrase Structure Rules or PS Rules:

(1) $S \rightarrow NP - VP$

(2) $VP \rightarrow V - NP$

(3) $NP \rightarrow DET - N$

(4) $V \rightarrow \text{followed}$

(5) $DET \rightarrow \text{the, a}$

(6) $N \rightarrow \text{man, girl}$

- PS rules different from traditional grammar
- These rules can generate more than one sentence.
- A girl followed the man.
- The girl followed a man
- A man followed a man.

Topic: 110: Disadvantages of Phrase structure grammar

A phrase structure grammar is essentially a grammar of segmentation and categorization, it is a taxonomy

Very strong in giving structural description of the language, is deficient in generative capacity. Incapable of accounting for all the intentions of native speakers. It fails to disambiguate all the ambiguities and understand all the synonymies

The processes that pose problems to PS grammar are:

1. ambiguities
2. synonymies
3. permutations
4. discontinuous constituents (e.g. particles)
5. remote relationship (e.g. those of cases)
6. concord phenomena
7. co-ordination.

A PS grammar runs into difficulties in describing syntactic structures of questions, negatives, passives, relatives, etc. easily. It fails to capture the deep meaning. It cannot discover the crucial notions, nor can it prevent the assignment of false, ungrammatical structure. PS rules are incapable of accounting for the multiplicity of relations existing either between elements in the same sentence, or between different

sentences.

For example:

1. The police diverted the traffic.
 2. The traffic was diverted by the police.
 3. The traffic was diverted by a country road.
- PS rules fail to show the relationship that connects 1 to 2. In sentence 2 'by the police' will be shown as a prepositional phrase consisting of a preposition, a determiner and a noun, and in sentence 3 'by a country road' too will be shown as a prepositional phrase (prep + NP).
 - Thus, it would ignore semantic considerations and case relations.
 - PS grammar does not have the precision, simplicity, elegance, power, insight, and competence of the TG grammar.
 - For sentences like the following both IC analysis and PS these are identical.
 - John is eager to please
 - John is easy to please
 - Phrase Structure Rules cannot help us distinguish among ambiguous sentences

Topic: 111: Automatic Grammatizator

- Every normal human being can talk.

So the average person tends to think that there is little or nothing mysterious about language. As Noam Chomsky has pointed out: We lose sight of the need for explanation when phenomena are too familiar and 'obvious'. Easily to assume that explanations must be transparent and close to the surface . . . As native speakers, we have a vast amount of data available to us, the trap of believing that there is nothing to be explained.

Nothing could be further from the truth . (Chomsky 1972a: 25–6). But the mysterious nature of human language becomes more apparent when one realizes that no one has yet managed to simulate the language ability of a human being. Computers can play chess, sort bank statements, and even talk about limited topics such as cubes, squares and cones. But we are far from producing a 'great automatic grammatizator' which could unaided hold conversations on any topic.

- Why is this?
- Perhaps we should think about language more carefully.

When people start thinking about language, the first question which often occurs to them is this: is language natural to humans? – in the same way that barking comes naturally to dogs. Clearly, in one sense, children 'learn' whatever language they are exposed to, be it Chinese, Nootka or English. So no one would deny that 'learning' is very important.

But the crucial question is whether children are born with ‘blank sheets’ in their head as far as language is concerned – or whether humans are ‘programmed’ with an outline knowledge of the structure of languages in general. This question of whether language is partly due to nature or wholly due to learning or nurture is often referred to as the nature–nurture controversy, and has been discussed for centuries.

This particular issue resurfaced in linguistics in 1959 when the linguist Noam Chomsky wrote a devastating and witty review of *Verbal Behavior*, a book by the Harvard psychologist B.F. Skinner (Skinner 1957; Chomsky 1959). This book claimed to ‘explain’ language as a set of habits gradually built up over the years.

According to Skinner, no complicated innate or mental mechanisms are needed. All that is necessary is the systematic observation of the events in the external world which prompt the speaker to utter sounds. Skinner’s claim to understand language was based on his work with rats and pigeons. He had proved that, given time, rats and pigeons could be trained to perform an amazing variety of seemingly complex tasks, provided two basic principles were followed. This type of ‘trial-and-error’ learning was called operant conditioning by Skinner, which can be translated as ‘training by means of voluntary responses’ (the word ‘operant’ means a voluntary response rather than an automatic one).

Skinner suggested that it is by means of this mechanism that the vast majority of human learning takes place, including language learning:

Recent work has shown that the methods can be extended to human behaviour without serious modification. (Skinner 1957: 3). All one needed to do in order to understand language, he said, was to identify the ‘controlling variables’, which would enable us to predict specific utterances.

Please pass the bread and butter.’

Or the presence of a beautiful painting might call forth the exclamation, ‘Oh how beautiful.’ Or a bad smell might cause one to exclaim ‘Oh what a terrible smell.’

In theory, Skinner saw no difficulty in linking up any particular set of words which a human might wish to produce with an identifiable external happening. In practice, the matter is far from simple, as Chomsky pointed out. Chomsky made two major criticisms of Skinner’s work. First, The behaviour of rats in boxes is irrelevant to human language. Second, Skinner fundamentally misunderstood the nature of language.

Topic: 112: [Missing]

Topic: 113: Behaviouristic view: The irrelevance of rats

Chomsky pointed out that the simple and well-defined sequence of events observed in the boxes of rats is just not applicable to language. The terminology used in the rat experiments cannot be re-applied to human language without becoming hopelessly vague. For example, how do you know that someone is likely to say ‘Oh what a beautiful picture’ (when looking at a beautiful painting)?

- They might say instead,

- ‘It clashes with the wallpaper’,
- ‘It’s hanging too low’,
- ‘It’s hideous.’

Skinner would say that instead of the utterance being ‘controlled’ by the beauty of the picture, it was ‘controlled’ by its clash with the wallpaper, or hanging too low, its hideousness. But this reduces the idea of ‘control’ to being meaningless, because you have to wait until you hear the utterance before you know what controlled it. This is quite unlike the predictable behaviour of rats which could be relied upon to respond to certain stimuli such as a flashing light with a fixed response.

- Another problem

The rats were repeatedly rewarded. Children do not receive pellets of food when they make a correct utterance. The idea of reward or reinforcement (since it reinforces the behaviour that is being learned) can in humans be naturally extended to approval or disapproval.

Skinner suggested that in these cases children automatically ‘reinforce’ themselves because they know they are producing sounds which they have heard in the speech of others. A poet who is uttering words aloud in an empty room will be ‘reinforced’ by the knowledge that others will be influenced by the poetry. So reinforcement seems a very woolly notion, since an actual reward need not exist, it need only be imagined or hoped for. A notion is certainly not comparable to the food pellets given to rats when they make a correct response.

Studies by Roger Brown and his associates provided even more problems for Skinner’s notion of reinforcement. After observing mother–child interactions they pointed out that parents tend to approve statements which are true rather than those which are grammatically correct. So a boy who said ‘*Teddy sock on*’ and showed his mother a teddy bear wearing a sock would probably meet with approval. But if the child said the grammatically correct utterance ‘*Look, Teddy is wearing a sock*’, and showed his mother a bear without a sock, he would meet with disapproval. In other words, if approval and disapproval worked in the way Skinner suggested, you would expect children to grow up telling the truth, but speaking ungrammatically.

In fact the opposite seems to happen (Brown et al. 1968). Chomsky used these and similar arguments to show the irrelevance of Skinner’s experiments to the problem of understanding language. Perhaps ‘irrelevance’ is too strong a word, since there are areas of language where habit forming works. Of course, just because Skinner’s ideas were over-simple does not automatically mean that Chomsky’s ideas were right. Maybe both Skinner’s and Chomsky’s views are outdated. Now, in the twenty first century, we know a lot more about language and its special qualities, partly because Chomsky in particular inspired so many to take language seriously as a key to understanding the human mind, to work on it further.

Topic: 114: Is language restricted to humans

- Judging by newspapers and popular books, there appear to be a vast number of animals which ‘talk’ – talking budgerigars, talking dolphins– ,h, etc.

- Clearly, the word ‘talk’ can be used in two totally different senses.
- ‘to utter words’, as in ‘Archibald’s got a talking parrot which says *Damn* if you poke it.’
- ‘to use language in a meaningful way’.
- Animals such as budgerigars can ‘talk’ in the first sense of the word.
- Psycholinguists: Can animals ‘talk’ in the second sense also.
- Interested in this problem because they want to know the answer to the following question: are we the only species which possesses language?
- If so, are we the only species capable of acquiring it?
- Are we biologically singled out as ‘articulate mammals’ or not?
- Do animals talk naturally?

In order to answer this question, we must compare human language with animal communication. A comparison presents a number of perhaps unsolvable problems. The first problem is this: are we comparing systems which differ quantitatively or qualitatively? On the one hand, human language may have gradually evolved from a more primitive animal means of communication in a continuous line of growth – a viewpoint sometimes known as a ‘continuity’ theory

On the other hand, human language may be something quite different from our basic animal heritage, and superimposed on it - a ‘discontinuity’ theory. The second major problem we face is that it is not always easy to decide what counts as communication in animals. As one researcher notes: Students of animal behaviour have often noted the extreme difficulty of restricting the notion of communication to anything less than every potential interaction between an organism and its environment.(Marshall 1970: 231)

- A useful first step might be to attempt to define ‘language’.
- This is not as easy as it sounds.

Many definitions found in elementary textbooks are too wide. For example: ‘A language is a system of arbitrary vocal symbols by means of which a social group cooperates’. This definition might equally well apply to a pack of wolves howling in chorus. A superficially promising approach was that suggested by the linguist Charles Hockett in the 1960s. The various ‘design features’ which characterize language.

These are: use of the vocal-auditory channel, arbitrariness, semanticity, cultural transmission, spontaneous usage, turn-taking, duality, displacement, structure-dependence, creativity, ability to read intentions. Finally, we come to birds. They also have failed to give any evidence of creativity. We might expect them to communicate about a multiplicity of situations, since the individual notes of a bird’s song can be combined in an indefinite number of ways.

But as far as researchers can judge, birdsong deals above all with just two aspects of life: courting a mate, and the marking of territory (Nottebohm 1975; Marler 1991). A final, crucial feature of language has come to the forefront in recent years. This is intention-reading (Tomasello 2003), or mind-reading. Some limited awareness of it has been detected among apes, especially chimps. But humans are the best at this skill. Teaching sign language to apes: Washoe and Nim

Over the past 50 or so years, several attempts have been made to teach human language to chimpanzees. Chomsky may be right, therefore, when he points out that the higher apes ‘apparently lack the capacity to develop even the rudiments of the computational structure of human language’ (Chomsky 1980: 57). Chomsky (1968) claimed that language is specific to humans as only humans possess a language acquisition device to acquire language – the universal grammar.

Lesson 20

Constituency Tests

Topic: 115: Biological evidence for innate language capacity

If an animal is innately programmed for some type of behaviour, then there are likely to be biological clues. No accident that fish have bodies which are streamlined and smooth, with fins and a powerful tail structurally adapted for moving fast through the water. Biological clues are not essential. It is impossible to guess from their bodies that birds make nests, and, sometimes, animals behave in a way quite contrary to what might be expected from their physical form. Inexplicable divergences between species obvious behaviour. Such unpredictability is not universal, and need not discourage us from looking for biological clues connected with speech – though we must realize that we are unlikely to find the equivalent of a large box labelled ‘language’.

Changes in the form of the body or structural changes are the most direct indications of innate programming. But we must also take into consideration physiological adaptations – changes in the bodily functions, such as rate of heartbeat and breathing. If we look at the organs used in speech, humans seem to be somewhere in the middle between the obvious structural adaptation of birds to flying, and the apparent lack of correlation between birds and nest building. The human brain and vocal tract have a number of slightly unusual features. By themselves, these features are not sufficient to indicate that people can talk. But if we first assume that all humans speak a language, then a number of puzzling biological facts fall in place.

They can be viewed as partial adaptations of the body to the production of language. Human teeth are unusual compared with those of other animals even in height, and form an unbroken barrier. Another important difference between humans and monkeys concerns the larynx, which contains the ‘voice box’ or ‘vocal cords’ simpler in structure than that of other primates. This is an advantage as air can move freely past and then out through the nose and mouth without being hindered by other appendages. Our brain: very different in appearance from that of other animals. Heavier, with more surface folding of the cortex, the outer layer of ‘grey matter’ which surrounds the inner core of nerve fibres – though grey

matter is actually pink in live humans, it goes grey after death. Superficially, the brains of a chimp and a human have certain similarities.

A further related discovery is that the location of speech centres in the left hemisphere seems to be linked to right-handedness. That is, most humans are right-handed, and most people's speech is controlled by the left hemisphere. This lateralization or localization of language in one half of the brain is a definite, biological characteristic of the human race. Humans are physically adapted to language in a way that snails, sheep and even apes are not. Their vocal organs, lungs and brains are 'preset' to cope with the intricacies of speech in much the same way that monkeys are pre-set to climb trees, or bats to squeak.

Topic: 116: A preordained language programme

Language emerges at about the same time in children all over the world. 'Why do children normally begin to speak between their eighteenth and twenty-eighth month?' asks one researcher:

Surely it is not because all mothers on earth initiate language training at that time. No evidence that any conscious and systematic teaching of language takes place, just as there is no special training for stance or gait (Lenneberg 1967: 125).

This regularity of onset suggests that language may be set in motion by a biological clock, similar to the one which causes kittens to open their eyes when they are a few days old, until relatively recently, few people had considered language within the framework of biological maturation. The characteristics of biologically triggered Behaviour: Behaviour that is triggered off biologically has a number of special characteristics. If it can be shown that speech, like sexual activities and the ability to walk, falls into the category of biologically scheduled behaviour, then we shall be rather clearer about what is meant by the claim that language is 'innate'. Exactly how many 'hallmarks' of biologically controlled behaviour should itemize is not clear.

The six listed below were obtained mainly by subdividing Lenneberg's four:

The behaviour emerges before it is necessary.

Its appearance is not the result of a conscious decision.

Its emergence is not triggered by external events (though the surrounding environment must be sufficiently 'rich' for it to develop adequately).

Direct teaching and intensive practice have relatively little effect.

There is a regular sequence of 'milestones' as the behaviour develops, and these can usually be correlated with age and other aspects of development.

There may be a 'critical period' for the acquisition of the behaviour.

Although children differ enormously in their ability to knit or play the violin, their language proficiency varies to a much lesser extent.

- 'Tomorrow I am going to start to learn to talk.'
- Children acquire language without making any conscious decision about it.
- Children begin to talk even when their surroundings remain unchanged.

Most of them live in the same house, eat the same food, have the same parents, and follow the same routine. No specific event or feature in their surroundings suddenly starts them off talking. Children do not manufacture any new brain cells after birth. They are born with millions, perhaps billions. At first the cells are not all interconnected, and the brain is relatively light (about 300g). From birth to around 2 years, many more cells interconnect, and brain weight increases rapidly. By the age of 2, it weighs nearly 1000g (Lenneberg 1967).

Although no external event causes the behaviour, the surrounding environment must be sufficiently 'rich' for it to develop adequately. 'Direct teaching and intensive practice have relatively little effect.' In activities such as typing or playing tennis, a person's achievement is often directly related to the amount of teaching they receive and the hours of practice they put in. Not 'naturally' superb athletes can win. We may conclude, then, that mere practice – in the sense of direct repetition and imitation – does not affect the acquisition of language in a significant way

Child's likely progress chart

• Language stage	• Beginning age
• Crying	• Birth
• Cooing	• 6 weeks
• Babbling	• 6 months
• Intonation patterns	• 8 months
• One-word utterances	• 1 year
• Two-word utterances	• 18 months
• Word inflection	• 2 years

- | | |
|--|--|
| <ul style="list-style-type: none"> • Questions, negatives • Rare or complex constructions • Mature speech | <ul style="list-style-type: none"> • 2 1/4 years • 5 years • 10 years |
|--|--|

Topic: 117: Conceivable innate grammatical information

- It is relatively easy to show that humans are innately predisposed to acquire language.
- The hard part is finding out exactly what is innate.
- People have indulged in speculation about this for centuries.
- The Egyptian king Psammetichus experiment

Nobody takes Psammetichus's theory seriously today – especially as the few reliable accounts we have of children brought up without human contact indicate that they were totally without speech when they were found. Although the speculations of Psammetichus can safely be ignored, the ideas of Noam Chomsky on the topic of innateness were for a long time taken seriously. He claimed that for language acquisition to be possible, a child must be endowed with a 'rich internal structure'.

- His ideas cannot be summarily dismissed.

Chomsky's notion of a rich innate schema contrasted strongly with the point of view popularly held earlier in the century that children are born with 'blank sheets' as far as language is concerned. Consequently, some people considered Chomsky as someone who had set out to shock the world with outrageous and novel proposals. Chomsky denied and pointed out that he was following in the footsteps of eighteenth-century 'rationalist' philosophers, who believed in the existence of 'innate ideas'.

Such philosophers held that 'beyond the peripheral processing mechanisms, there are innate ideas and principles of various kinds that determine the form of the acquired knowledge in what may be a rather restricted and highly organized way' (Chomsky 1965: 48). Chomsky gave an explicit account of his early views in his (now outdated) linguistic classic *Aspects of the Theory of Syntax* (1965), though he has repeated them in a number of other places with minor variations. Recently he has changed his mind on various points, quite fundamentally. His later views were set out in *Knowledge of Language: Its Nature, Origin and Use* (1986), later ones still in *The Minimalist Program* (1995) and further views in *On Nature and Language* (2002).

He regards his previous ideas unsatisfactory, and outlines his more recent ideas. Nevertheless, his ideas were specific enough to be interesting. Chomsky pointed out that children are to some extent in the same situation as a linguist faced with an unknown language. Both child and linguist are surrounded by a superficially unintelligible confusion of sound which they must somehow sort out. According to Chomsky, children would 'know' in advance how their internalized grammar must be organized.

They must have sets of the following rules:

- SEMANTIC RULES

- SYNTACTIC RULES
- PHONOLOGICAL RULES

He assumed that children were endowed with an innate hypothesis-making device, which enabled them to make increasingly complex theories about the rules which would account for the language they heard going on around them. In making these hypotheses, children were guided by an inbuilt knowledge of language universals. These provided a 'blueprint' for language, so that the child would know in outline what a possible language looked like. This involved, first, information about the 'building blocks' of language, set of sounds. Second, it entailed information about the way in which the components of a grammar were related to one another, and restrictions on the form of the rules.

Chomsky argued that children automatically knew that language involved two levels of syntax – a deep and a surface level, linked by 'transformations'. And (as he later argued) children also knew about some innately inbuilt constraints on the form sentences could take. With this help a child could speedily sift through the babble of speech he heard around him, and hypothesize plausible rules which would account for it. Children needed to be equipped with this information, he claimed, because the 'primary linguistic data' (the data children are exposed to) was likely to be 'deficient in various respects' (1965: 201). According to Michael Tomasello, 'how children learn language is not a logical problem but an empirical problem.' (Tomasello 2003: 328).

In his opinion, we need to turn to a usage-based approach, one which explores how human children combine inherited talents and learned skills as they acquire language. 'The human capacity for language is best seen as a conspiracy of many different cognitive, social-cognitive, information- processing, and learning skills, some of which humans share with primates and some of which are unique products of human evolution'.

Topic: 118: Transformational Grammar

In linguistics, transformational grammar (TG) or transformational-generative grammar (TGG) is part of the theory of generative grammar, especially of natural languages. It considers grammar to be a system of rules that generate exactly those combinations of words that form grammatical sentences in a given language and involves the use of defined operations (called transformations) to produce new sentences from existing ones. The theory largely originated and developed by Noam Chomsky. It is without question the most influential theory of linguistics in modern times, and one that no serious scholar can afford to neglect.

But the student of TG will often experience frustration and puzzlement, for it is a curious mixture of quite brilliant insights and of complex, technical but often seemingly artificial devices. The primary literature, apart from Chomsky's *Syntactic Structures*, the first published work, is often difficult to read and its full intelligibility depends on knowing all the background. The notion of PS grammar, although much of what was said comes from Chomsky's works, but his fame does not rest on it.

Much more important is the notion of transformation, together with ideas that are associated with it, or have developed out of a transformationally based theory. Although *Syntactic Structures* (1957) first introduced the world to this important theory, *The Logical Structure of Linguistic Theory*, which was not published until eighteen years later, was written some years before it. The theory was, undoubtedly,

revolutionary and even greeted with total skepticism by many scholars, but, as with all revolutions, some of it had been foreshadowed in earlier works, particularly in the writings of Chomsky's own teacher Zellig Harris.

Chomsky has considerably modified his ideas since 1957, and in his philosophical views has vastly distanced himself from The structuralists, and indeed from the whole of the empiricist tradition. Undoubtedly the best known, in some ways neatest and most discussed theoretical position is that of Aspects of the Theory of Syntax, a position that Chomsky himself has called the 'Standard Theory' .

This introduced and expounded the notion of Deep Structure, an essential part of the theory. Several important changes were made in the 1970s, to produce the 'Extended Standard Theory'(EST). Chomsky's position at the beginning of the 1980s is to be found in Lectures on Government and Binding.

Development of basic concepts

Though transformations continue to be important in Chomsky's current theories, he has now abandoned the original notion of Deep Structure and Surface Structure.

Initially, two additional levels of representation were introduced (LF — Logical Form, and PF — Phonetic Form), but in the 1970s, Chomsky sketched out a new program of research known as Minimalism, in which Deep Structure and Surface Structure are no longer featured and PF and LF remain as the only levels of representation. To complicate the understanding of the development of Chomsky's theories, the precise meanings of Deep Structure and Surface Structure have changed over time. By the 1970s, they were normally referred to simply as D-Structure and S-Structure by Chomskyan linguists. In particular, the idea that the meaning of a sentence was determined by its Deep Structure (taken to its logical conclusions by the generative semanticists during the same period) was dropped for good by Chomskyan linguists when LF took over this role. Previously, Chomsky and Ray Jackendoff had begun to argue that meaning was determined by both Deep and Surface Structure.

Topic: 119: The founder and the theory

Avram Noam Chomsky (born December 7, 1928) is an American linguist, philosopher, cognitive scientist, historian, political activist, and social critic. Sometimes called "the father of modern linguistics", Chomsky is also a major figure in analytic philosophy and one of the founders of the field of cognitive science. He holds a joint appointment as Institute Professor Emeritus at the Massachusetts Institute of Technology (MIT) and laureate professor at the University of Arizona, and is the author of over 100 books on topics such as linguistics, war, politics, and mass media.

Born to middle-class Jewish immigrants in Philadelphia, Chomsky developed an early interest in linguistics. He began studying at the University of Pennsylvania at age 16, taking courses in linguistics, mathematics, and philosophy. From 1951 to 1955, he was appointed to Harvard University's Society of Fellows.

While at Harvard, he developed the theory of transformational grammar; for this, he was awarded his doctorate in 1955. Chomsky began teaching at MIT in 1957 and emerged as a significant figure in the field of linguistics for his landmark work Syntactic Structures, which remodelled the scientific study of

language. From 1958 to 1959, he was a National Science Foundation fellow at the Institute for Advanced Study.

Chomsky is credited as the creator or co-creator of the universal grammar theory, the generative grammar theory, and the Chomsky hierarchy. He also played a pivotal role in the decline of behaviorism, being particularly critical of the work of B. F. Skinner. One of the most cited scholars in history, Chomsky has influenced a broad array of academic fields. He brought about a major revolution in the human sciences, contributing to the development of a new cognitivistic framework for the study of language and the mind.

Topic: 120: Deep and surface structure

The structuralist approach to language was *Corpus bound* as both phoneme and morpheme are units of form rather than of meaning. Their *method was inductive* and it also failed to account for the *ambiguities* in sentences and their interrelatedness. To answer these questions an adequate grammar needed that is productive, complex and arbitrary. It must establish the relationship of various sentences and account for the deep and surface structure of the sentences.

It must explain the relationship between sets of sentences:

(i) He wrote a letter.

(ii) A letter was written by him.

(i) Why did you go there?

(ii) You went there for some reason.

(i) Where do you live?

(ii) You live somewhere.

- TG grammar is both transformational and generative.

It goes a step further from the structural grammar. It analyses the sentences, divides them into parts and shows the functions of various parts and rearranges them and shows the inter-relatedness between sentences. It takes up the basic or kernel sentence first. A basic or kernel sentence is simple, assertive, and active in form.

Here are some examples:

Kernel sentences:	
	• John is playing football. • I wrote a letter. • You spoke the truth.

Non-kernel sentences:

- Is John playing football? (Interrogative)
- John is not playing football. (Negative)
- A letter was written by me. (Passive)
- Who wrote the letter? (Question form)
- Why did you speak the truth? (Question form)
- You spoke the truth when you were forced. (Complex sentence)
- You did not speak the truth. (Negative)
- You spoke the truth but told a new story. (Compound sentence)

All the non-kernel structures or *complicated structures* can be thought of as having been derived from their kernel forms with the help of some transformations. A sentence like: *Where did you go yesterday?* can be derived from the kernel sentence: *You went somewhere yesterday* by applying certain transformations. Transformational analysis not only shows the inter-relatedness between sentences but also explains the ambiguities between sentences that appear identical but are transforms from different kernels.

Flying planes can be dangerous.	(a)Some people fly planes. This can be dangerous. (b)Planes fly. They can be dangerous.
(a)I expected John to hit a six. (b)I asked John to hit a six.	(a)(i)I expected (something). (ii)John hits a six. (b)(i)I asked John (something). (ii)He hits a six.
(a)John is eager to please. (b)John is easy to please.	(a)(i)John is eager (for something). (ii)He pleases (someone). (b) (i)It is easy. (ii)Someone please John.

A kernel sentence is deep structure to which a string of transformation rules is applied to generate the surface structure.

Constituents I: Words Order

Topic: 121: Introduction

- Some aspects of grammar
- Grammar is traditionally subdivided into two different but interrelated areas of study – morphology and syntax.
- morphological operations
- Antidisestablishment-arianism,
- Syntax is the study of the way in which phrases and sentences are structured out of words, and so addresses questions like ‘What is the structure of a sentence like the following:

What’s the president doing??’

What is the nature of the grammatical operations by which its component words are combined together to form the overall sentence structure. A brief look at the approach to the study of syntax taken in traditional grammar: this also provides an opportunity to introduce some useful grammatical terminology. Within traditional grammar, the syntax of a language is described in terms of a taxonomy (i.e. classificatory list) of the range of different types of syntactic structures found in the language.

The central assumption underpinning syntactic analysis in traditional grammar is that phrases and sentences are built up of a series of constituents (i.e. syntactic units), each of which belongs to a specific grammatical category and serves a specific grammatical function. Given this assumption, the task of the linguist in analysing the syntactic structure of any given type of sentence is to identify each of the constituents in the sentence, and to say what category it belongs to and what function it serves.

Sentence	Categories
Students protested	It would traditionally be said that the sentence consists of <i>two constituents</i>
Students protested	Each of these constituents belongs to a <i>specific grammatical category</i> (students being a plural noun and protested a past tense verb)
Students protested	Each serves a <i>specific grammatical function</i> (students being the subject of the sentence, and protested being the predicate).

Students protested	The overall sentence has the categorial status of <i>a clause</i> which is <i>finite in nature</i> (by virtue of denoting an event taking place at a specific time),
Students protested	The semantic function of expressing a proposition which is <i>declarative in force</i> (in that it is used to make a statement rather than e.g. ask a question).

Accordingly, a traditional grammar of English would tell us that the simplest type of finite declarative clause found in English is a sentence like *The students protested* in which a *nominal subject* is followed by a *verbal predicate*.

Topic: 122: Grammatical Categories

- In traditional grammar, words are assigned to grammatical categories (called parts of speech) on the basis of their:
 - semantic properties (i.e. meaning),
 - morphological properties (i.e. the range of different forms they have)
 - syntactic properties (i.e. word-order properties relating to the positions they can occupy within sentences):
- A set of words which belong to the same category thus have a number of semantic, morphological and syntactic properties in common.
- There are traditionally said to be two different types of word, namely
 - Content words/connectives (= words which have substantive lexical content) on the one hand,
 - function words/functors (= words which essentially serve to mark grammatical properties) on the other.
- The differences contentive like *car* with a functor like *they*.
- A noun like *car* has substantive lexical content in that it denotes an object which typically has four wheels and an engine, and it would be easy enough to draw a picture of a typical car.
- By contrast, a pronoun such as *they* has no descriptive content (e.g. you can't draw a picture of they),

A functor which simply marks grammatical (more specifically, person, number and case) properties in that it is a third person plural nominative pronoun.

- Because they have lexical semantic content, content words often (though not always) have antonyms e.g. the adjective *tall* has the antonym *short*,

- The verb *increase* has the antonym decrease,
- The preposition *inside* has the antonym outside:
- By contrast, a typical function word like e.g. the pronoun *me* has no obvious antonym.
- Corresponding to these two different types of words *content and function*
- Two different kinds of grammatical category – namely lexical/substantive categories (= categories whose members are content words)
- Functional categories (= categories whose members are function words).

Topic: 123: Lexical/substantive categories

The main lexical/substantive categories found in English. Noun, verb, adjective, adverb and preposition. Conventionally abbreviated to N, V, A, ADV and P in order to save space

	Nouns (= N)
Semantic property	denote entities: bottle (since it denotes a type of object used to contain liquids),water (since it denotes a type of liquid) John (since it denotes a specific person).
Distinct subtypes of noun	Count noun: like chair (cf. one chair, two chairs . . .), Mass noun: like furniture is a mass noun in that it denotes an uncountable mass (*one furniture, *two furnitures).
Common and proper nouns	Boy a common noun like boy (which can be modified by a determiner like the – as in The boy is lying) Andrew (which cannot be used in the same way in English, as *The Andrew is lying).
	Nouns (= N)

Number	Count nouns generally have the morphological property that they have two different forms: a singular form (like horse in one horse) used to denote a single entity, a plural form (like horses in two horses) used to denote more than one entity.
Common nouns have the syntactic property	Only (an appropriate kind of) noun can be used to end a sentence such as They have no . . . (car, or a plural count noun like friends or a mass noun like money, but not other types of word (e.g. not see or slowly or up, as these are not nouns).

	Verb (= V).
semantic property	denote actions or events: so, eat, sing, pull and resign are all (action-denoting) verbs.
	From a syntactic point of view, verbs have the property that only an appropriate kind of verb (in its <i>uninflected infinitive form</i>) can be used to complete a sentence such as They/It can . . . So, words like stay, leave, hide, die, starve and cry are all verbs words like apple, under, pink and if aren't.
Morphological point	regular verbs like cry in English have the property that they have four distinct forms: e.g. alongside the bare (i.e. uninflected) form cry we find the present tense form cries, the past tense/perfect participle/passive participle form cried and the progressive participle form crying.

	Adjective (= A).
semantic property	denote states or attributes (cf. ill, happy, tired, conscientious, red, cruel, old etc.).
Syntactic properties	Can occur after be to complete a sentence like They may be . . . (as with They may be tired/ill/happy etc.),

	denote a gradable property varying degrees fast, faster, fastest
Modified by a degree word	very/rather/somewhat (cf. She is very happy). Many (but not all) adjectives have the morphological property that they have comparative forms ending in -er and superlative forms ending in -est (cf. big /bigger/biggest).

	Adverb (= ADV)
Semantic property	denote the manner in which an action is performed (as with well in She sings well).
Morphological property	Regular adverbs have the that they are formed from adjectives by the addition of the suffix -ly. the adjective sad we have the adverb sadly
Syntactic property	an adverb (like e.g. badly) is the only kind of word which could be used to end sentences such as She behaved . . ., He treats her . . .or He worded the statement . . .

	Preposition (=P)
Semantic property	of marking location (cf. in/on/off/inside/outside/under/above/below).
Syntactic property	with the appropriate kind of meaning can be modified by <i>right</i> in the sense of 'completely', or by <i>straight</i> in the sense of 'directly' down in He fell right down the stairs to in He went straight to bed.
Morphological property	they are invariable/uninflected forms (e.g. the preposition off has no past tense form *offed, no superlative form *offest and so on).

Topic: 124: Functional category

Functional categories

Determiner (= D) traditionally said to include the *definite article* **the** and the *demonstrative determiners* **this/that/these/those**.

	Determiner (= D)
Includes	definite article <i>the</i> and the demonstrative determiners <i>this/that/these/those</i>
Semantic property	determine specific semantic properties of the noun expression that they introduce, marking it as a definite referring expression Shall we take the car? (specific) car which is assumed to be familiar to the hearer/addressee.
	Quantifier (= Q),
Function	denoting expressions of quantity, such as some/all/no/any/each/every/most/much/many.
Indefinite article	the indefinite article <i>a</i> to be a quantifier – one which quantifies over a single entity a car

	Pronouns(=PRN)
Items	said to ‘stand in place of’ (the meaning of the prefix <i>pro-</i>) or ‘refer back to’ noun expressions.
Types	a number of different types of pronoun found in English and other languages He/his/him
The word <i>one</i>	John has a red car and Jim has a blue <i>one</i> has no lexical semantic content of its own, rather takes its content from its antecedent (i.e. <i>one</i> refers back to the noun <i>car</i> and so <i>one</i> is interpreted as having the same meaning as <i>car</i>).
Morphological	pronoun <i>one</i> behaves like a regular count noun in that it has a plural form

perspective	ending in -s (as in I'll take the green apples if you haven't got any red ones)
Status of <i>one</i>	N-pronoun /or pronominal noun

	Pronouns(=PRN)
Some	sentence like Many miners were rescued, but some died, the word some seems to function as a Q-pronoun (i.e. a pronominal quantifier)
Those	These apples are ripe, but those aren't, the word <i>those</i> seems to be a D-pronoun (i.e. a pronominal determiner).
Personal pronouns as D-pronouns	I/me/we/us/you/he/him/she/her/it/they/them the rationale: <i>We</i> republicans don't trust <i>you</i> democrats, <i>we</i> could be argued to be a determiner modifying the noun republicans, <i>you</i> could be seen as a determiner modifying the noun democrats

	Auxiliary (verb).
Semantic property	marking grammatical properties such as tense, aspect, voice or mood
Syntactic property	unlike lexical/main verbs they can be inverted with their subject in questions (so that corresponding to a statement like <i>It is raining</i> we have the question <i>Is it raining?</i> (the auxiliary <i>is</i> has moved in front of the subject <i>it</i> and is said to have been inverted).

The items *italicized* below are traditionally categorised as auxiliaries taking a [bracketed] complement :

(a) He has/had [gone]	have/be in (a,b) are (perfect/progressive) aspect auxiliaries
(b) She is/was [staying at home]	
(c) They are/were [taken away for questioning]	be in (c) is a (passive) voice auxiliary
(d) He really does/did [say a lot]	do in (d) is an expletive or dummy auxiliary (i.e. one with no intrinsic lexical semantic content)
(e) You can/could [help us]	can/could/may/might/will/would/shall/should in (e–h) are modal auxiliaries
(f) They may/might [come back]	
(g) He will/would [get upset]	
(h) I shall/should [return]	

Auxiliaries in sentences have been said to belong to the category T (= tense-marked auxiliary).

- So is the infinitive particle *to*, in sentences such as:
- *They are now expecting the president to be impeached tomorrow*
- In the sentence infinitival *to* seems to have future time-reference
- The act of impeachment will take place at some time in the future.
- Why we can use the word tomorrow in the *to*-clause.
- Infinitival *to* seems to have much the same function as the auxiliary *will* in:
- *They are now expecting that the president will be impeached tomorrow,*
- The infinitival *to* is an infinitival tense marker, and so belongs to the same category T as present/past tense auxiliaries such as *is/was*.

The difference between auxiliaries and infinitival *to* is that most auxiliaries overtly inflect for present/past tense (though this is not true of the invariable auxiliaries *must* and *ought*), whereas infinitival *to* is invariable in form. We can thus say that an auxiliary like *will* is a *finite T constituent*, whereas infinitival *to* is a *nonfinite T*.

Topic: 125:

- (a) I think [*that you may be right*]
- (b) I doubt [*if you can help me*]
- (c) I'm anxious [*for you to receive the best treatment possible*]

Each bracketed clause is a *complement clause*, the complement of the word immediately preceding it (think/doubt/anxious); The italicised (*that, if, for*) word which introduces each clause is a *complementiser* (= C). Complementisers are functors in the sense that they encode particular sets of grammatical properties. Complementisers encode whether they are finite or nonfinite.

For examples

I think [<i>that you may be right</i>]	I doubt [<i>if you can help me</i>]	I'm anxious [<i>for you to receive the best treatment possible</i>]
<i>That</i> is inherently finite in the sense that it can only be used to introduce a finite clause	<i>If</i> is also inherently finite in the sense that it can only be used to introduce a finite clause.	<i>For</i> is an inherently infinitival complementiser, and so can be used to introduce a clause containing <i>infinitival to</i>
<i>That</i> introduces a declarative clause (a statement),	<i>If</i> introduces an interrogative clause (i.e. a question)	<i>For</i> introduces an irrealis clause (i.e. one relating to a hypothetical event)
<i>That</i> is a finite declarative complementiser	<i>If</i> is a finite interrogative complementiser	<i>For</i> is an infinitival irrealis complementiser

Using the set of syntactic categories outlined above, we can employ the traditional labelled bracketing technique to categorise words in a way which describes how they are being used in a particular sentence.

(a) The president is clearly feeling angry that Congress has refused to negotiate with him

(b) [D The] [N president] [T is] [ADV clearly] [V feeling] [A angry] [C that] [N Congress] [T has] [V refused] [T to] [V negotiate] [P with] [PRN him]

Grammatical differences between individual words belonging to the same category are traditionally described in terms of sets of grammatical features, and these features are enclosed in square brackets.

- *She* invited *us* to dinner.

For example, both *she* and *us* are pronouns.

- *She* carries the features [third-person, singular-number, feminine-gender, nominative-case],
- *Us* carries the features [first-person, plural-number, accusative-case].

Traditional grammarians are also concerned to describe the grammatical functions

(a) John laughs

- The president laughs
- The president of Utopia laughs

The former president of the island paradise of Utopia laughs

(a) John reads novels S V C(DO)	• The complement novels describes the entity on which the act of reading is being performed • A verb which has a noun or pronoun expression as its direct object complement is traditionally said to be transitive. • From a semantic perspective, subjects and complements generally represent entities directly involved in the particular action or event described by the predicate: • we can say that subjects and complements are arguments of the predicate with which they are associated. • Predicates may have one or more arguments
(b) John reads <u>English novels</u> S V C(DO)	
(c) John reads <u>English novels</u> S V <u>imported from England</u> C(DO)	

(d) John reads English novels			
S C(DO)	V		
<u>imported by a friend of his from England</u>			
(a) [John] resigned	A predicate like resign in (a) which has a single argument is said to function as a one-place predicate		
(b)[John] felt [remorse]	(b) which has two arguments is a two-place predicate;		
(c) [John] sent [Mary] [flowers]	(8) which has three arguments is a three-place predicate.		

In addition to predicates and arguments, sentences can also contain *adjuncts*, as we can illustrate in relation to below:

(a) The president reads a novel after dinner

- The president reads a novel in his office
- *after dinner /in his office* serve to provide additional information and are both adjuncts.

We also find complex sentences which contain more than one clause

- Mary knows John reads

The press clearly thinks the president deliberately lied to Congress.

Topic: 126: Types of clauses

Finite and non finite clauses

Finite clauses must contain a verb which shows tense. They can be main clauses or subordinate clauses: *Is it raining?* Non-finite clauses contain a verb which does not show tense. We *usually* use non- finite verbs only in subordinate clauses.

Finite and non finite verbs

She <i>writes</i> a letter.	Bare infinitive form	does show tense/present time
She <i>wrote</i> a letter.	Past Tense form	does show tense/past time
*She <i>to write</i> a letter.	Infinitive form/ Infinitival To	does not show tense
*She <i>writing</i> a letter.	Present participle form	does not show tense
*She <i>written</i> a letter.	Past participle form	does not show tense

In linguistics, grammatical mood is a grammatical feature of verbs, used for signaling *modality*. It is the use of verbal inflections that allow speakers to express their attitude toward what they are saying (e.g. a statement of fact, of desire, of command, etc.). A clause containing a verb in the indicative mood denotes a real (or realis) event or state occurring at a specific point in time; A Subjunctive clause by contrast denotes a hypothetical or unreal (= irrealis) event or state which has not yet occurred and which may never occur.

(a) She was glad <i>that he apologised</i>	The verb <i>apologised</i> is finite by virtue of being inflected for past tense and indicative mood, and by virtue of having a nominative subject (he)
(b) She demanded <i>that he apologise</i>	The verb <i>apologise</i> is finite by virtue of being inflected for subjunctive mood (and perhaps present tense), and by virtue of having a nominative subject (he).
(c) I can't imagine him <i>apologising</i>	Nonfinite clause/ a tense less and mood less gerund form, and has an accusative subject him
(d) It would be sensible <i>for him to apologise</i>	Nonfinite clause/a tenseless and moodless infinitive form/an accusative subject him
(e) It's important <i>to know when to apologise</i>	Nonfinite clause/a tenseless and moodless infinitive form a 'silent' (implicit) subject in

A finite clause determines the kind of subject it can have. Finite clauses can have a nominative pronoun like *he* as their subject, but nonfinite clauses cannot. Whether a particular clause is finite or not is to see whether it can have a nominative pronoun (like I/we/he/she/they) as its subject. In addition to being finite or nonfinite, each clause within a sentence has a specific force.

Specific force

(a) He went home	Declarative in force, in that it is used to make a statement.
(b) Are you feeling OK?	Interrogative in force in that it is used to ask a question.
(c) You be quiet!	Imperative in force, by virtue of being used to issue an order or command.
(d) What a great idea that is!	Exclamative in force, in that it is used to exclaim surprise or delight.

In complex sentences, each clause has its own force

(a) He asked where she had gone	The main (asked) clause is declarative, whereas the complement(gone) clause is interrogative.
(b) Did you know that he has retired?	The main (know) clause is interrogative, whereas the complement (retired) clause is declarative.
(c) Tell her what a great time we had!	The main (tell) clause is imperative, whereas the complement (had) clause is exclamative.

From the perspective of traditional grammar, the syntax of a language is described in terms of a taxonomy (i.e. a classificatory list) of the range of different phrase-, clause- and sentence types found in the language. This reflects the fact that the primary goal of traditional grammar is description rather than explanation.

Lesson 22

Constituents II: Phrases I

Topic: 127: Syntax: Introduction

What is Syntax?

In linguistics, syntax is the set of rules, principles, and processes that govern the structure of sentences in a given language, usually including word order. The term syntax is also used to refer to the study of such principles and processes. The goal of many syntacticians is to discover the syntactic rules common to all languages. ‘Syntax’ also means ‘sentence construction’: how words group together to make phrases and sentences.

Some people use the term grammar means syntax, Most linguists follow the more recent practice whereby the grammar of a language includes all of its organizing principles: information about the sound system, about the form of words, how we adjust language according to context, and so on; syntax is only one part of this grammar. The term ‘syntax’ is also used to mean the study of the syntactic properties of languages. In this sense, it’s used in the same way as we use ‘stylistics’ to mean the study of literary style.

The scope of syntax study includes the classification of words, the order of words in phrases and sentences, the structure of phrases and sentences, and the different sentence constructions that languages use. One basic description of a language's syntax is the sequence in which the subject (S), verb (V), and object (O) usually appear in sentences. Over 85% of languages usually place the subject first, either in the sequence SVO or the sequence SOV.

The other possible sequences are VSO, VOS, OVS, and OSV, the last three of which are rare. In most generative theories of syntax, these surface differences arise from a more complex clausal phrase structure, and each order may be compatible with multiple derivations. Usually, we think of syntax as "word order." However, syntax is also achieved in some languages such as Latin by inflectional case endings. A sentence could make no sense and still be correct from the syntax point of view as long as words are in their appropriate spots and agree with each other.

Here is a classic example by Noam Chomsky, a linguist that illustrates a case in which a sentence is correct but does not make sense:

Colorless green ideas sleep furiously. In a language such as English, the main device for showing the relationship among words is word order; e.g., in “The girl loves the boy,” the subject is in initial position, and the object follows the verb.

- Transposing them changes the meaning.

In Latin, for example, “The girl loves the boy” may be

- *puella puerum amat* “the girl” the boy loves
- *puerum puella amat*
- *amat puella puerum,*
- *amat puerum puella,*
- *puella amat puerum.*
- The meaning remains constant .

In the sentence “My dog is playing in the yard” there is a closer relationship between the words “is playing,” which together form the verb, than between the words “playing in the,” which form only part of the verb and part of the phrase indicating the location of the playing. The study of syntax also includes the investigation of the relations among sentences that are similar, such as “John saw Mary” and “Mary was seen by John.” Syntax received much attention after 1957, when the American linguist Noam Chomsky proposed a radically new theory of language, transformational grammar.

Topic: 128: Some Concepts and Misconceptions

It is parents or by teachers at school. All developing children in every culture learn their native language or languages to perfection without any formal teaching. The simple exposure is required to ordinary, live, human language within a society. No culture or cultures teach their children ‘grammar’.

All children must be capable of constructing a mental grammar of their native languages without any formal instruction. Most linguists now believe that human infants are born pre-programmed to learn language the way as they are pre-programmed to walk upright. Children are not taught their native grammar the way they are not taught by their parents how to walk. He’s more happier than what I am, or when your school teachers tried to stop you from using a preposition to end a sentence with? In just the same way, some people have the idea that certain forms of language are more beautiful, or classier, or are simply ‘correct’.

But the belief that some forms of language are better than others has no linguistic basis

Since we often make social judgements about people based on their accent or dialect, we tend to transfer these judgements to their form of language. We may then think that some forms are undesirable, that some are ‘good’ and some ‘bad’. For a linguist, however, dialectal forms of a language don’t equate to ‘bad grammar’. Some people think examples *I didn’t do nothing wrong* are non-standard English are sloppy speech, or perhaps illogical.

- Middle English (the English of roughly 1100 to 1500) also had a double negative.

Ironically for the ‘logic’ argument, the variety of French that has the double negative is the most formal and prestigious variety, whereas colloquial French typically drops the initial negative word. Socially stigmatized forms of language are potentially just as ‘logical’ as standard English.

Speakers of non-standard dialects are, of course, following a set of mental rules, in just the same way that speakers of the most prestigious dialects are. The various dialects of a language in fact share the majority of their rules, and diverge in very few areas, but the extent of the differences tends to be exaggerated because they arouse such strong feelings. In sum, speakers of prestige dialects may feel that only their variety of English is ‘grammatically correct’, but these views cannot be defended on either logical or linguistic grounds.

Topic: 129: Use of Linguistic Examples

We have used examples from a wide variety of languages, including English. It is difficult to study examples from unfamiliar languages, and perhaps you wonder why we don't just use examples from English.

- Two main reasons for using foreign-language examples:
- to learn about the differences between languages,
- to learn about the similarities between them.



First, languages don't all look the same, and examining just English language and its immediate relatives doesn't show how much languages can differ. Imagine two languages, English and German, two closely related Germanic languages from northern Europe.

- a word-for-word translation.
- der schone Wasserfall (German)
- the pretty waterfall

ENGLISH and German languages are closely related while Spanish find difference in its translation

The translation of this phrase would look the same in any language: But this is not so. You might imagine that the translation of this phrase would look the same in any language: first a word for 'the', then a word for 'pretty' or 'beautiful', then a word for 'waterfall'. But this is not In Spanish, for instance, we'd get:

- la cascada Hermosa
- the waterfall beautiful

'the beautiful waterfall'

- The world's languages have many interesting and important syntactic features that you should know about.
- English has some but not all of these features, so if we only looked at English we'd miss out on the rest.

Let's see one example, from Spanish.

- Es nuevo. (Spanish)

Complete sentence

- is new

Not complete

It's new.

It literally means 'is new' – an impossible sentence in English. Spanish typically drops the subject pronoun meaning 'it' in such examples; for this reason, it's known as a pro-drop language. Many languages have examples parallel to this, but confining the discussion to English would never reveal that. In yet other languages, such as Arabic and Indonesian, the three-word English sentence It is new translates as 'It new'.

These simple examples show that we can't expect sentences in other languages to be word-for-word. The second reason is that languages share common properties that languages share – their homogeneity or sameness. One of the most important discoveries of modern linguistics is that languages don't vary from each other at random, but are remarkably alike. Most linguists want to uncover the central patterns common to all languages.

Although specific constructions are not universal (= common to all languages), all languages use a sub-set of the same basic tools of grammar. Each language has a wordlist or lexicon which all its speakers share, and that wordlist always contains words from several different classes. All languages combine these words into phrases and sentences. Languages manipulate the order of the phrases for various purposes –perhaps to ask questions, or to emphasize different parts of a sentence, or to show who's doing what to whom. This is syntax, and it forms the subject matter of our discussion.

Topic: 130

Speakers manipulate sentences in all sorts of ways because they're trying to convey different meanings. Syntax allows speakers to express all the meanings that they need to put across.

In the simplest cases, this might mean altering the basic word order of a sentence, to emphasize or downplay a particular phrase, or to ask a question, or else grouping words together in different ways to modify the meaning. A preliminary idea in languages demonstrates that languages really do have syntactic structure. Each of the word orders in the following sentences is attested among the world's languages, though some are much more common than others. The three most common basic word orders in languages other than English are those of (a,b and (c).

Kim drank the tea.

Kim, the one drinking the tea;	Drank, the verb, which expresses what Kim did;	The tea, expressing what is being drunk.
	a. Kim drank the tea.	
	b. *Kim the tea drank.	Japanese has this basic word order of (b).
	c. *Drank Kim the tea.	Welsh has this basic order of (c).
	d. *Drank the tea Kim.	Malagasy, spoken in Madagascar, has this basic order in(d).

e. *The tea drank Kim.	Two word orders in (e and f) are the rarest basic word orders in the languages of the , found in the Carib language family of the Amazon basin. For example, Hixkaryana
f. The tea Kim drank.	

- Possible to determine the basic, neutral word order in a language,
- But the flexibility or rigidity of the basic word order differs widely among the world's languages.
- English has a fixed basic word order,
- Russian a very flexible word order, ✓
- Japanese allows many different orders
- *Drank Kim the tea.

In English, this ungrammatical word orders might be permissible in poetry, but not in the spoken language or in prose.

f. The tea Kim drank.

- May sound odd be used to focus on what it was that Kim drank; the phrase the *tea* is fronted from its usual position as given
- *a Kim drank the tea.*

Try adding a bit of context: Kim visits an eccentric aunt who makes tea and beer out of strange garden plants: The tea, Kim drank____, but the home-made nettle beer, she really hated____.

- The gaps are used to show the normal position of the fronted phrases
- The last exercise, a stall while climbing, I didn't do ____ well.
- Object-fronting is, in fact, quite rare in English.

It's known as a marked (= unusual) construction, while the usual basic word order as in (a) is termed unmarked. Often there are stylistic reasons for changing basic word order. The fronted phrase in is rather long, and sounds clumsy in the usual object position: I didn't do the last exercise, a stall while climbing, well. A different kind of word order change, which involves breaking up a rather long phrase by

- Estimates____ vary greatly about the number of fluent speakers (i.e. of Esperanto).

This avoids the clumsiness of a long initial phrase *estimates about the number of fluent speakers* before the short phrase *vary greatly*. (Compare the normal word order in *Estimates about this vary greatly*.) As showed, English has a generally inflexible word order in the

sentence, but optional modifying phrases can be reordered quite easily, as is the case for the *about ... phrase* which modifies (= expands on) the word *estimates*.

Topic: 131: All Languages have Structure

All languages, whether living or dead, have syntactic structure, including sign language ; (such as British Sign Language). A language doesn't just consist of strings of words, but that the words group together to form larger Linguists describe this phrases-within-phrases pattern as hierarchical structure. One kind of hierarchical structure is seen in embedded sentences, a construction in which sentences occur within other sentences, such as *Chris told Lee [Kim couldn't swim]*.

This property is known as recursion. Here, the sentence in brackets – *Kim couldn't swim* – is the embedded sentence.

- It serves to tell you what it was that Chris told Lee.
- I wonder [*if Lee will arrive late*].
- The claim [*that she doesn't like Kim*] is very surprising.
- [*That we've no coffee left*] isn't my fault.
- We asked [*how to get to the station*].
- No limits to the number of embedded sentences that can be strung together.
- Kim couldn't swim
- Lee thought *that Kim couldn't swim*,
- I said *that Lee thought that Kim couldn't swim*.

a. I charged up the battery.

b. I charged up the street.

At first glance these sentences appear to be structurally identical charge means something different in (a) and (b), The only difference seems to be that street replaces battery. And yet the syntactic behaviour of the two sentences is entirely different. As always, the asterisks indicate ungrammatical sentences:

The only difference seems to be that street replaces battery. And yet the syntactic behaviour of the two sentences is entirely different.

a. I charged the battery up.

b. *I charged the street up.

a. *It was up the battery I charged.

b. It was up the street I charged.

a.*I charged up Lee's battery and (then) up Kim's too.

a. I [charged up] the battery.

b. I charged [up the street].

Charge up is phrasal verb the only thing you can do with a battery is charge it up; you can't charge it down, over, across, or anything else. In (b), it doesn't: instead, there's a syntactic unit *up the street*, which can be moved around the sentence for focus: *It was up the street I charged*.

- As a second demonstration of syntactic structure, possessive -'s as in *Lee's friend*.
- You might assume at first that this possessive ending simply attaches to a noun, a word such as *Lee* or *government*, as in *the government's dilemma*.

But consider:

a. I saw the woman next door's children.

b. What was that guy who retired last month's name?

When, as linguists, we try to figure out the syntactic structures of a language, we rely on the judgements of native speakers to tell us whether our example sentences are possible or impossible. These grammaticality judgements, along with examples that are collected from a spoken or written corpus of the language, form the data of the science of linguistics. It doesn't matter that native speakers usually can't tell us why they feel that a particular sentence is good or bad; the very fact that they have these intuitions shows up the structural differences and similarities between sentences.

Topic: 132: Promotion and Demotion

The syntactic variations involved simply reordering the elements of a sentence. Syntactic changes can have much more radical results than this. Let's discuss the idea of promotion and demotion processes—making a word or phrase more prominent or less prominent in the sentence. The demotion processes make part of the sentence less prominent.

Ali	menga-kirim	surat	itu	kepada	Hasan

Ali	send	letter	the	to	Hassan
-----	------	--------	-----	----	--------

Ali	sent	the	letter	to	Hassan
Ali	menga-kirima-kan	Hassan	surat	itu	
Ali	send	Hassan	letter	the	
Ali	sent	Hassan	the	letter	

In (1b), we find an ending -kan on the word for ‘send’: this ending indicates in Indonesian that the word Hasan has been promoted.

- English has no equivalent to-kan.
- Hasan can be promoted it to an even higher position in the following: Hasan was sent____the letter by Ali.
- Hasan was sent____the letter by Ali.
- We indicate the position that Hasan is understood to have moved from with the
- gap____.

To understand why a language would need to indicate this promotion of some part of the sentence, think about the difference in meaning between *Hasan sent the letter* and *Hasan was sent the letter*.

- Hasan was sent____the letter by Ali.

Here is another construction involving promotion and demotion – the passive – in English and Japanese.

(1) The women and boys with crates converged on the boats **and their catch was counted out by the market boss.**

(2) His normal work was filing girls’ teeth to points, although **pointed gnashers were considered a bit old-fashioned by the girls here.**

Compare these passive constructions with the sentences in (3) and (4), which are their counterparts in meaning, but are both active constructions:

(3) The market boss counted out **their catch**.

(4) The girls here considered **pointed gnashers** a bit old fashioned.

First, assume that active sentences are the more basic; they are learnt much earlier by children than are passives.

Two properties of the passive occur in any language which has the construction:

(i) the passive involves promotion of an object phrase to a new position in the sentence, known as the subject position, and

(ii) the phrase that used to be in the subject position undergoes demotion.

- Demotion of the subject phrase to the object position.
- The by-phrase is entirely optional.
- It could be omitted Their catch was counted out.
- Compare *The market boss counted out their catch* and *Their catch was counted out*.
- Both the subject *the market boss* and the object *their catch* are in the sentence, and neither can be omitted.



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